

Vector Bundles on Curves

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Introduction

The course aims at introducing vector bundles on curve and Koszul cohomology. You may benefit from the following references

- *Ample Vector Bundles*, Hartshorne
- *Koszul Cohomology and Algebraic Geometry*, Aprodu-Nagel
- *Lectures on Vector Bundles*, J.Le Potier
- *The Geometry of Moduli Space of Sheaves*, Huybrechts-Lehn.

1 Vector Bundles

1.1 Sheaf theory

Recall the definition of sheaves.

Definition 1.1. Let X be a topological space and $\mathcal{C}$ be a category. Assume $op(X)$ is the category of open subsets of X with morphisms are inclusions. A presheaf on X valued on $\mathcal{C}$ (or say of $\mathcal{C}$) is a contravariant functor $\mathcal{F} : op(X) \rightarrow \mathcal{C}$.

Remark 1.1. Though we can consider this general definition of presheaf, we would only consider presheaves at least valued on Ab , where Ab is the category of abelian groups. In addition, we need these presheaves map $\emptyset$ to $\{0\}$.

Definition 1.2. Let X be a topological space, $\mathcal{F}$ presheaf of abelian groups on X . For $V \subseteq U$, we call the correspondence homomorphism $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ a restriction map, denoted by $\cdot \rho_{UV}$. Elements in $\mathcal{F}(U)$ is called a section of $\mathcal{F}$ on U . In particular, if $U = X$, we call elements of $\mathcal{F}(X)$ global sections, otherwise local sections.

Remark 1.2. For convenience, for $s \in \mathcal{F}(U)$, we would write $s|_V$ to denote image of s instead of $\rho_{UV}(s)$. Also, sometimes $\mathcal{F}(U)$ can be rewritten as $\Gamma(U, \mathcal{F})$.

Definition 1.3. Let X be a topological space, $\mathcal{F}$ presheaf of abelian groups on X . $\mathcal{F}$ is a sheaf on X if it satisfies that

- (1) For each open subset U with open covering $\{U_i\}_{i \in I}$ and $s \in \mathcal{F}(U)$, if $s|_{U_i} = 0$ for all i , then $s = 0$.
- (2) For each open subset U with open covering $\{U_i\}_{i \in I}$, if there exists $s_i \in \mathcal{F}(U_i)$ for all i , satisfying $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j , then there exists $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i .

Remark 1.3. The two conditions in fact give a exact sequence for each open subset U with open covering $\{U_i\}_{i \in I}$.

$$0 \longrightarrow \mathcal{F}(U) \longrightarrow \prod_{i \in I} \mathcal{F}(U_i) \longrightarrow \prod_{i, j \in I} \mathcal{F}(U_i \cap U_j) \quad (1)$$

Example 1.1. (1) Let X be a topological space. Then $U \mapsto \{\text{continuous functions } U \rightarrow \mathbb{R}\}$ give a sheaf on X .

(2) Let $X = \mathbb{R}^n$. Then $U \mapsto \{C^\infty\text{-functions } U \rightarrow \mathbb{R}\}$ gives a sheaf on X .

(3) Let X be a topological space and A be an abelian group. View A as a topological space with discrete topology. Then $U \mapsto \{\text{locally constant functions } U \rightarrow A\}$ gives a sheaf on X , which is called the constant sheaf of A on X .

Definition 1.4. Let X be a topological space, $\mathcal{F}$ (pre)sheaf of abelian groups on X . For $x \in X$, define the stalk of $\mathcal{F}$ at x to be $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$. More concretely, an element of $\mathcal{F}_x$ is represented by a pair $(U, s) = s_x$ where U is an open neighbourhood of x and $s \in \mathcal{F}(U)$, called a germ. Two germs (U, s) and (V, t) represent the same element if there exists open neighbourhood W of x such that $W \subseteq U \cap V$ and $s|_W = t|_W$.

Proposition 1.1. Let X be a topological space, $\mathcal{F}$ sheaf of abelian groups on X . Suppose U is an open subset of X and $s, t \in \mathcal{F}(U)$. Then $s = t$ if and only if $s_x = t_x$ for all $x \in U$.

Proof. “ $\Rightarrow$ ”: Obviously.

“ $\Leftarrow$ ”: Since for all $x \in U$, we have $s_x = t_x$, there exists open neighbourhood W_x of x such that $W_x \subseteq U$ and $s|_{W_x} = t|_{W_x}$. Note that $\{W_x\}_{x \in U}$ form an open covering of U . By property of sheaf, we get $s = t$. $\square$

Definition 1.5. Let X be a topological space, $\mathcal{F}, \mathcal{G}$ (pre)sheaves of abelian groups on X . A morphism φ from $\mathcal{F}$ to $\mathcal{G}$ is a family of group homomorphisms $\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ such that for $V \subseteq U$, there is a commutative diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi(U)} & \mathcal{G}(U) \\ \downarrow \rho_{UV} & & \downarrow \rho_{UV} \\ \mathcal{F}(V) & \xrightarrow{\varphi(V)} & \mathcal{G}(V) \end{array} \quad (2)$$

Remark 1.4. By property of direct limit, morphism φ gives homomorphism $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ for all $x \in X$.

Definition 1.6. Let X be a topological space, $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ morphism of (pre)sheaves of abelian groups on X . We say φ is isomorphic if there exists morphism $\psi : \mathcal{G} \rightarrow \mathcal{F}$ such that for each open subsets U , $\psi(U)\varphi(U) = \text{id}_{\mathcal{F}(U)}$ and $\varphi(U)\psi(U) = \text{id}_{\mathcal{G}(U)}$.

Proposition 1.2. Let X be a topological space, $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ morphism of sheaves of abelian groups on X . Then φ is isomorphic if and only if φ_x is isomorphic for all $x \in X$.

Proof. “ $\Rightarrow$ ”: For all $(U, t) \in \mathcal{F}_x$, since $\varphi(V)$ is isomorphic, there exists $s \in \mathcal{F}(U)$ such that $\varphi(V)(s) = t$. Thus $\varphi_x((U, s)) = (U, t)$, get φ_x is surjective. For all $(U, s) \in \ker(\varphi_x)$, $(U, \varphi(U)(s)) = 0$ in $\mathcal{G}_x$. Then there exists open neighbourhood V of x such that $V \subseteq U$ and $\varphi(U)(s)|_V = 0$. By commutative diagram, $\varphi(U)(s)|_V = \varphi(V)(s|_V)$. Since $\varphi(V)$ is isomorphic, get $s|_V = 0$ and so $(U, s) = 0$ in $\mathcal{F}_x$. Get φ_x is isomorphic.

“ $\Leftarrow$ ”: For each open subset U , want to show that $\varphi(U)$ is isomorphic. For $s \in \ker \varphi(U)$, $\varphi(U)(s) = 0$. Then for all $x \in U$, $(U, \varphi(U)(s)) = 0$ in $\mathcal{G}_x$. As $(U, \varphi(U)(s)) = \varphi_x((U, s))$ and

φ_x is isomorphic, get $(U, s) = 0$ in $\mathcal{F}_x$. Thus there exists open neighbourhood W_x of x such that $W_x \subseteq U$ and $s|_{W_x} = 0$. By property of sheaf, get $s = 0$ and so $\varphi(U)$ is injective.

For all $t \in \mathcal{G}(U)$, (U, t) is an element of $\mathcal{G}_x$ for all $x \in U$. Then there exists $(V_x, s(x)) \in \mathcal{F}_x$ such that $\varphi_x((V_x, s(x))) = (U, t)$ and $\varphi(V_x)(s(x)) = t|_{V_x}$. Note that $\varphi(V_x \cap V_y)(s(x)|_{V_x \cap V_y}) = (t|_{V_x})|_{V_x \cap V_y} = (t|_{V_y})|_{V_x \cap V_y} = \varphi(V_x \cap V_y)(s(y)|_{V_x \cap V_y})$. Since $\varphi(V_x \cap V_y)$ is injective, get $s(x)|_{V_x \cap V_y} = s(y)|_{V_x \cap V_y}$. By property of sheaf, there exists $s \in \mathcal{F}(U)$ such that $s|_{V_x} = s(x)$. So $\varphi(U)(s)|_{V_x} = \varphi(V_x)(s(x)) = t|_{V_x}$. By property of sheaf, get $\varphi(U)(s) = t$ and so $\varphi(U)$ is isomorphic. $\square$

Definition 1.7. Let X be a topological space, $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ morphism of presheaves of abelian groups on X . We have several notions

- Define the kernel of φ to be $\ker(\varphi) : op(X) \rightarrow Ab \quad U \mapsto \ker(\varphi(U))$.
- Define the image of φ to be $im(\varphi) : op(X) \rightarrow Ab \quad U \mapsto im(\varphi(U))$.
- Define the cokernel of φ to be $coker(\varphi) : op(X) \rightarrow Ab \quad U \mapsto coker(\varphi(U))$.

Remark 1.5. The definition of cokernel in fact induce the definition of quotient presheaf.

Definition 1.8 (Sheafification). Let X be a topological space, $\mathcal{F}$ presheaf of abelian groups on X . Define a sheaf $\mathcal{F}^+$ of abelian groups on X by $U \mapsto \mathcal{F}^+(U)$, where $\mathcal{F}^+(U) = \{(s_x)_{x \in U} \mid \text{for all } x \in U, \exists \text{ open } V_x \ni x \text{ and } t(x) \in \mathcal{F}(V_x) \text{ such that } t(x)_y = s_y, \forall y \in V_x\}$ is a subset of $\prod_{x \in U} \mathcal{F}_x$.

Proposition 1.3. Let X be a topological space, $\mathcal{F}$ presheaf of abelian groups on X . Show that there is a natural morphism of presheaves $\varphi : \mathcal{F} \rightarrow \mathcal{F}^+$ satisfies that

- φ_x is identity homomorphism.
- If $\mathcal{F}$ is sheaf, then φ is isomorphism.
- For any morphism of presheaves $\psi : \mathcal{F} \rightarrow \mathcal{G}$ where $\mathcal{G}$ is a sheaf, there exists unique morphism $\phi : \mathcal{F}^+ \rightarrow \mathcal{G}$ such that the following diagram commutes.

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\psi} & \mathcal{G} \\ \searrow \varphi & & \nearrow \phi \\ & \mathcal{F}^+ & \end{array} \quad (3)$$

Reason 1.1. The construction of φ is just $\varphi(U) : s \mapsto (s_x)_{x \in U}$.

Definition 1.9. Let X be a topological space, $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ morphism of sheaves of abelian groups on X .

- (1) Define the kernel of φ to be $\ker(\varphi) : op(X) \rightarrow Ab \quad U \mapsto \ker(\varphi(U))$.
- (2) Define the image of φ to be sheafification of $op(X) \rightarrow Ab \quad U \mapsto im(\varphi(U))$, denoted by $im(\varphi)$.
- (3) Define the cokernel of φ to be sheafification of $op(X) \rightarrow Ab \quad U \mapsto coker(\varphi(U))$, denoted by $coker(\varphi)$.

Remark 1.6. The definition of cokernel in fact induce the definition of quotient sheaf.

Definition 1.10. Let X be a topological space, $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ morphism of sheaves of abelian groups on X .

(1) We say that φ is injective if $\ker(\varphi) = 0$.

(2) We say that φ is surjective if $\text{im}(\varphi) = \mathcal{G}$

Proposition 1.4. Let X be a topological space, $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ morphism of sheaves of abelian groups on X . Then

(1) φ is injective if and only if $\varphi(U)$ is injective for all open subset U .

(2) φ is injective if and only if φ_x is injective for all $x \in X$.

(3) φ is surjective if and only if φ_x is surjective for all $x \in X$.

(4) φ is isomorphic if and only if φ is injective and surjective.

Remark 1.7. The proof is similar to proof of Proposition 1.2. In addition, Hartshorne exercise II 1.2 gives a more explicit sketch.

Definition 1.11 (Direct Image and Inverse Image). Let $f : X \rightarrow Y$ be continuous map of topological spaces, $\mathcal{F}$ sheaf of abelian groups on X , $\mathcal{G}$ sheaf of abelian groups on Y .

(1) Define direct image to be the sheaf $f_*\mathcal{F} : V \mapsto f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(V))$.

(2) Define inverse image to be the sheafification of $U \mapsto \varinjlim_{V \supseteq f^{-1}(U)} \mathcal{G}(V)$, denoted by $f^{-1}\mathcal{G}$.

Remark 1.8. In general, we don't have $(f_*\mathcal{F})_{f(x)} \neq \mathcal{F}_x$, while there is a canonical homomorphism $(f_*\mathcal{F})_{f(x)} \rightarrow \mathcal{F}_x$. On the other hand, we always have $(f^{-1}\mathcal{G})_x = \mathcal{G}_{f(x)}$.

Example 1.2. Let $i : Z \hookrightarrow X$ be an inclusion of topological spaces, $\mathcal{F}$ sheaf of abelian groups on X . We often write $i^{-1}\mathcal{F}$ as $\mathcal{F}|_Z$.

1.2 Sheaf modules

Definition 1.12. Let X be a scheme or $(X, \mathcal{O}_X)$ be a ringed space. A sheaf of $\mathcal{O}_X$ -modules is a sheaf of abelian groups $\mathcal{F}$ on X such that for all $U \subseteq X$ open, $\mathcal{F}(U)$ is a module over $\mathcal{O}_X(U)$ which is compatible with restriction maps.

Definition 1.13. Let X be a scheme or $(X, \mathcal{O}_X)$ be a ringed space, $\mathcal{F}, \mathcal{G}$ two $\mathcal{O}_X$ -modules. A morphism φ from $\mathcal{F}$ to $\mathcal{G}$ is a morphism of sheaves compatible with the module structure i.e. for all $U \subseteq X$ open, $\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is a homomorphism of $\mathcal{O}_X(U)$ -modules.

Remark 1.9. Kernel, image and cokernel of a morphism of $\mathcal{O}_X$ -modules are still $\mathcal{O}_X$ -modules. Can also define submodule, quotient module and direct sum.

Definition 1.14. Let X be a scheme or $(X, \mathcal{O}_X)$ be a ringed space. An sheaf ideal on X is an $\mathcal{O}_X$ -submodule of $\mathcal{O}_X$.

Definition 1.15 (Tensor Products). Let X be a scheme or $(X, \mathcal{O}_X)$ be a ringed space, $\mathcal{F}, \mathcal{G}$ two $\mathcal{O}_X$ -modules. Define tensor product $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ to be the sheafification of $U \mapsto \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$.

Remark 1.10. The stalk of $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ at $x \in X$ is just $\mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{G}_x$.

Definition 1.16 (Free $\mathcal{O}_X$ -modules). Let X be a scheme or $(X, \mathcal{O}_X)$ be a ringed space, $\mathcal{F}$ an $\mathcal{O}_X$ -module. We say that $\mathcal{F}$ is free if $\mathcal{F}$ is isomorphic to a direct sum of $\mathcal{O}_X$. In addition, we say that $\mathcal{F}$ is free of rank r if $\mathcal{F} \cong \mathcal{O}_X^r$.

Definition 1.17 (Locally Free $\mathcal{O}_X$ -modules). Let X be a scheme or $(X, \mathcal{O}_X)$ be a ringed space, $\mathcal{F}$ an $\mathcal{O}_X$ -module. We say that $\mathcal{F}$ is locally free if X can be covered by open subsets $\{U_i\}_{i \in I}$ such that $\mathcal{F}|_{U_i}$ is free as $\mathcal{O}_{U_i}$ -module. In addition, we say that $\mathcal{F}$ is locally free of rank r if X can be covered by open subsets $\{U_i\}_{i \in I}$ such that $\mathcal{F}|_{U_i} \cong \mathcal{O}_{U_i}^r$ as $\mathcal{O}_{U_i}$ -module.

Definition 1.18 (Invertible $\mathcal{O}_X$ -modules). Let X be a scheme or $(X, \mathcal{O}_X)$ be a ringed space, $\mathcal{F}$ an $\mathcal{O}_X$ -module. We say that $\mathcal{F}$ is invertible if it is locally free of rank 1.

Remark 1.11. If take $\mathcal{G} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{O}_X)$, then $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G} \cong \mathcal{O}_X$.

Lemma 1.1. Let $f : X \rightarrow Y$ be a morphism of schemes or ringed spaces, $\mathcal{F}$ an $\mathcal{O}_X$ -module, $\mathcal{G}$ an $\mathcal{O}_Y$ -module. Then the direct image $f_*\mathcal{F}$ of $\mathcal{F}$ is an $\mathcal{O}_Y$ -module induced by $f^\sharp$. On the other hand, the inverse image $f^*\mathcal{G} = f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$ of $\mathcal{G}$ is an $\mathcal{O}_X$ -module where $f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$ is induced by $f^\sharp$.

1.3 Quasi-coherent and coherent sheaves

Definition 1.19. Let A be a ring, $X = \text{Spec } A$, $M \in \mathbf{Mod}_A$. Define an $\mathcal{O}_X$ -module $\widetilde{M}$ on X by the same process as $\mathcal{O}_X$ replacing A by M .

Remark 1.12. The definition gives a fully faithful functor from $\mathbf{Mod}_A$ to category of $\mathcal{O}_{\text{Spec}(A)}$ -modules $M \mapsto \widetilde{M}$.

Proposition 1.5. Let A be a ring, $X = \text{Spec } A$, $M \in \mathbf{Mod}_A$. Then

- (1) $\widetilde{M}(X) = M$,
- (2) $\widetilde{M}(D(f)) = M_f = M \otimes_A A_f$
- (3) $\widetilde{M}_{\mathfrak{p}} = M_{\mathfrak{p}} = M \otimes_A A_{\mathfrak{p}}$.

Proposition 1.6. Let A be a ring, $X = \text{Spec } A$. Then

- (1) Let M_i be A -modules. Then $\widetilde{\bigoplus_i M_i} \cong \bigoplus_i \widetilde{M_i}$.
- (2) Let M, N be A -modules. Then $\widetilde{M \otimes_A N} \cong \widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N}$.
- (3) Let L, M, N be A -modules. Then $L \rightarrow M \rightarrow N$ is exact if and only if $\widetilde{L} \rightarrow \widetilde{M} \rightarrow \widetilde{N}$ is exact.
- (4) Let $f : \text{Spec}(B) \rightarrow \text{Spec } A$ be a morphism of affine schemes corresponding to ring homomorphism $\varphi : A \rightarrow B$, $M \in \text{Mod}_A$, $N \in \text{Mod}_B$. Then $f_*\widetilde{N} \cong \widetilde{N}$, where N is viewed as A -module, and $f^*\widetilde{M} \cong \widetilde{M \otimes_A B}$.

Definition 1.20 (Quasi-coherent Sheaves). Let X be a scheme, $\mathcal{F}$ a sheaf of abelian groups on X . We say that $\mathcal{F}$ is quasi-coherent if for all affine open subset $U = \text{Spec } A \subseteq X$, $\mathcal{F}|_U \cong \widetilde{M}$ for some A -module M .

Definition 1.21 (Coherent Sheaves). Let X be a noetherian scheme, $\mathcal{F}$ a sheaf of abelian groups on X . We say that $\mathcal{F}$ is coherent if for all affine open subset $U = \text{Spec } A \subseteq X$, $\mathcal{F}|_U \cong \widetilde{M}$ for some finite A -module M .

Theorem 1.1. Let X be a scheme, $\mathcal{F}$ an $\mathcal{O}_X$ -module. Suppose X admits an affine open subset covering $\{U_i = \text{Spec}(A_i)\}_i$ such that for all i , $\mathcal{F}|_{U_i} \cong \widetilde{M_i}$ for some A_i -module M_i . Then $\mathcal{F}$ is quasi-coherent. If moreover X is noetherian and M_i are finite over A_i , then $\mathcal{F}$ is coherent.

Proof. Let $U = \text{Spec } A \subseteq X$ be an affine open subset. Cover each $U \cap U_i$ with $\{U_{ik}\}$ such that $U_{ik} \subseteq U_i$ of the form $D(g_{ik})$. Then $\mathcal{F}|_{U_{ik}} \cong \widetilde{M_{ig_{ik}}}$. Can assume $X = U = \text{Spec } A$ and finitely many affine open subset U_i such that $\mathcal{F}|_{U_i} \cong \widetilde{M_i} = \widetilde{\mathcal{F}(U_i)}$.

To prove $\mathcal{F} \cong \widetilde{\mathcal{F}(X)}$, it suffices to show that for all $f \in A$, $\mathcal{F}(X)_f = \mathcal{F}(X) \otimes_A A_f \rightarrow \mathcal{F}(D(f))$ is isomorphic. Set $V_i = U_i \cap D(f) = D(f|_{U_i})$. As tensor product commutes with finite direct sum, there is commutative diagram with exact rows,

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}(X)_f & \longrightarrow & \prod_i \mathcal{F}(U_i)_f & \longrightarrow & \prod_{i,j} \mathcal{F}(U_i \cap U_j)_f \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{F}(D(f)) & \longrightarrow & \prod_i \mathcal{F}(V_i) & \longrightarrow & \prod_{i,j} \mathcal{F}(V_i \cap V_j) \end{array} \quad (4)$$

Similar to proof of Lemma 2.6, get $\mathcal{F}$ is quasi-coherent.

Further if A is noetherian, $\mathcal{F} = \widetilde{M}$. Can cover X by finitely many $\{D(f_i)\}$ such that $\mathcal{F}(D(f_i)) = M_{f_i}$ is finite over A_{f_i} . Assume $M_{f_i} = \sum_j \frac{m_{ij}}{1} A_{f_i}$. Take $N = \sum_{i,j} m_{i,j} A$. Then N is a finitely generated A -submodule of M and for all i , $N_{f_i} = M_{f_i}$. Thus there is a natural morphism $\widetilde{N} \rightarrow \widetilde{M}$ and $\widetilde{N}|_{D(f_i)} \xrightarrow{\sim} \widetilde{M}|_{D(f_i)}$ is isomorphic for all i . As $d(f)_i$ cover X , get $\widetilde{N} \rightarrow \widetilde{M}$ is isomorphic so that $N = M$ and M is finitely generated. Thus $\mathcal{F}$ is coherent. $\square$

Proposition 1.7. Let X be a scheme. Then

- (1) Kernel, image and cokernel of morphism of quasi-coherent sheaves are still quasi-coherent.
- (2) Direct sum of quasi-coherent sheaves is quasi-coherent.
- (3) Tensor product of two quasi-coherent sheaves is quasi-coherent.

Remark 1.13. If X is moreover noetherian, then (1) and (3) still hold for coherent and (2) is also ok if the direct sum is finite.

Proposition 1.8. Let X be an affine scheme. Then for all exact sequence of quasi-coherent sheaves

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0 \quad (5)$$

the sequence of global sections

$$0 \longrightarrow \mathcal{F}(X) \longrightarrow \mathcal{G}(X) \longrightarrow \mathcal{H}(X) \longrightarrow 0 \quad (6)$$

is exact.

Proposition 1.9. *Let X be a scheme, $\mathcal{F}$ an $\mathcal{O}_X$ -module. Then $\mathcal{F}$ is quasi-coherent if and only if for all $x \in X$, there exists open neighbourhood $U \ni x$ such that there is an exact sequence*

$$\mathcal{O}_U^{\oplus J} \longrightarrow \mathcal{O}_U^{\oplus I} \longrightarrow \mathcal{F}|_U \longrightarrow 0 \quad (7)$$

If moreover X is noetherian, then $\mathcal{F}$ is coherent if and only if for all $x \in X$, there exists open neighbourhood $U \ni x$ such that there is an exact sequence with I, J finite

$$\mathcal{O}_U^{\oplus J} \longrightarrow \mathcal{O}_U^{\oplus I} \longrightarrow \mathcal{F}|_U \longrightarrow 0 \quad (8)$$

Proof. " $\Rightarrow$ ": For $\mathcal{F}$ quasi-coherent and all $x \in X$, there is affine open neighbourhood $U = \text{Spec } A \ni x$ such that $\mathcal{F}|_U \cong \widetilde{M}$ for some A -module M . Note that there is an exact sequence for some index sets I, J

$$A^{\oplus J} \longrightarrow A^{\oplus I} \longrightarrow M \longrightarrow 0 \quad (9)$$

By Proposition 1.6 (3), get exact sequence

$$\mathcal{O}_U^{\oplus J} \longrightarrow \mathcal{O}_U^{\oplus I} \longrightarrow \mathcal{F}|_U \longrightarrow 0 \quad (10)$$

" $\Leftarrow$ ": For all $x \in X$, there exists open neighbourhood $U \ni x$ such that there is an exact sequence

$$\mathcal{O}_U^{\oplus J} \longrightarrow \mathcal{O}_U^{\oplus I} \longrightarrow \mathcal{F}|_U \longrightarrow 0 \quad (11)$$

Can assume that U is affine. By Proposition 1.9, get exact sequence

$$\mathcal{O}_U(U)^{\oplus J} \longrightarrow \mathcal{O}_U(U)^{\oplus I} \longrightarrow \mathcal{F}|_U(U) \longrightarrow 0 \quad (12)$$

By Proposition 1.6, get exact sequence

$$\mathcal{O}_U^{\oplus J} \longrightarrow \mathcal{O}_U^{\oplus I} \longrightarrow \widetilde{\mathcal{F}|_U(U)} \longrightarrow 0 \quad (13)$$

Thus $\mathcal{F}|_U \cong \widetilde{\mathcal{F}|_U(U)}$. By Theorem 1.1, $\mathcal{F}$ is quasi-coherent. For coherent, the same argument would make sense. $\square$

Example 1.3. (1) Let X be a scheme. Then locally free $\mathcal{O}_X$ -modules are quasi-coherent.
 (2) Let X be a noetherian scheme. Then locally free $\mathcal{O}_X$ -modules of rank $< \infty$ are coherent.
 (3) Let $i : Y \longrightarrow X$ be a closed immersion with X, Y noetherian. Then $i_*\mathcal{O}_Y$ is coherent on X .

(4) Let X be an integral noetherian scheme with function field K . Then constant sheaf K_X is quasi-coherent but not coherent in general.

(5) Let X be a scheme, $U \subseteq X$ open subset with inclusion $j : U \longrightarrow X$, $\mathcal{F}$ sheaf on U .

Define $j_!\mathcal{F}$ to be the sheafification of $V \mapsto \begin{cases} \mathcal{F}(V) & \text{if } V \subseteq U \\ 0 & \text{otherwise} \end{cases}$. Then stalk $(j_!\mathcal{F})_x =$

$\begin{cases} \mathcal{F}_x & \text{if } x \in U \\ 0 & \text{otherwise} \end{cases}$. Now let $X = \text{Spec } A$ be an affine scheme with A integral domain, $U \subsetneq X$

nonempty open subset with inclusion $j : U \longrightarrow X$. Then $j_!\mathcal{O}_U$ is an $\mathcal{O}_X$ -module but not quasi-coherent.

Now assume that S is a graded ring. Let $M = \bigoplus_{d \in \mathbb{Z}} M_d$ be a graded C -module. Define $\widetilde{M}$ on $X = \text{Proj}(S)$ by the same process on $\mathcal{O}$ replacing S by M . In particular, $\widetilde{M}|_{D_+(f)} = \widetilde{M(f)}$ and $\widetilde{M}_{\mathfrak{p}} = M_{(\mathfrak{p})}$ for $\mathfrak{p} \in X$. Then $\widetilde{M}$ is quasi-coherent on X and coherent if X is noetherian and M is finitely generated. In addition, we should note that $\widetilde{M}$ doesn't determine M . For instance, if $M = \bigoplus_{d \geq 0} M_d$ and $N = \bigoplus_{d \geq d_0} M_d$ for some $d_0 > 0$, then $\widetilde{M} = \widetilde{N}$.

Definition 1.22 (Twisting). Let S be a graded ring and $X = \text{Proj}(S)$, $\mathcal{F}$ an $\mathcal{O}_X$ -module. Set $\mathcal{O}_X(n) = \widetilde{S(n)}$ where $S(n)$ is the graded C -module given by $S(n)_d = S(n+d)$. Also set $\mathcal{F}(n) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$.

Proposition 1.10. Let S be a graded ring and $X = \text{Proj}(S)$. Assume that S is generated by S_1 as S_0 -algebra. Then $\mathcal{O}_X$ is an invertible sheaf. In addition, For all graded S -module M , we have $\widetilde{M(n)} = \widetilde{M}(n)$. In particular, $\mathcal{O}_X(n) \otimes_{\mathcal{O}_X} \mathcal{O}_X(m) \cong \mathcal{O}_X(m+n)$.

Proof. Since S is generated by S_1 as S_0 -algebra, $\{D_+(f)\}_{f \in S_1}$ cover X . Suffice to show for all $f \in S_1$, $S(n)_{(f)}$ is free of rank 1 over $S_{(f)}$. Note that there is a canonical isomorphism $S(n)_{(f)} \longrightarrow S_{(f)} \quad s \longmapsto f^{-n}s$, done! $\square$

Lemma 1.2 (qcqs Lemma). Let X be a scheme, $\mathcal{F}$ quasi-coherent sheaf on X , $\mathcal{L}$ invertible sheaf on X , $s \in \mathcal{L}(X)$. Assume that X is either noetherian or quasi-compact and separated over $\text{Spec}(\mathbb{Z})$. Then

- (1) Let $f \in \mathcal{F}(X)$. If $f|_{X_s} = 0$, then there exists $n > 0$ such that $f \otimes s^n = 0$ in $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)(X)$, where $X_s = \{x \in X \mid s_x \mathcal{O}_{X,x} = \mathcal{L}_x\}$ is open and $\mathcal{L}^n$ denotes $\mathcal{L}^{\otimes n}$.
- (2) Let $g \in \mathcal{F}(X_s)$. Then there exists $n_0 > 0$ such that for all $n \geq n_0$, $g \otimes (s^n|_{X_s})$ extends to a section in $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)(X)$.

Proof. Cover X by finitely many affine open subsets $\{U_i\}$ such that $\mathcal{L}|_{U_i}$ is free and generated by $e_i \in \mathcal{L}(U_i)$. Then $s_i = s|_{U_i} = h_i e_i$ for some $h_i \in \mathcal{O}_X(U_i)$. Get $X_s \cap U_i = D(h_i) \subseteq U_i$.

(1): Note that $\mathcal{F}|_{U_i} \cong \widetilde{\mathcal{F}(U_i)}$. Since $f|_{D(h_i)}$, there exists $n > 0$ such that for all i , $f_i = f|_{U_i}$ satisfies that $h_i^n f_i = 0$ in $\mathcal{F}(U_i)$. Thus $(f \otimes s)|_{U_i} = f_i \otimes s_i^n = h_i^n f_i \otimes e_i^n = 0$. As U_i cover X , get $f \otimes s^n = 0$.

(2): Set $g_i = g|_{D(h_i)}$, Then there exists $m > 0$ such that for all i , there exists $g'_i \in \mathcal{F}(U_i)$ such that $h_i^m g_i = g'_i$. Set $t_i = g'_i \otimes e_i^m \in (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^m)(U_i)$. Note that t_i and t_j coincide on $X_s \cap U_i \cap U_j$. Thus $(t_i|_{U_i \cap U_j} - t_j|_{U_i \cap U_j})|_{X_s \cap U_i \cap U_j} = 0$. As $U_i \cap U_j$ is either noetherian or affine, by (1), there exists $p > 0$ such that $(t_i|_{U_i \cap U_j} - t_j|_{U_i \cap U_j}) \otimes s|_{U_i \cap U_j}^p = 0$ in $((\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^m)|_{U_i \cap U_j} \otimes_{\mathcal{O}_{U_i \cap U_j}} \mathcal{O}_{U_i \cap U_j}^p)(\mathcal{L}|_{U_i \cap U_j})^p(U_i \cap U_j)$. While $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^m)|_{U_i \cap U_j} \otimes_{\mathcal{O}_{U_i \cap U_j}} (\mathcal{L}|_{U_i \cap U_j})^p \cong (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{m+p})|_{U_i \cap U_j}$, get $t_i \otimes s_i^p$ and $t_j \otimes s_j^p$ coincide on $U_i \cap U_j$. Thus we can glue them to obtain $t \in (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{m+p})(X)$ such that $t|_{X_s} = g \otimes (s|_{X_s}^{m+p})$. Take $n_0 = m + p$, done! $\square$

Definition 1.23. Let X be a scheme, $\mathcal{F}$ an $\mathcal{O}_X$ -module. We say that $\mathcal{F}$ is generated by global sections (or say globally generated) if there exists a family of global sections $\{s_i\}$ such that for all $x \in X$, $\{(s_i)_x\}$ generate $\mathcal{F}_x$.

Remark 1.14. Note that $\mathcal{F}$ is generated by global sections if and only if $\mathcal{F}$ is a quotient of a free $\mathcal{O}_X$ -module.

Example 1.4. (1) Let $X = \mathbb{P}_k^n$ be projective k -scheme. Then $\mathcal{O}_X(-1)$ is not generated by global sections since there is no global sections at all.

(2) Let $X = \operatorname{Spec} A$ be an affine scheme. Then all quasi-coherent sheaves on X are generated by global sections.

Definition 1.24. Let A be a ring and X be a projective scheme over A with commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{i} & \mathbb{P}_A^d \\ & \searrow & \downarrow \\ & & \operatorname{Spec} A \end{array} \quad (14)$$

Set $\mathcal{O}_X(n) = i^* \mathcal{O}_{\mathbb{P}_A^d}(n)$. For $\mathcal{F}$ an $\mathcal{O}_X$ -module, also set $\mathcal{F}(n) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$.

Lemma 1.3 (Projection Formula for Affine Case). Let $f : X \rightarrow Y$ be an affine morphism, $\mathcal{F}$ a quasi-coherent sheaf on X , $\mathcal{G}$ a quasi-coherent sheaf on Y . Then $f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^* \mathcal{G}) \cong f_* \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$.

Proof. Can assume that $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec} A$, $\mathcal{F} = \tilde{N}$ and $\tilde{M}$. Then $f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^* \mathcal{G}) \cong \tilde{P}$, where $P = N \otimes_B (M \otimes_A B)$ as A -module. On the other hand, $f_* \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G} \cong \tilde{Q}$, where $Q = N \otimes_A M$ as A -module. As $P \cong Q$, done! $\square$

Theorem 1.2 (Serre's Theorem). Let A be a noetherian ring and X be a projective scheme over A with commutative diagram,

$$\begin{array}{ccc} X & \xrightarrow{i} & \mathbb{P}_A^d \\ & \searrow & \downarrow \\ & & \operatorname{Spec} A \end{array} \quad (15)$$

Assume that $\mathcal{F}$ is a coherent sheaf on X . Then there exists $n_0 > 0$ such that for all $n \geq n_0$, $\mathcal{F}(n)$ is generated by finitely many global sections.

Proof. Since i is a closed immersion, i is finite so that $i_* \mathcal{F}$ is coherent. By Lemma 1.3, $i_*(\mathcal{F}(n)) \cong (i_* \mathcal{F})(n)$. Thus global sections of $i_*(\mathcal{F}(n))$ are equal to global sections of $(i_* \mathcal{F})(n)$ and stalks of $i_*(\mathcal{F}(n))$ are equal to stalks of $(i_* \mathcal{F})(n)$. It suffices to prove for $X = \mathbb{P}_A^d = \operatorname{Proj}(A[x_0, \dots, x_d])$.

Cover X by $\{U_i = D_+(x_i)\}$. Each $\mathcal{F}|_{U_i}$ is generated by finitely many sections $g_{ij} \in \mathcal{F}(U_i)$. By Lemma 1.2, there exists $n_0 > 0$ such that $g_{ij} \otimes x_i^{n_0}$ extends to a global section for all i, j . Thus $\mathcal{F}(n)$ is generated by these extensions. $\square$

Corollary 1.1. Let A be a noetherian ring and X be a projective scheme over A with commutative diagram,

$$\begin{array}{ccc} X & \xrightarrow{i} & \mathbb{P}_A^d \\ & \searrow & \downarrow \\ & & \operatorname{Spec} A \end{array} \quad (16)$$

Assume that $\mathcal{F}$ is a coherent sheaf on X . Then there exists $m \in \mathbb{Z}$ and $r < \infty$ such that $\mathcal{F}$ is a quotient of $\mathcal{O}_X(m)^{\oplus r}$.

Reason 1.2. Since $\mathcal{F}(n)$ is generated by global sections for large enough n , $\mathcal{F}(n)$ is quotient of $\mathcal{O}_X^{\oplus I}$ with finite index set. Tensor by $\mathcal{O}_X(-n)$, done!

1.4 Vector bundles

Let X be an algebraic variety which we mean a separated scheme of finite type over $\mathbb{C}$. And all morphisms mentioned are morphisms of varieties.

Definition 1.25. A complex linear fibration over X is a surjection $p : E \rightarrow X$ from variety E such that for all $x \in X$, the fiber E_x is an affine space. In convention, we call E the total space and X the base variety.

Example 1.5. We call $X \times \mathbb{A}_{\mathbb{C}}^r \rightarrow X$ the trivial fibration of rank r .

Lemma 1.4. Given a linear fibration $p : E \rightarrow X$, for any open subset $U \subseteq X$, the restriction $p^{-1}(U) \rightarrow U$ is also a linear fibration. Denote $E|_U = p^{-1}(U)$.

Definition 1.26. Given two linear fibrations $E \xrightarrow{p} X$ and $F \xrightarrow{q} X$, a morphism $f : E \rightarrow F$ is said to be a morphism of linear fibrations if the following diagram commutes

$$\begin{array}{ccc} E & \xrightarrow{f} & F \\ & \searrow p & \swarrow q \\ & X & \end{array} \quad (17)$$

and for all $x \in X$, $E_x \xrightarrow{f_x} F_x$ is a linear transformation.

Definition 1.27. An algebraic vector bundle of rank r on X is a linear fibration $E \xrightarrow{p} X$ which is locally trivial i.e. for all $x \in X$, there is an open neighbourhood U of x and isomorphism of linear fibrations φ_U such that the following diagram commutes

$$\begin{array}{ccc} E|_U & \xrightarrow{\varphi_U} & U \times \mathbb{A}_{\mathbb{C}}^r \\ & \searrow p & \swarrow \\ & U & \end{array} \quad (18)$$

where φ_U are called the local charts or local trivialization.

And for two such U_i and U_j with local charts φ_i and φ_j , there is an invertible matrix g_{ij} with entries regular functions on U_{ij} , called transition function, satisfying that $g_{ii} = I_r$, $g_{ij} \cdot g_{jk} = g_{ik}$ and for all affine open subset $W \subseteq U_{ij}$, $\varphi_i \circ \varphi_j^{-1}|_W$ is linear transformation defined by $g_{ij}|_W$. Equivalently, g_{ij} would correspond to a morphism $\tilde{g}_{ij} : U_{ij} \rightarrow \mathrm{GL}_r(\mathcal{O}_X(U_{ij}))$.

Lemma 1.5. Given data $\{g_{ij}\}$ satisfies that $g_{ii} = I_r$ and $g_{ij} \cdot g_{jk} = g_{ik}$, we can recover a vector bundle.

Proof. Construct E by gluing up $U_i \times \mathbb{A}_{\mathbb{C}}^r$ along $U_{ij} \times \mathbb{A}_{\mathbb{C}}^r \xrightarrow{\varphi_{ij}} U_{ij} \times \mathbb{A}_{\mathbb{C}}^r$ defined by g_{ij} . $\square$

Definition 1.28. Let E, F be two vector bundles on X of rank r and s respectively. We say a morphism $f : E \rightarrow F$ of linear fibrations is a morphism of vector bundles if there is a $r \times s$

matrix g_U with entries regular functions on U such that for all affine open subset $W \subseteq U$, restriction of the following morphism to W is linear transformation defined by $g_U|_W$

$$U \times \mathbb{A}_{\mathbb{C}}^r \xrightarrow{\varphi_U^{-1}} E|_U \xrightarrow{f|_U} F|_U \xrightarrow{\psi_U} U \times \mathbb{A}_{\mathbb{C}}^s \quad (19)$$

where φ_U and ψ_U are local charts. Equivalently, g_U would correspond to a morphism $\tilde{g}_U : U \rightarrow \mathrm{GL}_r(\mathcal{O}_X(U))$.

Remark 1.15. With Lemma 1.5, we can easily define $E \oplus F$, $E \otimes F$, $\mathcal{H}om(E, F)$, $E^\vee$, $\wedge^k E$ and $S^k E$.

Proposition 1.11. Let $f : E \rightarrow F$ be a morphism of vector bundles. Assume that for $x_0 \in X$, $f_{x_0} : E_{x_0} \rightarrow F_{x_0}$ is invertible. Then there exists an open neighbourhood U_0 of x_0 such that $f|_U : E|_U \rightarrow F|_U$ is an isomorphism.

Proof. Assume $x_0 \in U$ with local charts $\varphi_U : E|_U \rightarrow U \times \mathbb{A}_{\mathbb{C}}^r$ and $\psi_U : F|_U \rightarrow U \times \mathbb{A}_{\mathbb{C}}^s$. By definition, there is a matrix g_U and f_{x_0} is defined by $g_U(x_0)$. This immediately comes from the fact that there exists an open neighbourhood $U_0 \subseteq U$ of x_0 such that $g_U|_{U_0}$ is invertible. $\square$

Definition 1.29. Let E be a vector bundle of rank r on X . A subbundle of rank m of E is a subvariety $F \subseteq E$ such that fiber F_x is an m -dimensional affine linear subspace of E_x and the induced fibration $p|_F : F \rightarrow X$ is locally trivial.

Example 1.6. Let $X = \mathbb{P}_{\mathbb{C}}^n$ be n -dimensional projective space with homogeneous coordinate $x_0, \dots, x_n$. Consider the subvariety $\Sigma := V(x_i y_j - x_j y_i)$ of $X \times \mathbb{A}_{\mathbb{C}}^{n+1}$, where $y_0, \dots, y_n$ are coordinate of $\mathbb{A}_{\mathbb{C}}^{n+1}$. Then the fiber is a 1-dimensional affine space and hence Σ is a subbundle of rank 1 of the trivial bundle, denoted by $\mathcal{O}(-1)$.

Proposition 1.12. Let $F \subseteq E$ be a subbundle of rank m . Then there exists unique vector bundle structure on E/F satisfying the universal property that any map of vector bundles $E \rightarrow G$ vanishing over F would factor through E/F uniquely.

Proof. Firstly cover X with affine open subsets trivializing F and E , we may assume that $X = \mathrm{Spec} A$, $E = \mathbb{A}_A^n$ and $F = \mathbb{A}_A^m$. Since $F \subseteq E$ is a subbundle, by definition, for all $x \in X$, $E_x = \mathbb{A}_{k(x)}^n$ and $F_x = \mathbb{A}_{k(x)}^m$ with coordinates $x_1, \dots, x_n$ and $y_1, \dots, y_m$ respectively, satisfying that y_i can be linearly represented by x_i .

It is clear that these relations still hold over a neighbourhood $U = D(f)$ of x . Then we extend y_i to a basis $y_1, \dots, y_m, z_1, \dots, z_{n-m}$ and hence $\mathrm{Spec} A_f[x_1, \dots, x_n]/(y_1, \dots, y_m) \cong \mathrm{Spec} A_f[z_1, \dots, z_{n-m}]$ is a linear subspace. For different $D(f)$ and $D(g)$, since $z_1, \dots, z_{n-m}$ and $z'_1, \dots, z'_{n-m}$ span same linear subspace, there is an invertible transformation matrix $M \in \mathrm{Mat}_{n-m}(A_{fg})$ i.e. the transition function. Thus we defines a vector bundle structure on E/F .

Now given a map of vector bundles $E \rightarrow G$ vanishing over F . Still assume that $X = \mathrm{Spec} A$ trivializing all those bundles. Then $E \rightarrow G$ corresponds to a linear transformation $\varphi : A[w_1, \dots, w_r] \rightarrow A[x_1, \dots, x_n]$, which is defined by an $n \times r$ matrix with entries in A .

By assumption, we know y_i are not in the image of φ and hence $\text{im } \varphi$ lies in the subspace spanned by z_i . Thus φ factors a canonical map $A[w_1, \dots, w_r] \rightarrow A[z_1, \dots, z_{n-m}]$. Finally, by universal property, we know the vector bundle structure on E/F is also unique. $\square$

Proposition 1.13 (Rigidity Property). *Let $f : E \rightarrow F$ be a map of vector bundles. Suppose that rank of fiber morphisms f_x is constant as x varies over X . Then $\ker f$ and $\text{im } f$ are subbundles of E and F respectively.*

Proof. Here we only show $\ker f$ is a subbundle of E and similar argument makes sense for $\text{im } f$. We may assume that $X = \text{Spec } A$ trivializing both E and F . Now f corresponds to a linear transformation $\varphi : A[y_1, \dots, y_m] \rightarrow A[x_1, \dots, x_n]$ defined by an $n \times m$ matrix M with entries in A .

Since the rank of fiber morphism is constant, we know both the dimensions of $\text{im } \varphi_x$ are constant, denoted by $n - r$. Then we may choose $y_1, \dots, y_r$ spanning the complementary subspace of $\text{im } \varphi_x$, which can be linearly represented by x_i . And these relation still hold over an neighbourhood $U = D(f)$ of x . For different $D(f)$ and $D(g)$, since both $y_1, \dots, y_r$ and $y'_1, \dots, y'_r$ span complementary subspace of $\text{im } \varphi_{fg}$, there is an invertible transformation matrix $M \in \text{Mat}_r(A_{fg})$ i.e. the transition function. Thus we give a subbundle structure on $\ker f$. $\square$

Remark 1.16. *Though proof for $\text{im } f$ follows by similar argument, there is still a little tricky thing in that case. When we talk about $\ker f_x$, it is clear the points living outside $\text{im } \varphi_x$. But when we talk about $\text{im } f_x$, what are they? As X is locally noetherian, by a proposition in Some Useful Results, we know they are just the points living in coimage of φ_x .*

In addition, since this note is translated by me from language of varieties to language of schemes, there must be some ambiguity about the terminology. For example, here by $\ker f$, we possibly mean the subset of points whoes image under f are same to image of the origin.

Corollary 1.2. *Let $f : E \rightarrow F$ be a map of vector bundles. When f is surjective, $\ker f$ is a subbundle of E . And when f is injective, $\text{im } f$ is a subbundle of F .*

Reason 1.3. *When f is surjective, as remark above, we know $\ker \varphi = 0$ so that rank of f_x is constant. When f is injective, as proof above, we know $\text{im } \varphi = k[x_1, \dots, x_n]$ so that rank of f_x is constant.*

Definition 1.30. *Let $p : E \rightarrow X$ be a vector bundle on X of rank r . A (regular) section of E over open subset $U \subseteq X$ is a morphism $s : U \rightarrow E$ such that $(p \circ s)|_U = \text{id}_U$. Denote $\Gamma(U, E) = \{\text{regular sections of } E \text{ over } U\}$.*

Lemma 1.6. *Let $p : E \rightarrow X$ be a vector bundle on X of rank r . Then $\Gamma(U, E)$ naturally admits an $\mathcal{O}(U)$ -module structure*

$$\mathcal{O}(U) \times \Gamma(U, E) \longrightarrow \Gamma(U, E) \quad (20)$$

Moreover, $\mathcal{O}_X(E)(U) = \Gamma(U, E)$ gives a locally free sheaf $\mathcal{O}_X(E)$ of rank r .

Proof. For $\alpha \in \mathcal{O}(U)$ and section $s : U \rightarrow E$, locally s is defined by a homomorphism $\varphi : A[x_1, \dots, x_r] \rightarrow A$. Define $\alpha \cdot s$ by gluing up $\alpha \cdot \varphi : x_i \mapsto \alpha \cdot \varphi(x_i)$. In particular, locally $\Gamma(U, E) \cong A^{\oplus r}$ hence $\mathcal{O}_X(E)$ is a locally free sheaf of rank r . $\square$

Proposition 1.14. *Let $\mathcal{C}$ be the category of locally free sheaves of finite rank on X . There is a functor $\theta : \mathbf{Vect}_X \rightarrow \mathcal{C}$ sending E to $\mathcal{O}_X(E)$ and $E \rightarrow F$ to $\{\Gamma(U, E) \rightarrow \Gamma(U, F)\}$, which is an equivalence of category.*

Proof. First to prove θ fully faithful i.e. $\mathrm{Hom}_{\mathbf{Vect}_X}(E, F) \rightarrow \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X(E), \mathcal{O}_X(F))$ is bijection for all E, F . We may assume that $X = \mathrm{Spec} A$ and E, F are both trivial. Then $f : \mathbb{A}_A^n \rightarrow \mathbb{A}_A^m$ corresponds to a matrix $M \in \mathrm{Mat}_{n \times m}(A)$. Also, $(s : U \rightarrow E) \rightarrow (f \circ s : U \rightarrow F)$ also corresponds to such a matrix. Hence the bijection is clear.

Next to prove θ dense (or say essentially surjective) i.e. given $\mathcal{E} \in \mathcal{C}$, there exists $E \in \mathbf{Vect}_X$ with isomorphism $\mathcal{O}_X(E) \xrightarrow{\sim} \mathcal{E}$. Assume $U_i = \mathrm{Spec} A_i$ is an affine cover of X trivializing $\mathcal{E}$. Denote the trivialization $\varphi_i : \mathcal{E}|_{U_i} \rightarrow \mathbb{A}_{A_i}^n$. Consider the transition functions $g_{ij} = \varphi_i \circ \varphi_j^{-1}|_{U_{ij}}$. Then thoes transition functions give a vector bundle E .

Define $\psi_i : \mathcal{O}_X(E)(U_i) = \Gamma(U_i, E) \rightarrow \mathcal{E}|_{U_i}$ sending $s : U_i \rightarrow \mathbb{A}_{U_i}^n$ to $\varphi_i^{-1}(v_s)$, where v_s is the vector defining s . Then

$$\psi_i|_{U_{ij}}(s|_{U_{ij}}) = \varphi_i^{-1}(v_s) = \varphi_i^{-1}(g_{ij}v'_s) = \varphi_i^{-1} \circ \varphi_i \circ \varphi_j^{-1}(v'_s) = \varphi_j^{-1}(v'_s) = \psi_j|_{U_{ij}}(s|_{U_{ij}}) \quad (21)$$

Hence ψ_i glue up and get our desired isomorphism. $\square$

1.5 Bundles associated to a representation

Lemma 1.7. *Let $f : Y \rightarrow X$ be a morphism, $p : E \rightarrow X$ vector bundle of rank r on X . Consider the fibered product $F = Y \times_X E$. Then the first projection $q : F \rightarrow Y$ gives a bundle on Y of rank r and $\mathcal{O}_Y(F) = f^*\mathcal{O}_X(E)$.*

Proof. Locally we have trivializations $\varphi_i : E|_{U_i} \rightarrow U_i \times \mathbb{A}_{\mathbb{C}}^r$ and transition functions $g_{ij} : U_{ij} \times \mathbb{A}_{\mathbb{C}}^r \rightarrow U_{ij} \times \mathbb{A}_{\mathbb{C}}^r$ satisfying cocycle condition. Then after base change we get $\tilde{\varphi}_i : F|_{V_i} \rightarrow V_i \times \mathbb{A}_{\mathbb{C}}^r$ and $\tilde{g}_{ij} : U_{ij} \times \mathbb{A}_{\mathbb{C}}^r \rightarrow U_{ij} \times \mathbb{A}_{\mathbb{C}}^r$ still satisfying cocycle condition. Hence $q : F \rightarrow Y$ defines a bundle on Y of rank r .

Now for all open subset $V \subseteq Y$, define $\sigma(V) : f^*\mathcal{O}_X(E)(V) \rightarrow \mathcal{O}_Y(F)(V)$ sending $(U, s) \otimes 1$ to $\tilde{s}|_V$, where $\tilde{s} : f^{-1}(U) \rightarrow F$ is the pull back of $s : U \rightarrow E$. For all $y \in Y$, want to show that $\sigma_y : \mathcal{O}_X(E)_{f(y)} \otimes_{\mathcal{O}_{X, f(y)}} \mathcal{O}_{Y, y} \rightarrow \mathcal{O}_Y(F)_y$ is isomorphism.

For injectivity, if $\sum_{i=1}^r (U, s_i) \otimes b_i \in \ker \sigma_y$, then locally we get $\sum_{i=1}^r b_i \cdot f^\# \circ s_i^\#(x_j) = 0$ for all $1 \leq j \leq r$, so that also $\sum_{i=1}^r (s_i^\#(x_1), \dots, s_i^\#(x_r)) \otimes b_i = 1 \otimes (\sum_{i=1}^r b_i \cdot f^\# \circ s_i^\#(x_1), \dots, \sum_{i=1}^r b_i \cdot f^\# \circ s_i^\#(x_r)) = 0$. Then $(U, s) = \oplus_{i=1}^r a_i \otimes 1 = 0$. For surjectivity, given $(b_1, \dots, b_r) \in B^{\oplus r}$, clearly it is the image of $\sum_{i=1}^r (U, s_i) \otimes b_i$ where s_i is defined by $e_i \in A^{\oplus r}$. $\square$

Let $p : E \rightarrow X$ be a vector bundle of rank r . To avoid ambiguity, we denote the general linear group by G_r and the group variety by $\mathrm{GL}_r(\mathbb{C})$, which is the open subset $D(\det)$ of $\mathbb{A}_{\mathbb{C}}^{r^2}$.

Consider the variety $R = X \times \mathrm{GL}_r(\mathbb{C})$. We have a group action of G_r on R defined locally as following, for $M = (a_{ij})$

$$\mathrm{id} \times (\circ M^{-1}) : A \times \mathrm{GL}_r(\mathbb{C}) \rightarrow A \times \mathrm{GL}_r(\mathbb{C}) \quad (22)$$

where $\circ M^{-1}$ corresponds to the homomorphism defined by $(x_{ij}) \mapsto (\sum_k x_{ik} a_{kj})$. Then $R/G_r = X$ and the orbits can be identified with the fibers of $R \rightarrow X$. In addition, each matrix M gives a morphism $\mathbb{A}_{\mathbb{C}}^r \rightarrow E|_U$ satisfying that composition with trivialization φ_U is a linear transformation defined by M . Then it is natural to consider transition function g_{ij} which gives a $\mathrm{GL}_r(\mathbb{C})$ -bundle on X , denoted by $\mathrm{Fr}(E)$, the frame bundle of E . And there is an group action of G_r on $R \times \mathbb{A}_{\mathbb{C}}^r$ defined locally as following, for $M = (a_{ij})$

$$\mathrm{id} \times (\circ M^{-1}) \times M : A \times \mathrm{GL}_r(\mathbb{C}) \times \mathbb{A}_{\mathbb{C}}^r \rightarrow A \times \mathrm{GL}_r(\mathbb{C}) \times \mathbb{A}_{\mathbb{C}}^r \quad (23)$$

where M corresponds to the linear transformation defined by M . Consider $R \times \mathrm{GL}_r(\mathbb{C})/G_r$. It is clear that $A[x_{ij}]_{(\det)}[y_k]^{G_r} = A[\sum_{k=1}^r x_{ik} y_k]$ and hence the quotient scheme is isomorphic to $\mathbb{A}_{\mathbb{C}}^r$. With transition functions g_{ij} , gluing up we get original E .

Now given a representation $\psi : \mathrm{GL}_r(\mathbb{C}) \rightarrow \mathrm{GL}_m(\mathbb{C})$, which is in fact a morphism of group schemes, consider the group action of G_r on $R \times \mathbb{A}_{\mathbb{C}}^m$ defined locally as following, for $M = (a_{ij})$

$$\mathrm{id} \times (\circ M^{-1}) \times \psi(M) : A \times \mathrm{GL}_r(\mathbb{C}) \times \mathbb{A}_{\mathbb{C}}^m \rightarrow A \times \mathrm{GL}_r(\mathbb{C}) \times \mathbb{A}_{\mathbb{C}}^m \quad (24)$$

Since the representation corresponds to a homomorphism $A[y_{ij}] \rightarrow A[x_{ij}]$, similarly the invariant subring is generated by $\psi(x_{ij})(y_k)$, where the (i, j) -entry of $\psi(x_{ij})$ is the image of y_{ij} in $A[x_{ij}]$. Then locally the quotient scheme is isomorphic to $\mathbb{A}_{\mathbb{C}}^m$ with natural transition functions $\psi(g_{ij})$. These information give a vector bundle E_{ψ} of rank m on X , called the bundle associated to representation ψ .

Remark 1.17. *The idea of frame bundle also comes from moduli theory. Consider functor $\mathcal{F} : \mathrm{Sch}_X \rightarrow \mathrm{Set}$ defined as following*

$$(f : T \rightarrow X) \mapsto \{f^* \mathcal{O}(E) \xrightarrow{\sim} \mathcal{O}_T^r\} \quad (25)$$

And in fact this functor is representable by frame bundle above.

2 Sheaf Cohomology

2.1 Cohomology group

There are lots of abelian categories in Algebraic Geometry.

Example 2.1. (1) $\mathbf{Ab}$ is the category of abelian groups.

(2) $\mathbf{Mod}_A$ is the category of A -modules.

(3) $\mathbf{Ab}(X)$ is the category of sheaves of abelian groups on a topological space X .

(4) $\mathbf{Mod}(\mathcal{O}_X)$ is the category of $\mathcal{O}_X$ -modules on a ringed space $(X, \mathcal{O}_X)$.

(5) $\mathbf{Qcoh}(X)$ is the category of quasi-coherent sheaves on a scheme X .

(6) $\mathbf{Coh}(X)$ is the category of coherent sheaves on a noetherian scheme X .

There are some special functors in Algebraic Geometry.

- Example 2.2.** (1) $\text{Hom}_{\mathcal{C}}(A, \cdot) : \mathcal{C} \longrightarrow \mathbf{Ab}$ is left exact covariant functor.
 (2) $\text{Hom}_{\mathcal{C}}(\cdot, A) : \mathcal{C} \longrightarrow \mathbf{Ab}$ is left exact contravariant functor.
 (3) $\Gamma(X, \cdot) : \mathbf{Ab}(X) \longrightarrow \mathbf{Ab}$ is left exact covariant functor.
 (4) $M \otimes_A \cdot : \mathbf{Mod}_A \longrightarrow \mathbf{Mod}_A$ is right exact covariant functor. In particular, $M \otimes_A \cdot$ is exact if and only if M is flat A -module.

Definition 2.1. Let $\mathcal{C}$ be a category with enough injective objects, $\mathcal{F}$ a left exact covariant functor, $J \in \mathcal{C}$. We say that J is acyclic for the functor $\mathcal{F}$ (or say $\mathcal{F}$ -acyclic) if $R^i\mathcal{F}(J) = 0$ for all $i > 0$.

Proposition 2.1. Let $\mathcal{C}$ be a category with enough injective objects, $\mathcal{F}$ a left exact covariant functor, $A \in \mathcal{C}$. Assume that there is an $\mathcal{F}$ -acyclic resolution J^* of A i.e. $0 \longrightarrow A \longrightarrow J_0 \longrightarrow J_1 \longrightarrow \cdots$ exact with each J_i $\mathcal{F}$ -acyclic. Then $R^i\mathcal{F}(A) \cong H^i(\mathcal{F}(J^*))$.

Proposition 2.2. Let $(X, \mathcal{O}_X)$ be a ringed space. Then $\mathbf{Mod}(\mathcal{O}_X)$ has enough injective objects. In particular, $\mathbf{Ab}(X)$ has enough injective objects if viewing X as $(X, \mathbb{Z}_X)$ ringed space, where $\mathbb{Z}_X$ is the constant sheaf.

Proof. Let $\mathcal{F}$ be an $\mathcal{O}_X$ -module. For all $x \in X$, can embed $\mathcal{F}_x$ in an injective $\mathcal{O}_{X,x}$ -module I_x since $\mathbf{Mod}_A$ has enough injective objects for all ring A . View I_x as a sheaf on $\{x\}$. Set $\mathcal{I} = \prod_{x \in X} j_* I_x$ where $j : \{x\} \longrightarrow X$ is inclusion. For all $\mathcal{G}$ $\mathcal{O}_X$ -module, we have $\text{Hom}_{\mathcal{O}_X}(\mathcal{G}, \mathcal{I}) = \prod_{x \in X} \text{Hom}_{\mathcal{O}_X}(\mathcal{G}, j_* I_x) = \prod_{x \in X} \text{Hom}_{\mathcal{O}_{X,x}}(\mathcal{G}_x, I_x)$. With $\mathcal{F}_x \hookrightarrow I_x$, get a monomorphism $\mathcal{F} \longrightarrow \mathcal{I}$ with $\mathcal{I}$ injective. $\square$

Definition 2.2 (Sheaf Cohomology). Let X be a topological space, $\Gamma(X, \cdot) : \mathbf{Ab}(X) \longrightarrow \mathbf{Ab}$. Define the cohomology functor $H^i(X, \cdot) = R^i\Gamma(X, \cdot)$. For $\mathcal{F} \in \mathbf{Ab}(X)$, we call $H^i(X, \mathcal{F})$ the i th cohomology group of $\mathcal{F}$.

Definition 2.3. Let X be a topological space, $\mathcal{F} \in \mathbf{Ab}(X)$. We say that $\mathcal{F}$ is flasque if for all $V \subseteq U$ inclusion of open subsets, the restriction map $\mathcal{F}(U) \longrightarrow \mathcal{F}(V)$ is surjective.

- Proposition 2.3.** (1) Direct products of flasque sheaves are flasque.
 (2) Direct image of flasque sheaves are flasque.
 (3) quotient of flasque sheaves by flasque sheaves are flasque.
 (4) If $0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0$ is exact with $\mathcal{F}$ flasque, then $0 \longrightarrow \mathcal{F}(U) \longrightarrow \mathcal{G}(U) \longrightarrow \mathcal{H}(U) \longrightarrow 0$ is exact for all open subset $U \subseteq X$.

Proposition 2.4. Let $(X, \mathcal{O}_X)$ be a ringed space. Then every injective $\mathcal{O}_X$ -module is flasque.

Proof. Consider $U \subseteq X$ is an inclusion of open subsets. Set $\mathcal{G}_U = j_! \mathcal{O}_U$. Let $\mathcal{F}$ be an injective $\mathcal{O}_X$ -module, $V \subseteq U$ open subset. Then the natural morphism $\mathcal{G}_V \longrightarrow \mathcal{G}_U$ is monic so that $0 \longrightarrow \mathcal{G}_V \longrightarrow \mathcal{G}_U$ is exact. Since $\mathcal{F}$ is injective, get $\text{Hom}_{\mathcal{O}_X}(\mathcal{G}_U, \mathcal{F}) \longrightarrow \text{Hom}_{\mathcal{O}_X}(\mathcal{G}_V, \mathcal{F}) \longrightarrow 0$ exact. Note that $\text{Hom}_{\mathcal{O}_X}(\mathcal{G}_U, \mathcal{F})|_U = \text{Hom}_{\mathcal{O}_U}(\mathcal{O}_U, \mathcal{F}|_U) = \mathcal{F}(U)$. Thus $\mathcal{F}(U) \longrightarrow \mathcal{F}(V) \longrightarrow 0$ is exact so that $\mathcal{F}$ is flasque. $\square$

Corollary 2.1. *Let X be a topological space. Then every injective object in $\mathbf{Ab}(X)$ is flasque by viewing X as ringed space $(X, \mathbb{Z}_X)$ where $\mathbb{Z}_X$ is the constant sheaf.*

Proposition 2.5. *Let X be a topological space. Then every flasque sheaf on X is acyclic for the functor $\Gamma(X, \cdot)$.*

Proof. Let $\mathcal{F}$ be a flasque sheaf on X . Consider exact sequence $0 \rightarrow \mathcal{F} \rightarrow \mathcal{I} \rightarrow \mathcal{G} \rightarrow 0$ where $\mathcal{I}$ is an injective object in $\mathbf{Ab}(X)$. By Corollary 2.1, $\mathcal{I}$ is flasque. Thus $\mathcal{G}$ is flasque since it is quotient of flasque sheaf by flasque sheaf. Take the long exact sequence $0 \rightarrow \mathcal{F}(X) \rightarrow \mathcal{I}(X) \rightarrow \mathcal{G}(X) \rightarrow H^1(X, \mathcal{F}) \rightarrow H^1(X, \mathcal{I}) \rightarrow \dots$. While $\mathcal{F}$ is flasque, we have $0 \rightarrow \mathcal{F}(X) \rightarrow \mathcal{I}(X) \rightarrow \mathcal{G}(X) \rightarrow 0$ is exact. Also, as $\mathcal{I}$ is injective, $H^i(X, \mathcal{I}) = 0$ for all $i > 0$. Thus $H^i(X, \mathcal{F}) = 0$ for all $i > 0$ so that $\mathcal{F}$ is acyclic for the functor $\Gamma(X, \cdot)$. $\square$

Corollary 2.2. *Let $(X, \mathcal{O}_X)$ be a ringed space. Then the right derived functor of $\Gamma(X, \cdot) : \mathbf{Mod}(\mathcal{O}_X) \rightarrow \mathbf{Ab}$ coincide with $H^i(X, \cdot)$.*

Reason 2.1. *Take injective resolution in $\mathbf{Mod}(\mathcal{O}_X)$. By Proposition 2.4 and Proposition 2.5, it is a $\Gamma(X, \cdot)$ -acyclic resolution. Thus by Proposition 2.1, the two right derived functors coincide.*

Corollary 2.3. *Let X be a topological space, $Y \subseteq X$ closed subset, $i : Y \rightarrow X$ inclusion. Then for all $\mathcal{F} \in \mathbf{Ab}(Y)$, we have $H^i(Y, \mathcal{F}) \cong H^i(X, j_*\mathcal{F})$.*

Reason 2.2. *Take injective resolution J^* of $\mathcal{F}$ in $\mathbf{Ab}(Y)$. Then f_*J^* is a flasque resolution of $f_*\mathcal{F}$.*

Remark 2.1 (Gabber Theorem). *Let X be a scheme. Then $\mathbf{Qcoh}(X)$ has enough injective objects.*

Remark 2.2. *Let X be a scheme. Suppose that X is either noetherian or quasi-compact and separated. Then the right derived functor of $\Gamma(X, \cdot) : \mathbf{Qcoh}(X) \rightarrow \mathbf{Ab}$ coincide with $H^i(X, \cdot)$. In fact, this property is equivalent to each injective object is $\Gamma(X, \cdot)$ -acyclic.*

Theorem 2.1 (Grothendieck Vanishing). *Let X be a noetherian topological space of dimension $n < \infty$. Then for all $\mathcal{F} \in \mathbf{Ab}(X)$, we have that $H^i(X, \mathcal{F}) = 0$ for all $i > n$.*

2.2 Cohomology of noetherian affine schemes

Theorem 2.2 (Artin-Rees). *Let A be a noetherian ring, $M \subseteq N$ finitely generated A -modules, $\mathfrak{a} \subseteq A$ ideal. Then for all $n > 0$, there exists $m > n$ such that $\mathfrak{a}M \supseteq M \cap \mathfrak{a}^m N$.*

Remark 2.3. *This theorem says that the $\mathfrak{a}$ -adic topology of M is compatible with the induced topology of $\mathfrak{a}$ -adic topology of N on M .*

Lemma 2.1. *Let A be a noetherian ring, I an injective A -module, $\mathfrak{a} \subseteq A$ ideal. Assume $J \subseteq I$ is the A -submodule consist of $x \in I$ such that $\mathfrak{a}^n x = 0$ for some $n > 0$. Then J is injective.*

Proof. It suffices to show that for all $\mathfrak{b} \subseteq A$ ideal and $\varphi : \mathfrak{b} \rightarrow J$ homomorphism of A -modules, there exists $\psi : A \rightarrow J$ extending φ satisfies the following diagram commutes

$$\begin{array}{ccc} \mathfrak{b} & \hookrightarrow & A \\ \downarrow \varphi & \nearrow \psi & \\ J & & \end{array} \quad (26)$$

Since A is noetherian, B is finitely generated. Thus there exists $n > 0$ such that $\mathfrak{a}^n \varphi(\mathfrak{b}) = 0$. By Artin-Rees, there exists $m > n$ such that $\mathfrak{a}^m \mathfrak{b} \supseteq \mathfrak{b} \cap \mathfrak{a}^m$. Thus $\varphi(\mathfrak{b} \cap \mathfrak{a}^m) = 0$ so that φ factors through $\mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^m)$. Consider $\varphi_I : \mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^m) \rightarrow J \hookrightarrow I$. As I is injective, φ_I extends to $\psi_I : A/\mathfrak{a}^m \rightarrow I$. Note that $\text{im}(\psi_I) \subseteq J$ since $\mathfrak{a}^m \text{im}(\psi_I) = 0$, get ψ . $\square$

Lemma 2.2. *Let A be a noetherian ring, I an injective A -module. Then for all $f \in A$, the localization $\theta : I \rightarrow I_f$ is surjective.*

Proof. For all $i > 0$, set $\mathfrak{b}_i = \text{ann}_A(f^i)$. Get an increasing sequence of ideals $\mathfrak{b}_1 \subseteq \mathfrak{b}_2 \subseteq \dots$. As A is noetherian, $\{\mathfrak{b}_i\}$ stabilize from $\mathfrak{b}_r$ for some $r < \infty$. For $x \in I_f$, we can write $x = \frac{\theta(y)}{f^n}$ where $y \in I$ and $n > 0$. There is a canonical homomorphism $\varphi : (f^{n+r}) \rightarrow I \quad f^{n+r} \mapsto f^r y$. Easy to check that φ is well defined. Since I is injective, φ extends to $\psi : A \rightarrow I$. Set $z = \psi(1)$, then $f^{n+r} z = f^r y$. Thus $\theta(z) = \frac{z}{1} = \frac{f^{n+r} z}{f^{n+r}} = \frac{f^r y}{f^{n+r}} = \frac{\theta(y)}{f^n} = x$ so that θ is surjective. $\square$

Proposition 2.6. *Let A be a noetherian ring, I an injective A -module. Then $\tilde{I}$ is flasque on $X = \text{Spec } A$.*

Proof. Set $Y = \text{Supp}(\tilde{I}) = \{x \in X \mid \tilde{I}_x \neq 0\} \subseteq X$ closed. If $Y = \emptyset$, then $\tilde{I}$ is flasque. If $Y \neq \emptyset$, since A is noetherian, we can assume for all injective A -module J with support $\text{Supp}(\tilde{J}) \subsetneq Y$, $\tilde{J}$ is flasque. Want to show $\tilde{I}(X) \rightarrow \tilde{I}(U)$ is surjective for all open subset $U \subseteq X$.

If $Y \cap U = \emptyset$, then $\tilde{I}(U) = 0$, done! If $Y \cap U \neq \emptyset$, take standard open subset $D(f) \subseteq U$ and $D(f) \cap Y \neq \emptyset$. Set $Z = X \setminus D(f)$. Let $s \in \tilde{I}(U)$, then $s|_{D(f)} \in \tilde{I}(D(f)) = I_f$. By Lemma 2.2, we can lift $s|_{D(f)}$ to $t \in \tilde{I}(X) = I$. Thus $(s-t)|_{D(f)} = 0$ so that $(s-t)|_U \in \Gamma_{Z \cap U}(U, \tilde{I}|_U)$ i.e. support of it is in Z . Remains to show $\Gamma_Z(X, \tilde{I}) \rightarrow \Gamma_{Z \cap U}(U, \tilde{I}|_U)$.

Note that $J = \Gamma_Z(X, \tilde{I})$ is exactly the submodule consist of $x \in I$ such that $f^n x = 0$ for some $n > 0$. By Lemma 2.1, J is injective A -module with $\text{Supp}(\tilde{J}) \subsetneq Y$. By induction hypothesis, $\tilde{J}$ is flasque. Thus $\tilde{J}(X) \rightarrow \tilde{J}(U)$ is surjective. While $\tilde{J}(X) = \Gamma_Z(X, \tilde{I})$ and $\tilde{J}(U) = \Gamma_{Z \cap U}(U, \tilde{I}|_U)$, done! $\square$

Theorem 2.3 (Serre Vanishing). *Let A be a noetherian ring and $X = \text{Spec } A$. Then for all $\mathcal{F}$ quasi-coherent sheaf in X , we have that $H^i(X, \mathcal{F}) = 0$ for all $i > 0$.*

Proof. For $\mathcal{F}$ quasi-coherent, set $M = \mathcal{F}(X)$. Take injective resolution I^* of M in Mod_A . Thus $0 \rightarrow \mathcal{F} \rightarrow \tilde{I}^0 \rightarrow \dots$ is exact, which is a $\Gamma(X, \cdot)$ -acyclic resolution by Proposition 2.6. As $\tilde{I}^i$ is $\Gamma(X, \cdot)$ -acyclic, can apply $\Gamma(X, \cdot)$ to calculate $H^i(X, \mathcal{F})$. But $\Gamma(X, \cdot)$ gives back to $0 \rightarrow M \rightarrow I^0 \rightarrow \dots$ exact, get $H^i(X, \mathcal{F}) = 0$ for all $i > 0$. $\square$

Corollary 2.4. *Let X be a noetherian affine scheme. Then for all exact sequence $0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$ of $\mathcal{O}_X$ -module with $\mathcal{F}$ quasi-coherent, we have that $0 \rightarrow \mathcal{F}(X) \rightarrow \mathcal{G}(X) \rightarrow \mathcal{H}(X) \rightarrow 0$ is exact.*

Corollary 2.5. *Let X be a noetherian scheme. Suppose that $0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$ is an exact sequence of $\mathcal{O}_X$ -module. If $\mathcal{F}$ and $\mathcal{H}$ are both quasi-coherent (resp. coherent), then $\mathcal{G}$ is quasi-coherent (resp. coherent).*

Proof. Can assume that $X = \text{Spec } A$ is affine. We have a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \widetilde{\mathcal{F}(X)} & \longrightarrow & \widetilde{\mathcal{G}(X)} & \longrightarrow & \widetilde{\mathcal{H}(X)} \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow & & \downarrow \sim \\ 0 & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{G} & \longrightarrow & \mathcal{H} \longrightarrow 0 \end{array} \quad (27)$$

Thus by 5 Lemma, $\mathcal{G}$ is quasi-coherent.

If moreover $\mathcal{F}$ and $\mathcal{H}$ are coherent, then $\mathcal{F}(X)$ and $\mathcal{H}(X)$ are finitely generated. Then $\mathcal{G}(X)$ is also finitely generated so that $\mathcal{G}$ is coherent. $\square$

Lemma 2.3. *Let X be a noetherian scheme. Assume that $H^1(X, \mathcal{I}) = 0$ for all $\mathcal{I}$ coherent ideal sheaf. Then we can cover X by affine open subsets of the form X_f for $f \in A = \Gamma(X, \mathcal{O}_X)$.*

Proof. Suffice to show that every closed point $x \in X$ admits an affine open neighbourhood X_f . For $x \in X$ closed point, choose affine open neighbourhood U of x and set $Y = X \setminus U$. Get $0 \rightarrow \mathcal{I}_{Y \cup \{x\}} \rightarrow \mathcal{I}_Y \rightarrow i_* \mathcal{O}_{\{x\}} \rightarrow 0$ is exact with reduced scheme structures on each set, where $i : \{x\} \rightarrow X$ is inclusion. By assumption, $\Gamma(X, \mathcal{I}_Y) \rightarrow \Gamma(X, i_* \mathcal{O}_{\{x\}})$ is surjective. Take $f \in \Gamma(X, \mathcal{I}_Y)$ such that f is mapped to 1. Then $x \in X_f$. On the other hand, $X_f \subseteq U$, get X_f is affine. $\square$

Lemma 2.4. *Let X be a noetherian scheme. Assume that $H^1(X, \mathcal{I}) = 0$ for all $\mathcal{I}$ coherent ideal sheaf. Then for all $e > 0$ and $\mathcal{F} \subseteq \mathcal{O}_X^{\oplus r}$ coherent, we have that $H^1(X, \mathcal{F}) = 0$.*

Proof. Define filtration $\mathcal{F} \supseteq \mathcal{F} \cap \mathcal{O}_X^{\oplus(r-1)} \supseteq \dots \supseteq \mathcal{F} \cap \mathcal{O}_X \supseteq 0$ such that the associated quotients are coherent ideal sheaves. By long exact sequence and induction, get $H^1(X, \mathcal{F} \cap \mathcal{O}_X^i) = 0$ for all $i > 0$ so that $H^1(X, \mathcal{F}) = 0$. $\square$

Theorem 2.4 (Serre, Cohomology Criterion of Affineness). *Let X be a noetherian scheme. Then the following conditions are equivalent:*

- (1) X is affine
- (2) $H^i(X, \mathcal{F}) = 0$ for all $i > 0$ and $\mathcal{F}$ quasi-coherent.
- (3) $H^1(X, \mathcal{I}) = 0$ for all $\mathcal{I}$ coherent ideal sheaf.

Proof. See in the Hartshorne Chapter III p.215-216. $\square$

Remark 2.4. *In fact, noetherian condition is not necessary.*

2.3 Čech cohomology

Definition 2.4. Let X be a topological space, $\mathcal{U} = \{U_i\}_{i \in I}$ open covering of X with well-ordered index set I , $\mathcal{F}$ sheaf of abelian groups on X . For $i_0, \dots, i_p \in I$, set $U_{i_0, \dots, i_p} = U_{i_0} \cap \dots \cap U_{i_p}$. Define complex $C^*(\mathcal{U}, \mathcal{F})$ to be

$$C^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0, \dots, i_p} \mathcal{F}(U_{i_0, \dots, i_p}) \quad (28)$$

with coboundary map

$$d^p : C^p(\mathcal{U}, \mathcal{F}) \longrightarrow C^{p+1}(\mathcal{U}, \mathcal{F})$$

$$\alpha = (\alpha_{i_0, \dots, i_p})_{i_0, \dots, i_p} \longmapsto \left(\sum_{k=0}^{p+1} (-1)^k \alpha_{i_0, \dots, \widehat{i_k}, \dots, i_{p+1}} \Big|_{U_{i_0, \dots, i_{p+1}}} \right)_{i_0, \dots, i_{p+1}} \quad (29)$$

Remark 2.5. If we don't assume that I is well-ordered, we need to set $U_{i_0, \dots, i_p} = 0$ if there are repeated index and $\alpha_{i_0, \dots, i_p} = (-1)^{\text{sgn}(\sigma)} \alpha_{i_{\sigma(0)}, \dots, i_{\sigma(p)}}$.

Definition 2.5. Let X be a topological space, $\mathcal{U} = \{U_i\}_{i \in I}$ open covering of X with well-ordered index set I , $\mathcal{F}$ sheaf of abelian groups on X . Define $\check{H}^p(\mathcal{U}, \mathcal{F}) = h^p(C^*(\mathcal{U}, \mathcal{F}))$, called the p th Čech cohomology of $\mathcal{F}$ with respect to $\mathcal{U}$.

Example 2.3. (1) For any $\mathcal{F}$ and $\mathcal{U}$, $\check{H}^0(\mathcal{U}, \mathcal{F}) = \Gamma(X, \mathcal{F})$.

(2) For $\mathcal{U} = \{X\}$, $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for all $p > 0$.

Lemma 2.5. Let X be a topological space, $\mathcal{U} = \{U_i\}_{i \in I}$ open covering of X with well-ordered index set I , $\mathcal{F}$ sheaf of abelian groups on X . Suppose that $\mathcal{U}$ contains $U_i = X$ for some i . Then $0 \longrightarrow \mathcal{F}(X) \longrightarrow C^*(\mathcal{U}, \mathcal{F})$ is exact.

Proof. Obviously, exactness at $\mathcal{F}(X)$ and $C^0(\mathcal{U}, \mathcal{F})$ is ok. For $p \geq 1$, without loss of generality, may assume that $i = 0$ which is the smallest element. Define homotopy $k^p : C^p(\mathcal{U}, \mathcal{F}) \longrightarrow C^{p-1}(\mathcal{U}, \mathcal{F})$ $\alpha \longmapsto (\alpha_{0, i_0, \dots, i_{p-1}})_{i_0, \dots, i_{p-1}}$. Check that $d^{p-1} \circ k^p + k^{p+1} \circ d^p = \text{id}_{C^*(\mathcal{U}, \mathcal{F})}$. Thus identity is homotopic to zero map, which inducing $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for all $p > 0$. $\square$

2.4 Comparison Theorem

Definition 2.6. Let X be a topological space, $\mathcal{U} = \{U_i\}_{i \in I}$ open covering of X with well-ordered index set I , $\mathcal{F}$ sheaf of abelian groups on X . Define a complex of sheaves on X , denoted by $\mathcal{C}^*(\mathcal{U}, \mathcal{F})$, as following

$$\mathcal{C}^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0, \dots, i_p} f_*(\mathcal{F}|_{U_{i_0, \dots, i_p}}) \quad (30)$$

where f is the inclusion, with coboundary map

$$d^p : \mathcal{C}^p(\mathcal{U}, \mathcal{F}) \longrightarrow \mathcal{C}^{p+1}(\mathcal{U}, \mathcal{F}) \quad (31)$$

defined accordingly.

Remark 2.6. Obviously, $\Gamma(X, \mathcal{C}^p(\mathcal{U}, \mathcal{F})) = C^p(\mathcal{U}, \mathcal{F})$.

Proposition 2.7. *Let X be a topological space, $\mathcal{U} = \{U_i\}_{i \in I}$ open covering of X with well-ordered index set I , $\mathcal{F}$ sheaf of abelian groups on X . Then $0 \rightarrow \mathcal{F} \rightarrow \mathcal{C}^*(\mathcal{U}, \mathcal{F})$ is exact.*

Reason 2.3. *Since exactness of complex of sheaves is equivalent to exactness at all stalks. Set $x \in X$, we can represent elements in stalks by sections on V , which is small enough to be contained in U_j for some j . Apply Lemma 2.5, get exactness of complex of sections on V .*

Proposition 2.8. *Let X be a topological space, $\mathcal{U} = \{U_i\}_{i \in I}$ open covering of X with well-ordered index set I , $\mathcal{F}$ flasque sheaf of abelian groups on X . Then $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for all $p > 0$.*

Proof. Since flasque is stable under restriction to open subset, direct limit and product, $\mathcal{C}^p(\mathcal{U}, \mathcal{F})$ is flasque for all p . Thus $\mathcal{C}^*(\mathcal{U}, \mathcal{F})$ is a flasque resolution of $\mathcal{F}$. By Proposition 2.5 and Proposition 2.1, get

$$\check{H}^p(\mathcal{U}, \mathcal{F}) = h^p(\Gamma(X, \mathcal{C}^*(\mathcal{U}, \mathcal{F}))) = H^p(X, \mathcal{F}) = 0 \quad (32)$$

hence we are done! $\square$

Lemma 2.6. *Let X be a topological space, $\mathcal{U} = \{U_i\}_{i \in I}$ open covering of X with well-ordered index set I , $\mathcal{F}$ sheaf of abelian groups on X . Then for all $p \geq 0$, there is a natural homomorphism $\check{H}^p(\mathcal{U}, \mathcal{F}) \rightarrow H^p(X, \mathcal{F})$.*

Proof. Let $0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}^*$ be an injective resolution of $\mathcal{F}$ in $\text{Ab}(X)$. By Comparison Theorem in Homological Algebra, there exists a morphism of complexes which is unique up to homotopy, inducing the natural homomorphisms we want. $\square$

Theorem 2.5 (Comparison Theorem). *Let X be a topological space, $\mathcal{U} = \{U_i\}_{i \in I}$ open covering of X with well-ordered index set I , $\mathcal{F}$ sheaf of abelian groups on X . Suppose that for all p and all $i_0, \dots, i_p$, we have that $H^q(U_{i_0, \dots, i_p}, \mathcal{F}|_{U_{i_0, \dots, i_p}}) = 0$ for all $q > 0$. Then the natural homomorphism $\check{H}^r(\mathcal{U}, \mathcal{F}) \rightarrow H^r(X, \mathcal{F})$ is isomorphic for all $r \geq 0$.*

Proof. For $r = 0$, the natural homomorphism is identity map of global sections of $\mathcal{F}$. For $r > 0$, embed $\mathcal{F}$ into an injective sheaf $\mathcal{I}$. There is an exact sequence

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{I} \rightarrow \mathcal{G} \rightarrow 0 \quad (33)$$

On the one hand, by definition of Čech sheaf resolution, we have an exact sequence of complexes

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{I} & \longrightarrow & \mathcal{G} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{C}^\bullet(\mathcal{U}, \mathcal{F}) & \longrightarrow & \mathcal{C}^\bullet(\mathcal{U}, \mathcal{I}) & \longrightarrow & \mathcal{C}^\bullet(\mathcal{U}, \mathcal{G}) \longrightarrow 0 \end{array} \quad (34)$$

Taking products, get

$$0 \rightarrow C^*(\mathcal{U}, \mathcal{F}) \rightarrow C^*(\mathcal{U}, \mathcal{I}) \rightarrow C^*(\mathcal{U}, \mathcal{G}) \rightarrow 0 \quad (35)$$

On the other hand, by Horseshoe Lemma, we may take injective resolution $\mathcal{I}_{\mathcal{F}}^{\bullet}$, $\mathcal{I}_{\mathcal{I}}^{\bullet}$ and $\mathcal{I}_{\mathcal{G}}^{\bullet}$ of $\mathcal{F}$, $\mathcal{I}$ and $\mathcal{G}$ respectively such that

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{I} & \longrightarrow & \mathcal{G} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{I}_{\mathcal{F}}^{\bullet} & \longrightarrow & \mathcal{I}_{\mathcal{I}}^{\bullet} & \longrightarrow & \mathcal{I}_{\mathcal{G}}^{\bullet} \longrightarrow 0 \end{array} \quad (36)$$

is an exact sequence of complexes and $\mathcal{I}_{\mathcal{I}}^i = \mathcal{I}_{\mathcal{F}}^i \oplus \mathcal{I}_{\mathcal{G}}^i$ for all $i \geq 0$. Then by Comparison Theorem for injective resolution, we get morphism of chain complexes $\alpha : \mathcal{C}^{\bullet}(\mathcal{U}, \mathcal{F}) \rightarrow \mathcal{I}_{\mathcal{F}}^{\bullet}$, $\beta : \mathcal{C}^{\bullet}(\mathcal{U}, \mathcal{I}) \rightarrow \mathcal{I}_{\mathcal{I}}^{\bullet}$ and $\gamma : \mathcal{C}^{\bullet}(\mathcal{U}, \mathcal{G}) \rightarrow \mathcal{I}_{\mathcal{G}}^{\bullet}$, which are unique up to homotopy.

Consider the following diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{C}^i(\mathcal{U}, \mathcal{F}) & \xrightarrow{f} & \mathcal{C}^i(\mathcal{U}, \mathcal{I}) & \xrightarrow{g} & \mathcal{C}^i(\mathcal{U}, \mathcal{G}) \longrightarrow 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & \mathcal{I}_{\mathcal{F}}^i & \xrightarrow{u} & \mathcal{I}_{\mathcal{I}}^i & \xrightarrow{v} & \mathcal{I}_{\mathcal{G}}^i \longrightarrow 0 \end{array} \quad (37)$$

Want to modify α , β and γ . Since the bottom row is split, we have inclusion $\sigma : \mathcal{I}_{\mathcal{G}}^{\bullet} \rightarrow \mathcal{I}_{\mathcal{I}}^{\bullet}$ and projection $\pi : \mathcal{I}_{\mathcal{I}}^{\bullet} \rightarrow \mathcal{I}_{\mathcal{F}}^{\bullet}$. Take $\beta' = \beta + \pi \circ (\gamma \circ g - v \circ \beta)$ and $\alpha' = \sigma \circ (\beta' \circ f - u \circ \alpha)$. Then by our choice of α , β and γ , it is clear that the new diagram is commutative.

By assumption, for all $i_0, \dots, i_p$,

$$0 \longrightarrow \mathcal{F}(U_{i_0, \dots, i_p}) \longrightarrow \mathcal{I}(U_{i_0, \dots, i_p}) \longrightarrow \mathcal{G}(U_{i_0, \dots, i_p}) \longrightarrow 0 \quad (38)$$

is exact. Taking global sections and by Homological Algebra, there is commutative daigram between two long exact sequences

$$\begin{array}{ccccccc} \dots & \longrightarrow & \check{H}^r(\mathcal{U}, \mathcal{F}) & \longrightarrow & \check{H}^r(\mathcal{U}, \mathcal{I}) & \longrightarrow & \check{H}^r(\mathcal{U}, \mathcal{G}) \longrightarrow \check{H}^{r+1}(\mathcal{U}, \mathcal{F}) \longrightarrow \dots \\ & & \downarrow & & \downarrow & & \downarrow \\ \dots & \longrightarrow & H^r(X, \mathcal{F}) & \longrightarrow & H^r(X, \mathcal{I}) & \longrightarrow & H^r(X, \mathcal{G}) \longrightarrow H^{r+1}(X, \mathcal{F}) \longrightarrow \dots \end{array} \quad (39)$$

Since $\mathcal{I}$ is injective, get $\mathcal{I}$ flasque and $\check{H}^r(\mathcal{U}, \mathcal{I}) = 0$ for all $r > 0$. Thus $\check{H}^{r+1}(\mathcal{U}, \mathcal{F}) = \check{H}^r(\mathcal{U}, \mathcal{G})$ and $H^{r+1}(X, \mathcal{F}) = H^r(X, \mathcal{G})$ for all $r \geq 1$. In particular, we have commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \check{H}^0(\mathcal{U}, \mathcal{F}) & \longrightarrow & \check{H}^0(\mathcal{U}, \mathcal{I}) & \longrightarrow & \check{H}^0(\mathcal{U}, \mathcal{G}) \longrightarrow \check{H}^1(\mathcal{U}, \mathcal{F}) \longrightarrow 0 \\ & & \parallel & & \parallel & & \parallel \\ \dots & \longrightarrow & H^0(X, \mathcal{F}) & \longrightarrow & H^0(X, \mathcal{I}) & \longrightarrow & H^0(X, \mathcal{G}) \longrightarrow H^1(X, \mathcal{F}) \longrightarrow 0 \end{array} \quad (40)$$

Hence by 5 Lemma, we get the natural map $\check{H}^1(\mathcal{U}, \mathcal{F}) \rightarrow H^1(X, \mathcal{F})$ is isomorphic. Finally, by induction, done! $\square$

Remark 2.7. Here our modifying assumption is in fact same as the proof of injectivity of a split exact sequence consists of injective objects.

Corollary 2.6. Let X be a (noetherian) separated scheme, $\mathcal{U}$ affine open covering of X , $\mathcal{F}$ quasi-coherent sheaf on X . Then the natural homomorphism $\check{H}^p(\mathcal{U}, \mathcal{F}) \rightarrow H^p(X, \mathcal{F})$ is isomorphic for all p .

Remark 2.8. This corollary immediately comes from Serre Vanishing Theorem.

Corollary 2.7. Let X be a (noetherian) separated scheme covered by $r+1$ affine open subsets. Then for all $\mathcal{F}$ quasi-coherent on X , we have that $H^i(X, \mathcal{F}) = 0$ for all $i > r$. In particular, when $r = 0$, this specializes to Serre Vanishing Theorem.

2.5 Cohomology of projective space

Theorem 2.6. Let A be a noetherian ring, $S = A[x_0, \dots, x_r]$, $X = \text{Proj}(S) = \mathbb{P}_A^r$, $r \geq 1$. Then we have that

- (1) $H^i(X, \mathcal{O}_X(n)) = 0$ for all $0 < i < r$ and all $n \in \mathbb{Z}$
- (2) $H^r(X, \mathcal{O}_X(-r-1)) \cong A$ is free A -module of rank 1
- (3) the natural pairing $H^0(X, \mathcal{O}_X(n)) \times H^r(X, \mathcal{O}_X(-n-r-1)) \longrightarrow H^r(X, \mathcal{O}_X(-r-1))$ is a perfect pairing of finite free A -modules for all $n \in \mathbb{Z}$.

Remark 2.9. In fact, we have already known that $H^0(X, \mathcal{O}_X(n)) = \begin{cases} S_n & n \geq 0 \\ 0 & n < 0 \end{cases}$ and

$H^i(X, \mathcal{O}_X(n)) = 0$ for all $i > r$ and all $n \in \mathbb{Z}$. Thus with this theorem, we have figured out all the cohomological groups of $\mathbb{P}_A^r$. Especially, all of them are finite free A -modules.

Proof. Consider $\mathcal{F} = \bigoplus_{n \in \mathbb{Z}} \mathcal{O}_X(n)$ quasi-coherent. Since cohomology commutes with direct sum, $H^i(X, \mathcal{F}) = \bigoplus_{n \in \mathbb{Z}} H^i(X, \mathcal{O}_X(n))$. In particular, $H^0(X, \mathcal{F}) = \Gamma_*(\mathcal{O}_X) \cong S$. Note that X is noetherian, separated and covered by $r+1$ affine open subsets. By Corollary 2.7, get $\check{H}^p(\mathcal{U}, \mathcal{F}) \cong H^p(X, \mathcal{F})$ for all p .

For all $i_0, \dots, i_p$, note that $U_{i_0, \dots, i_p} = D_+(x_{i_0}, \dots, x_{i_p})$. Get $\mathcal{F}(U_{i_0, \dots, i_p}) \cong S_{x_{i_0}, \dots, x_{i_p}}$. Thus $C^*(\mathcal{U}, \mathcal{F})$ is given by

$$0 \longrightarrow \prod_{i_0} S_{x_{i_0}} \longrightarrow \dots \longrightarrow S_{x_0, \dots, x_r}. \quad (41)$$

Since $\check{H}^r(\mathcal{U}, \mathcal{F})$ is the cokernel of $d^{r-1} : \prod_k S_{x_0, \dots, \widehat{x_k}, \dots, x_r} \longrightarrow S_{x_0, \dots, x_r}$. Note that $S_{x_0, \dots, x_r}$ is free A -module spanned by $\{x_0^{l_0} \cdots x_r^{l_r}\}$, where $l_0, \dots, l_r \in \mathbb{Z}$. In addition, $\text{im}(d^{r-1})$ is free A -module spanned by $\{x_0^{l_0} \cdots x_r^{l_r}\}$, where $l_0, \dots, l_r \in \mathbb{Z}$ and at least one $l_i \geq 0$. Get $H^r(\mathcal{U}, \mathcal{F})$ is free A -module spanned by $\{x_0^{l_0} \cdots x_r^{l_r}\}$, where $l_i < 0$. While the only "monomial" of degree $-r-1$ is $x_0^{-1} \cdots x_r^{-1}$, get $H^r(X, \mathcal{O}_X(-r-1)) \cong A$.

For (3), we also find $H^r(X, \mathcal{O}_X(-n-r-1)) = 0$ if $n < 0$. On the other hand, $H^0(X, \mathcal{O}_X(n)) = 0$ if $n < 0$. Thus, (3) is trivially true for $n < 0$. For $n \geq 0$, $H^0(X, \mathcal{O}_X(n))$ has basis $\{x_0^{m_0} \cdots x_r^{m_r}\}$, where $m_i \geq 0$ and $\sum_{i=0}^r m_i = n$. The natural pairing is generated by $(x_0^{m_0} \cdots x_r^{m_r}, x_0^{l_0} \cdots x_r^{l_r}) \mapsto x_0^{m_0+l_0} \cdots x_r^{m_r+l_r}$, here that $x_i^{m_i+l_i} = 0$ if $m_i + l_i \geq 0$. Then it is a perfect pairing since it is easy to see that dual basis of $x_0^{m_0} \cdots x_r^{m_r}$ is $x_0^{-m_0-1} \cdots x_r^{-m_r-1}$.

For (1), take induction on r . When $r = 1$, there is nothing to prove. Suppose that $r \geq 2$. Consider localization of $C^*(\mathcal{U}, \mathcal{F})$ with respect to x_r . By Theorem 2.1, the localization gives the Čech complex of $\mathcal{F}|_{U_r}$ with respect to $\mathcal{U}|_{U_r}$. As localization is exact and U_r is affine, $H^i(X, \mathcal{F})_{x_r} = H^i(U_r, \mathcal{F}|_{U_r}) = 0$. Suffice to show that for all $0 < i < r$, x_r acts injectively on

$H^i(X, \mathcal{F})$. View x_r as a homomorphism of A -modules like below

$$x_r : H^i(X, \mathcal{F}(-1)) \longrightarrow H^i(X, \mathcal{F}) \quad (42)$$

here $\mathcal{F}(-1)$ is same sheaf on $\mathcal{F}$ with grading shifted. Consider exact sequence of graded C -modules

$$0 \longrightarrow S(-1) \xrightarrow{x_r} S \longrightarrow S/(x_r) \longrightarrow 0 \quad (43)$$

Get exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_X(-1) \xrightarrow{x_r} \mathcal{O}_X \longrightarrow j_*\mathcal{O}_H \longrightarrow 0 \quad (44)$$

where H is the hyperplane given by $x_r = 0$ and $j : H \hookrightarrow X$ is closed immersion. Take direct sum, get exact sequence

$$0 \longrightarrow \mathcal{F}(-1) \xrightarrow{x_r} \mathcal{F} \longrightarrow \mathcal{F}_H \longrightarrow 0 \quad (45)$$

where $\mathcal{F}_H = \oplus_{n \in \mathbb{Z}} j_*(\mathcal{O}_H(n))$. Get long exact sequence

$$\cdots \longrightarrow H^{i-1}(X, \mathcal{F}_H) \longrightarrow H^i(X, \mathcal{F}(-1)) \xrightarrow{x_r} H^i(X, \mathcal{F}) \longrightarrow H^i(X, \mathcal{F})_H \longrightarrow \cdots \quad (46)$$

Now $H \cong \mathbf{P}_A^{r-1}$ and since j is closed immersion, $H^i(X, \mathcal{F}_H) = H^i(H, \oplus_{n \in \mathbb{Z}} \mathcal{O}_H(n))$. By induction hypothesis, $H^i(X, \mathcal{F}_H) = 0$ for all $0 < i < r - 1$. Thus x_r is injective on $H^i(X, \mathcal{F}(-1))$ for all $1 < i < r$. When $i = 1$, find that $H^0(X, \mathcal{F}) = S \longrightarrow H^0(X, \mathcal{F}_H) = S/(x_r)$ is surjective. Get x_r is injective on $H^i(X, \mathcal{F}(-1))$ for all $0 < i < r$, done! $\square$

Theorem 2.7 (Serre). *Let A be a noetherian ring, X projective scheme over A , $\mathcal{F}$ coherent sheaf on X . Then*

(1) *for all $i \geq 0$, $H^i(X, \mathcal{F})$ is a finite A -module.*

(2) *for $j : X \hookrightarrow \mathbb{P}_A^r$ closed immersion, there exists integer n_0 relative to $\mathcal{F}$ such that $H^i(X, \mathcal{F}(n)) = 0$ for all $i > 0$ and $n \geq n_0$.*

Proof. Since j is finite, $j_*(\mathcal{F})$ is coherent on $\mathbb{P}_A^r$ and by Projection Formula, $j_*(\mathcal{F}(n)) = (j_*(\mathcal{F}))(n)$. What's more, since j is closed immersion, $H^i(X, \mathcal{F}) = H^i(\mathbb{P}_A^r, j_*(\mathcal{F}))$. Thus we can reduce this problem to $X = \mathbb{P}_A^r$. For $\mathcal{F} = \mathcal{O}_X(m)$, this is just what Theorem 2.6 said. Also, for general $\mathcal{F}$, as $\mathbb{P}_A^r$ can be covered by $r + 1$ affine open subsets, $H^i(X, \mathcal{F}(n)) = 0$ for all $i > r$ and $n \in \mathbb{Z}$.

For (1), by induction, suppose theorem is correct for $i + 1$. By Corollary 1.1, there exists $m \in \mathbb{Z}$ and $r < \infty$ such that $\mathcal{F}$ is quotient of $\mathcal{O}_X(m)^r$, inducing the following exact sequence of coherent sheaves

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{O}_X(m)^r \longrightarrow \mathcal{F} \longrightarrow 0 \quad (47)$$

Consider the long exact sequence

$$\cdots \longrightarrow H^i(X, \mathcal{O}_X(m)^r) \longrightarrow H^i(X, \mathcal{F}) \longrightarrow H^{i+1}(X, \mathcal{G}) \longrightarrow \cdots \quad (48)$$

By hypothesis, $H^{i+1}(X, \mathcal{G})$ is finite A -module. As A is noetherian, $H^i(X, \mathcal{F})$ is also finite.

For (2), by above discuss, we only need to check for $0 < i \leq r$. Induct on i , assume for $i + 1$ and all coherent sheaf $\mathcal{F}$, there exists $n_{i+1}(\mathcal{F})$ such that $H^{i+1}(X, \mathcal{F}(n)) = 0$ for all

$n \geq n_{i+1}(\mathcal{F})$. Take same short exact sequence. Tensor the short exact sequence by $\mathcal{O}_X(n)$. Get exact sequence

$$0 \longrightarrow \mathcal{G}(n) \longrightarrow \mathcal{O}_X(m+n)^r \longrightarrow \mathcal{F}(n) \longrightarrow 0 \quad (49)$$

Consider the long exact sequence

$$\cdots \longrightarrow H^i(X, \mathcal{O}_X(m+n)^r) \longrightarrow H^i(X, \mathcal{F}(n)) \longrightarrow H^{i+1}(X, \mathcal{G}(n)) \longrightarrow \cdots \quad (50)$$

By hypothesis, there exists $n_{i+1}(\mathcal{G})$ such that $H^{i+1}(X, \mathcal{G}(n)) = 0$ for all $n \geq n_{i+1}(\mathcal{G})$. By Theorem 2.6, $H^i(X, \mathcal{O}_X(m+n)^r) = 0$ for large enough n . Thus there exists $n_i(\mathcal{F})$ such that $H^i(X, \mathcal{F}(n)) = 0$ for all $n \geq n_i(\mathcal{F})$. As there are finitely many i , we can take a uniform upper $n_0(\mathcal{F})$ bound for $n_i(\mathcal{F})$. $\square$

Corollary 2.8. *Let $f : X \longrightarrow Y$ be a projective morphism with X, Y noetherian, $\mathcal{F}$ coherent sheaf on X . Then $f_*\mathcal{F}$ is coherent on Y .*

Reason 2.4. *Can assume $Y = \text{Spec } A$. Then $f_*\mathcal{F} \cong \widetilde{H^0(X, \mathcal{F})}$ which is a finite A -module by Theorem 2.7.*

Corollary 2.9. *Let X be a geometrically integral scheme projective over field k . Then $\Gamma(X, \mathcal{O}_X) = k$.*

Reason 2.5. *By Theorem 2.7, $L = \Gamma(X, \mathcal{O}_X)$ is finite over k . While X is integral, get L is integral domain. Thus by Noetherian Normalization Lemma, L/k is finite field extension. Note that geometrically integral if and only if k is algebraically closed in function field $k(X)$. Get $L = k$.*

3 Riemann-Roch Formula

3.1 Serre's duality

Definition 3.1. *Let X be a variety, $\mathcal{F} \in \mathbf{Ab}(X)$. Define the Euler characteristic of $\mathcal{F}$ to be $\chi(\mathcal{F}) := \sum_{i \geq 0} (-1)^i \cdot h^i(X, \mathcal{F})$, which is in fact a finite sum by Grothendieck Vanishing Theorem.*

Lemma 3.1. *Let X be a variety. Then for all short exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ in $\mathbf{Ab}(X)$, we have that $\chi(\mathcal{F}) = \chi(\mathcal{F}') + \chi(\mathcal{F}'')$.*

Proof. Applying global section functor, we get a long exact sequence

$$0 \rightarrow H^0(X, \mathcal{F}') \rightarrow H^0(X, \mathcal{F}) \rightarrow H^0(X, \mathcal{F}'') \rightarrow H^1(X, \mathcal{F}') \rightarrow \cdots \quad (51)$$

By Grothendieck Vanishing Theorem, this sequence would end for some n large enough. Then we can split it into several short exact sequences. Summing the equations of dimensions given by these short exact sequence, we are done! $\square$

Theorem 3.1 (Serre's Duality). *Let X be a smooth projective variety of dimension n , ω_X canonical sheaf. Then for all locally free sheaf $\mathcal{F}$ on X , we have canonical isomorphism $H^i(X, \mathcal{F}^\vee \otimes \omega_X) \xrightarrow{\sim} H^{n-i}(X, \mathcal{F})^\vee$.*

Remark 3.1. In class, Professor Du gave a proof as following. First take injective resolutions $\mathcal{I}^\bullet$ of $\mathcal{F}$, then tensor with ω_X . Since ω_X is flat, we $0 \rightarrow \mathcal{F} \otimes \omega_X \rightarrow \mathcal{I}^\bullet \otimes \omega_X$ is still exact. However, we do not know if $\mathcal{I} \otimes \omega_X$ is still injective.

Corollary 3.1. Let X be a smooth projective variety of dimension n , ω_X canonical sheaf. Then for all locally free sheaf $\mathcal{F}$ on X , $\chi(\mathcal{F}) = (-1)^n \cdot \chi(\mathcal{F}^\vee \otimes \omega_X)$.

Proof. Immediately comes from Grothendieck Vanishing and Serre's duality. $\square$

3.2 Divisors

Proposition 3.1. Let X be an integral variety, $\mathcal{F} \in \mathbf{Coh}(X)$. Then there exists open dense subset $U \subseteq X$ such that $\mathcal{F}|_U$ is locally free. Moreover, we can well define $\text{rank}(\mathcal{F}) := \text{rank}(\mathcal{F}|_U)$.

Proof. Consider the subset $\{x \in X \mid \dim_{k(x)} \mathcal{F}_x / \mathfrak{m}_x \mathcal{F}_x \leq d\}$, which by Nakayama's Lemma is an open subset. Then we may take d minimal so that $U := \{x \in X \mid \dim_{k(x)} \mathcal{F}_x / \mathfrak{m}_x \mathcal{F}_x = d\}$ is an open subset. Since X is integral, U is irreducible.

Now we want to show that $\mathcal{F}|_U$ is locally free. Take affine subset $W = \text{Spec } A$ such that $\mathcal{F}|_W = \widetilde{M}$ for some finitely generated A -module M . Then for all $\mathfrak{p} \in \text{Spec } A$, $M_{\mathfrak{p}}$ is generated by exact d elements.

Narrowing W , we may assume that there exists $m_1, \dots, m_d$ in M such that $M_{\mathfrak{p}}$ can be generated by them for all $\mathfrak{p} \in \text{Spec } A$. Then if there is some relation $\sum_{i=1}^d a_i m_i = 0$, then $a_i \in \mathfrak{R}(A) = (0)$ for all i . Otherwise, there exists $\mathfrak{q} \in \text{Spec } A$ such that $a_i \notin \mathfrak{q}$ for some i . And then this relation becomes a relation in $M_{\mathfrak{q}} \otimes A_{\mathfrak{q}} / \mathfrak{q} A_{\mathfrak{q}}$, contradicting to $\mathfrak{q} \in U$. Hence $\mathcal{F}|_W$ is free and $\mathcal{F}|_U$ is locally free. $\square$

Lemma 3.2. Let X be an integral variety. Then for all short exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ in $\mathbf{Coh}(X)$, we have that $\text{rank}(\mathcal{F}) = \text{rank}(\mathcal{F}') + \text{rank}(\mathcal{F}'')$.

Proof. We may take open dense subset $U \subseteq X$ such that $\mathcal{F}|_U, \mathcal{F}'|_U$ and $\mathcal{F}''|_U$ are all locally free. Then it is clear that $\text{rank}(\mathcal{F}|_U) = \text{rank}(\mathcal{F}'|_U) + \text{rank}(\mathcal{F}''|_U)$. $\square$

Definition 3.2. Let X be an integral variety, $\mathcal{F} \in \mathbf{Coh}(X)$. Define $\deg(\mathcal{F}) = \chi(\mathcal{F}) - \text{rank} \cdot \chi(\mathcal{O}_X)$.

Lemma 3.3. Let X be an integral variety. Then for all short exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ in $\mathbf{Coh}(X)$, we have that $\deg(\mathcal{F}) = \deg(\mathcal{F}') + \deg(\mathcal{F}'')$.

Reason 3.1. Immediately comes from Lemma 3.1 and Lemma 3.2

Now let C be a smooth connected projective curve. Define the (arithmetic) genus of C to be $g(C) := 1 - \chi(C) = h^1(C, \mathcal{O}_C)$.

Proposition 3.2. Let $C \subseteq \mathbb{P}_{\mathbb{C}}^2$ be a smooth curve of degree n . Then C is connected and $g(C) = \frac{(n-1)(n-2)}{2}$.

Proof. Assume $C = V(f)$ with $\deg f = n$. Consider exact sequence $0 \rightarrow \mathcal{O}_{\mathbb{P}^2}(-n) \xrightarrow{f} \mathcal{O}_{\mathbb{P}^2} \rightarrow \mathcal{O}_C \rightarrow 0$. Then $\chi(\mathcal{O}_C) = \chi(\mathcal{O}_{\mathbb{P}^2}) - \chi(\mathcal{O}_{\mathbb{P}^2}(-n))$. Since $\chi(\mathcal{O}_{\mathbb{P}^2}) = h^0(\mathcal{O}_{\mathbb{P}^2}) = 1$ and by Serre's duality, $\chi(\mathcal{O}_{\mathbb{P}^2}(-n)) = -h^1(\mathcal{O}_{\mathbb{P}^2}(-n)) = -h^1(\mathcal{O}_{\mathbb{P}^2}(-n))^\vee = -h^0(\mathcal{O}_{\mathbb{P}^2}(n-3)) = -\frac{(n-1)(n-2)}{2}$, we get that $g(C) = \frac{(n-1)(n-2)}{2}$. $\square$

Remark 3.2. *Connectedness comes from the fact that every two different curves in $\mathbb{P}^2_{\mathbb{C}}$ would intersect, in which case smoothness implies irreducibility. In general, we would just assume curve to be integral.*

Let $D = \sum_i a_i x_i$ be an effective Weil divisor on C . Define the degree of D to be $\deg D = \sum_i a_i$.

Proposition 3.3. *Every effective Weil divisor D would correspond to an invertible sheaf $\mathcal{O}_C(D)$. Moreover, $\mathcal{O}_C(D_1) \cong \mathcal{O}_C(D_2)$ if and only if $D_1 \sim D_2$.*

Proof. Consider $C_i = \text{Spec } \mathcal{O}_{C, x_i} / \mathfrak{m}_{x_i}^{a_i}$. Then there are closed immersions $\varphi_i : C_i \hookrightarrow \text{Spec } C$. Taking disjoint product $Y := \sqcup C_i$, we get a closed subscheme of C . Consider the ideal sheaf $\mathcal{O}_C(-D) := \mathcal{I}_{Y/C}$. And our desired $\mathcal{O}_C(D)$ can be defined as dual of $\mathcal{O}_C(-D)$.

Now we are supposed to show that $\mathcal{O}_C(-D)$ is locally free of rank 1. Affine locally, we have $U = \text{Spec } A$ and $C_i = \text{Spec } A / \mathfrak{p}_i^{a_i}$. Then $Y = \sqcup C_i = \text{Spec } \bigoplus A / \mathfrak{p}_i^{a_i} \cong \text{Spec } A / \prod \mathfrak{p}_i^{a_i}$ and $\mathcal{I}_{Y/C}|_U = \prod \mathfrak{p}_i^{a_i}$.

Note that C is smooth, for all i , $A_{\mathfrak{p}_i}$ is a DVR. Take uniformizers $\varpi_i \in A_{\mathfrak{p}_i}$. Then there exists open subset $W = D(s) \subseteq U$ such that $\varpi_i \in A_s$. Thus $\mathfrak{p}_i A_s$ is a principle ideal generated by ϖ_i and so is $\prod \mathfrak{p}_i^{a_i}$. Conclude that $\mathcal{O}_C(-D)$ is locally free of rank 1. $\square$

Remark 3.3. *For arbitrary Weil divisor D , we can write $D = D_1 - D_2$ with D_1 and D_2 effective. Then define $\mathcal{O}_C(D) = \mathcal{O}_C(D_1) \otimes \mathcal{O}_C(-D_2)$. In addition, the second part comes from careful argument about transition functions.*

Lemma 3.4. $\deg D = \deg(\mathcal{O}_C(D))$ for all Weil divisor D .

Proof. First prove for effective D . Take $\ell : Y \rightarrow C$ as above, then there is an exact sequence

$$0 \longrightarrow \mathcal{O}_C(-D) \longrightarrow \mathcal{O}_C \longrightarrow \ell_* \mathcal{O}_Y \longrightarrow 0 \quad (52)$$

Tensoring by $\mathcal{O}_C(D)$, we get exact sequence

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{O}_C(D) \longrightarrow \ell_* \mathcal{O}_Y(D) \longrightarrow 0 \quad (53)$$

Then by Lemma 3.1, $\deg(\mathcal{O}_C(D)) = \chi(\mathcal{O}_C(D)) - \chi(\mathcal{O}_C) = \chi(\ell_* \mathcal{O}_Y(D))$. It suffices to show $\chi(\ell_* \mathcal{O}_Y \otimes \mathcal{O}_C(D)) = \deg D$. Note that we can take affine open covering $\mathcal{U}$ of C trivializing $\mathcal{O}_C(D)$ such that for all i , there is a unique $U_i = \text{Spec } A_i$ in $\mathcal{U}$ containing x_i .

Use such a covering to compute cohomology. Since $\ell_* \mathcal{O}_Y = \bigoplus \ell_* \mathcal{O}_{C_i}$, $\chi(\ell_* \mathcal{O}_Y(D)) = \sum \chi(\ell_* \mathcal{O}_{C_i}(D))$. It is clear that $\Gamma(C, \ell_* \mathcal{O}_{C_i}(D)) \cong \Gamma(U_i, \ell_* \mathcal{O}_{C_i}(D))$. Thus $\chi(\ell_* \mathcal{O}_{C_i}(D)) = a_i - 0 = a_i$ and hence $\chi \ell_* \mathcal{O}_Y(D) = \sum a_i = \deg D$.

For arbitrary D , write $D = D_1 - D_2$ with D_1 and D_2 effective. Then $\deg D = \deg D_1 - \deg D_2 = \deg(\mathcal{O}_C(D_1)) - \deg(\mathcal{O}_C(D_2)) = \chi(\mathcal{O}_C(D_1)) - \chi(\mathcal{O}_C(D_2))$. Consider exact sequence

$$0 \longrightarrow \mathcal{O}_C(D_1 - D_2) \longrightarrow \mathcal{O}_C(D_1) \longrightarrow \mathcal{O}_{D_2}(D_1) \longrightarrow 0 \quad (54)$$

By similar argument, we know $\chi(\mathcal{O}_C(D)) = \chi(\mathcal{O}_C(D_1)) - \deg D_2 = \chi(\mathcal{O}_C(D_1)) - \chi(\mathcal{O}_C(D_2)) + \chi(\mathcal{O}_C)$. Thus $\deg D = \deg(\mathcal{O}_C(D))$. $\square$

Corollary 3.2. *The map $\deg : \text{Pic}(C) \rightarrow \mathbb{Z}$ is a group homomorphism.*

Reason 3.2. *Immediately comes from the fact that $\deg : \text{WeilDiv} \rightarrow \mathbb{Z}$ is a group homomorphism and Lemma 3.4.*

Lemma 3.5. *Let $\mathcal{L}$ be an invertible sheaf on C . Then for all nonzero global section s of $\mathcal{L}$, we have that $\mathcal{O}_C(\text{div}(s)) \cong \mathcal{L}$.*

Proof. Assume g_{ij} are transition functions of $\mathcal{L}$ and $s|_{U_i} = f_i e_i$ where e_i has image 1 under trivialization. Then we have that $f_j e_j = f_j g_{ij} e_i$ so that $f_i = g_{ij} f_j$ on $U_i \cap U_j$. By definition, $\text{div}(s) = (U_i, f_i)$ and $\mathcal{O}_C(\text{div}(s))$ is defined by transition functions $\frac{f_i}{f_j} = g_{ij}$. Hence $\mathcal{O}_C(\text{div}(s)) \cong \mathcal{L}$. $\square$

Corollary 3.3. *Let $\mathcal{L}$ be an invertible sheaf on C . If $H^0(C, \mathcal{L}) \neq 0$, then $\deg \mathcal{L} \geq 0$.*

Reason 3.3. *By Lemma 3.5, pick a nonzero global section s of $\mathcal{L}$, then we know $\mathcal{L} \cong \mathcal{O}_C(\text{div}(s))$. Note that s has no pole, $\text{div}(s)$ is an effective Weil divisor so that $\deg \mathcal{L} = \deg \mathcal{O}_C(\text{div}(s)) = \deg(\text{div}(s)) \geq 0$.*

Definition 3.3. *Let C be a smooth connected projective curve. Define a relation $\geq$ in $\text{Pic}(C)$ by $\mathcal{L} \geq \mathcal{L}'$ if $H^0(C, \mathcal{L} \otimes \mathcal{L}'^\vee) \neq 0$. Or equivalently, $\mathcal{L} \geq \mathcal{L}'$ if there is a nonzero morphism $\mathcal{L}' \hookrightarrow \mathcal{L}$.*

Corollary 3.4. *If $\mathcal{L} \geq \mathcal{L}'$, then $\deg \mathcal{L} \geq \deg \mathcal{L}'$.*

Proof. If $\mathcal{L} \geq \mathcal{L}'$, then $H^0(C, \mathcal{L} \otimes \mathcal{L}'^\vee) \neq 0$. Hence by Corollary 3.3, $\deg(\mathcal{L} \otimes \mathcal{L}'^\vee) \geq 0$ so that $\deg(\mathcal{L}) \geq \deg(\mathcal{L}')$. $\square$

3.3 Grothendieck group

Definition 3.4. *Let X be a variety. Define the Grothendieck group $K(X)$ (resp. $K_0(X)$) to be the quotient group of the free group spanned by the coherent sheaves (resp. locally free sheaves of finite rank) by the subgroup spanned by elements of the form $\mathcal{F} - \mathcal{F}' - \mathcal{F}''$ for short exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$.*

Remark 3.4. *There is a natural homomorphism $i : K_0(X) \rightarrow K(X)$ sending $\bar{\mathcal{E}}$ to $\bar{\mathcal{E}}$.*

Lemma 3.6. *Let X be a regular projective curve, $\mathcal{F} \in \mathbf{Coh}(X)$. Then there is an exact sequence $0 \rightarrow \mathcal{E}_1 \rightarrow \mathcal{E}_2 \rightarrow \mathcal{F} \rightarrow 0$ with $\mathcal{E}_1$ and $\mathcal{E}_2$ locally free of finite rank.*

Proof. Since X is projective, there is a very ample line bundle $\mathcal{O}_X(1)$. Then by Corollary 1.1, there is a surjection $\mathcal{O}_X(m)^{\oplus r} \twoheadrightarrow \mathcal{F}$. Set $\mathcal{E}_1 = \mathcal{O}_X(m)^{\oplus r}$, consider kernel $\mathcal{E}_2$. Want to show that $\mathcal{E}_2$ is locally free. In fact, since for all closed point $x \in X$, $\mathcal{O}_{X,x}$ is a DVR and hence PID. As $\mathcal{E}_{2,x}$ is a submodule of $\mathcal{E}_{1,x}$ which is free, we get $\mathcal{E}_{2,x}$ also free. Hence $\mathcal{E}_2$ is locally free of finite rank. $\square$

Proposition 3.4. *Let X be a regular projective curve. Then $i : K_0(X) \rightarrow K(X)$ is a group isomorphism.*

Proof. First to show that i is surjective. For all $\mathcal{F} \in \mathbf{Coh}(X)$, by Lemma 3.5, there is a short exact sequence

$$0 \longrightarrow \mathcal{E}_2 \xrightarrow{\varphi_2} \mathcal{E}_1 \xrightarrow{\varphi_1} \mathcal{F} \longrightarrow 0 \quad (55)$$

with $\mathcal{E}_1$ and $\mathcal{E}_2$ locally free of finite rank. Then we get $\overline{\mathcal{F}} = \overline{\mathcal{E}_1} - \overline{\mathcal{E}_2}$ so that i is surjective.

Conversely, it is natural to define $j : K(X) \rightarrow K_0(X)$ sending $\overline{\mathcal{F}}$ to $\overline{\mathcal{E}_1} - \overline{\mathcal{E}_2}$. Given another exact sequence

$$0 \longrightarrow \mathcal{E}'_2 \xrightarrow{\psi_2} \mathcal{E}'_1 \xrightarrow{\psi_1} \mathcal{F} \longrightarrow 0 \quad (56)$$

By universal property of resolution, there is a chain map

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{E}_2 & \xrightarrow{\varphi_2} & \mathcal{E}_1 & \xrightarrow{\varphi_1} & \mathcal{F} \longrightarrow 0 \\ & & \alpha_2 \uparrow & & \alpha_1 \uparrow & & \parallel \\ 0 & \longrightarrow & \mathcal{E}'_2 & \xrightarrow{\psi_2} & \mathcal{E}'_1 & \xrightarrow{\psi_1} & \mathcal{F} \longrightarrow 0 \end{array} \quad (57)$$

Then we can construct a commutative diagram with exact rows and exact columns except the medium column

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & \mathcal{E}_2 & \xrightarrow{\varphi_2} & \mathcal{E}_1 & \xrightarrow{\varphi_1} & \mathcal{F} \longrightarrow 0 \\ & & \parallel & & \varphi_2 + \alpha_1 \uparrow & & \psi_1 \uparrow \\ 0 & \longrightarrow & \mathcal{E}_2 & \longrightarrow & \mathcal{E}_2 \oplus \mathcal{E}'_1 & \longrightarrow & \mathcal{E}'_1 \longrightarrow 0 \\ & & \uparrow & & -\alpha_2 \oplus \psi_2 \uparrow & & \psi_2 \uparrow \\ & & 0 & \longrightarrow & \mathcal{E}'_2 & \xlongequal{\quad} & \mathcal{E}'_2 \longrightarrow 0 \\ & & & & \uparrow & & \uparrow \\ & & & & 0 & & 0 \end{array} \quad (58)$$

By diagram chasing, the medium column also exact. Hence $\overline{\mathcal{E}_2} + \overline{\mathcal{E}'_1} - \overline{\mathcal{E}_1} - \overline{\mathcal{E}'_2} = 0$ in $K_0(X)$ so that j is well defined. Left to show that j is a group homomorphism. For exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$, by Horseshoe Lemma, we can construct a locally free resolution of $\mathcal{F}$ as direct sum of thoes resolutions of $\mathcal{F}'$ and $\mathcal{F}''$. Thus j is indeed a group homomorphism. Note that $i \circ j = \text{id}_{K(X)}$ and $j \circ i = \text{id}_{K_0(X)}$, we conclude that i is isomorphic. $\square$

Corollary 3.5. *Let X be a regular projective curve. Then $K(X)$ admits a ring structure with multiplication induced by tensor product of locally free sheaves, called the Grothendieck ring of X .*

Corollary 3.6. *Let X be a regular projective curve. Then $K(X) \xrightarrow{j} K_0(X) \xrightarrow{\det} \text{Pic}(X)$ sending $\overline{\mathcal{L}_1 + \mathcal{L}_2}$ to $\det \mathcal{L}_1 \otimes \det \mathcal{L}_2$ is a group homomorphism.*

Theorem 3.2. *Let X be a regular projective curve, $\mathcal{F} \in \mathbf{Coh}(X)$. Then $\deg \mathcal{F} = \deg(\det \mathcal{F})$.*

Proof. Note that $\deg : K(X) \rightarrow \mathbb{Z}$ is a group homomorphism, suffices to check for generators of $K(X)$ i.e. locally free sheaves of finite rank. For each vector bundle $\mathcal{E}$ on X , we have a filtration of $\mathcal{E}$ by line bundle

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E} \quad (59)$$

such that $\mathcal{E}_{i+1}/\mathcal{E}_i = \mathcal{L}_i$ are line bundles. Then $\det \mathcal{E}_{i+1} = \det \mathcal{E}_i \otimes \mathcal{L}_i$ for all i . Combining together, we get $\det \mathcal{E} = \bigotimes_{i=0}^{r-1} \mathcal{L}_i$. Hence $\deg(\det \mathcal{E}) = \deg(\bigotimes_{i=0}^{r-1} \mathcal{L}_i) = \sum_{i=0}^{r-1} \deg \mathcal{L}_i$.

On the other hand, we also have that $\deg \mathcal{E}_{i+1} = \deg \mathcal{E}_i + \deg \mathcal{L}_i$ for all i . Summing together, we get $\deg \mathcal{E} = \sum_{i=0}^{r-1} \deg \mathcal{L}_i = \deg(\det \mathcal{E})$, done! $\square$

Lemma 3.7. *Let X be an arbitrary scheme, D, E effective Cartier divisors with disjoint supports. Then there is an exact sequence*

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_X(D) \oplus \mathcal{O}_X(E) \longrightarrow \mathcal{O}_X(D + E) \longrightarrow 0 \quad (60)$$

Proof. Construct the maps locally. Take affine cover U_i of X trivializing both $\mathcal{O}_X(D)$ and $\mathcal{O}_X(E)$. Assume $D|_{U_i}$ and $E|_{U_i}$ are defined by regular sections f_i and g_i respectively. Since $\text{Supp } D$ and $\text{Supp } E$ are disjoint, f_i and g_i are coprime.

Define $\mathcal{O}_{U_i} \rightarrow \mathcal{O}_X(D)|_{U_i} \oplus \mathcal{O}_X(E)|_{U_i}$ sending 1 to (f_i, g_i) , and $\mathcal{O}_X(D)|_{U_i} \oplus \mathcal{O}_X(E)|_{U_i} \rightarrow \mathcal{O}_X(D + E)|_{U_i}$ sending (a, b) to $g_i a - f_i b$. Obviously these maps give a short exact sequence and glue up to our desired exact sequence. $\square$

Proposition 3.5. *Let X be a regular projective curve. Then $K(X) \cong \text{Pic}(X) \oplus \mathbb{Z}$.*

Proof. Suffices to show $K_0(X) \cong \text{Pic}(X) \oplus \mathbb{Z}$. We may consider the rank map $\text{rank} : K_0(X) \rightarrow \mathbb{Z}$. Denote the kernel by $A(X)$. Then it is clear that $K_0(X) \cong A(X) \oplus \mathbb{Z}$.

Left to show that $A(X) \cong \text{Pic}(X)$. Construct map $\alpha : \text{Pic}(X) \rightarrow A(X)$ sending $\mathcal{L}$ to $\overline{\mathcal{L} - \mathcal{O}_X}$. Check that α is a group morphism. For divisors D and E , want to show $\alpha(\mathcal{O}_X(D + E)) = \alpha(\mathcal{O}_X(D)) + \alpha(\mathcal{O}_X(E))$.

In fact, as X is projective, for all different points x and y in X , $x - y$ is a principle divisor. Hence we can pick effective divisors D_1, D_2, E_1 and E_2 pairwise disjoint such that $D = D_1 - D_2$ and $E = E_1 - E_2$. Then by Lemma 3.7, there is an exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D_2 - E_2) \longrightarrow \mathcal{O}_X(D_1 + E_1 - D_2 - E_2) \oplus \mathcal{O}_X \longrightarrow \mathcal{O}_X(D_1 + E_1) \longrightarrow 0 \quad (61)$$

so that $\overline{\mathcal{O}_X(D + E)} = \overline{\mathcal{O}_X(-D_2 - E_2) + \mathcal{O}_X(D_1 + E_1) - \mathcal{O}_X}$. Similarly, we have

$$\begin{aligned} \overline{\mathcal{O}_X(D)} &= \overline{\mathcal{O}_X(-D_2) + \mathcal{O}_X(D_1) - \mathcal{O}_X} \\ \overline{\mathcal{O}_X(E)} &= \overline{\mathcal{O}_X(-E_2) + \mathcal{O}_X(E_1) - \mathcal{O}_X} \\ \overline{\mathcal{O}_X(D_1 + E_1)} &= \overline{\mathcal{O}_X(D_1) + \mathcal{O}_X(E_1) - \mathcal{O}_X} \\ \overline{\mathcal{O}_X(-D_2 - E_2)} &= \overline{\mathcal{O}_X(-D_2) + \mathcal{O}_X(-E_2) - \mathcal{O}_X} \end{aligned} \quad (62)$$

Combining these equation, we get

$$\begin{aligned}\overline{\mathcal{O}_X(D + E)} &= \overline{\mathcal{O}_X(-D_2 - E_2) + \mathcal{O}_X(D_1 + E_1) - \mathcal{O}_X} \\ &= \overline{\mathcal{O}_X(-D_2) + \mathcal{O}_X(-E_2) - \mathcal{O}_X + \mathcal{O}_X(D_1) + \mathcal{O}_X(E_1) - \mathcal{O}_X - \mathcal{O}_X} \\ &= \overline{\mathcal{O}_X(D) + \mathcal{O}_X(E) - \mathcal{O}_X}\end{aligned}\quad (63)$$

Thus α is a group homomorphism. And $\det \circ \alpha(\mathcal{L}) = \mathcal{L} \otimes \mathcal{O}_X = \mathcal{L}$ so that α is injective. On the other hand, $A(X)$ is spanned by elements of the form $\overline{\mathcal{E} - r\mathcal{O}_X}$, where $\mathcal{E}$ is of rank r . In fact, by same argument as proof 3.2, we know $\overline{\mathcal{E}} = \sum_{i=0}^{r-1} \overline{\mathcal{L}_i}$ in $K_0(X)$. Hence $\overline{\mathcal{E} - r\mathcal{O}_X} = \sum_{i=0}^{r-1} \overline{\mathcal{L}_i} - \mathcal{O}_X = \alpha(\bigotimes_{i=0}^{r-1} \mathcal{L}_i)$. Conclude that α is isomorphic. $\square$

Lemma 3.8. *Let X be a regular projective curve, $\mathcal{E}$ and $\mathcal{E}'$ vector bundles on X . Then $\deg(\mathcal{E} \otimes \mathcal{E}') = \text{rank } \mathcal{E} \cdot \deg \mathcal{E}' + \text{rank } \mathcal{E}' \cdot \deg \mathcal{E}$.*

Proof. Take filtration $(\mathcal{L}'_1, \dots, \mathcal{L}'_{r'})$ of $\mathcal{E}'$. Consider exact sequence $0 \rightarrow \mathcal{E}'_i \otimes \mathcal{E} \rightarrow \mathcal{E}'_{i+1} \otimes \mathcal{E} \rightarrow \mathcal{L}'_{i+1} \otimes \mathcal{E} \rightarrow 0$. Hence $\deg(\mathcal{E} \otimes \mathcal{E}') = \sum \deg(\mathcal{L}'_i \otimes \mathcal{E})$. Suffice to show that $\deg(\mathcal{E} \otimes \mathcal{L}) = \deg \mathcal{E} + r \cdot \deg \mathcal{L}$ for all line bundle $\mathcal{L}$. Take filtration $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ of $\mathcal{E}$. Similarly, we get $\deg(\mathcal{E} \otimes \mathcal{L}) = \sum \deg(\mathcal{L}_i \otimes \mathcal{L}) = \sum (\deg \mathcal{L}_i + \deg \mathcal{L}) = \deg \mathcal{E} + r \cdot \deg \mathcal{L}$. $\square$

Corollary 3.7. *Let X be a regular projective curve, $\mathcal{E}$ vector bundle on X . Then $\deg \mathcal{E}^\vee = -\deg \mathcal{E}$.*

Proof. Assume $\text{rank } \mathcal{E} = r$ and $g(X) = g$. Then we have that

$$\begin{aligned}\deg \mathcal{E}^\vee &= \chi(\mathcal{E}^\vee) - r(1 - g) \\ &= -\chi(\mathcal{E} \otimes \omega_X) - r(1 - g) \\ &= -\deg(\mathcal{E} \otimes \omega_X) - 2r(1 - g) \\ &= -\deg \mathcal{E} - r \cdot \deg \omega_X - 2r(1 - g) \\ &= -\deg \mathcal{E}\end{aligned}\quad (64)$$

$\square$

3.4 Riemann-Roch for curves

Let X be a smooth projective curve, $D = \sum_i n_i P_i$ a Weil divisor on X . Set $\ell(D) := \dim_k(\Gamma(X, \mathcal{O}_X(D))) = \dim_k(H^0(X, \mathcal{O}_X(D)))$. Since X is projective and $\mathcal{O}_X(D)$ is coherent, get $\ell(D) < \infty$.

Lemma 3.9. *Let X be a smooth projective curve, D a Weil divisor on X . If $\ell(D) \neq 0$, then $\deg(D) \geq 0$. If $\ell(D) \neq 0$ and $\deg(D) = 0$, then $D \sim 0$ i.e. $\mathcal{O}_X(D) \cong \mathcal{O}_X$.*

Proof. If $\ell(D) \neq 0$, then $\mathcal{O}_X(D) \neq 0$ and nonzero section s gives an effective divisor $\text{div}(s) \sim D$ so that $\deg(D) = \deg(\text{div}(s)) \geq 0$. Since $\text{div}(s)$ is effective, if moreover $\deg(D) = 0$, then s is invertible so that $D \sim 0$. $\square$

Definition 3.5 (Canonical Divisors). *A canonical divisor K_X is any Cartier divisor corresponding to the canonical sheaf ω_X . Set $g(X) (= p_g(X)) := \dim_k(\Gamma(X, \omega_X))$ to be the geometric genus of X .*

Theorem 3.3 (Riemann-Roch Theorem for Curve Case). *Let X be a smooth projective curve, D a Weil divisor on X . Then $\ell(D) - \ell(K_X - D) = \deg(D) + 1 - g(X)$.*

Remark 3.5. *By Serre's duality, $\ell(K_X - D) = \dim_k(H^1(X, \mathcal{O}_X(D)))$. In particular, assuming the theorem, for $D = 0$, we have that $g(X) = \ell(K_X) + 1 - \ell(0)$. Since $\ell(0) = \dim_k(\Gamma(X, \mathcal{O}_X)) = \dim_k(k) = 1$, get $g(X) = \ell(K_X)$, called the arithmetic genus of X .*

For $\mathcal{F}$ coherent sheaf on X , set $\chi(\mathcal{F}) := \sum_i (-1)^i \dim_k(H^i(X, \mathcal{F}))$, called the Euler characteristic of $\mathcal{F}$. By Serre duality, can rewrite Riemann-Roch Theorem as $\chi(\mathcal{O}_X(D)) = \deg(D) + 1 - g(X)$. Note that $1 - g(X) = \chi(\mathcal{O}_X)$, get $\chi(\mathcal{O}_X(D)) - \chi(\mathcal{O}_X) = \deg(D)$.

Proof. By Serre duality, $\ell(K_X - D) = h^1(X, \mathcal{O}_X(D))$ so that $\ell(D) - \ell(K_X - D) = \chi(D)$. Then by definition of degree and Lemma 3.4, $\chi(D) = \deg(D) + 1 - g(X)$, done! $\square$

Example 3.1. *Let X be a curve which smooth and proper over field k , D a Weil divisor on X .*

- (1) *If $\deg(D) > 0$, then dimension of the complete linear system of nD is $\dim_k(|nD|) = n \deg(D) - g(X)$ for n large enough, since $\ell(K_X - nD) = 0$ for n large enough.*
- (2) *Let $D = K_X$, then $\ell(K_X) - 1 = \deg(K_X) + 1 - g(X)$. Get $\deg(K_X) = 2g(X) - 2$.*
- (3) *If $g(X) = 1$, then $\deg(K_X) = 0$ and $\ell(K_X) = g(X) = 1$. Get $K_X \sim 0$.*
- (4) *Let X be an elliptic curve i.e. $g(X) = 1$ and the k -points of X isn't empty, $P_0 \in X(k)$. Take $\text{Pic}^0(X) \subseteq \text{Pic}(X)$ to be the subgroup corresponding to divisor classes of degree zero. Then $X(k) \longrightarrow \text{Pic}^0(X) \quad P \longmapsto \mathcal{O}_X(P - P_0)$ is bijective inducing a group structure over $X(k)$. Injection is obvious. For surjection, let D be a divisor of degree 0, then $\deg(D + P_0) = \deg(P_0) = 1$. By Riemann-Roch, $\ell(D + P_0) - \ell(K_X - D - P_0) = 1$. While $\deg(K_X - D - P_0) < 0$, get $\ell(K_X - D - P_0) = 0$ and $\ell(D + P_0) = 1$. Thus there exists effective divisor $P \sim D + P_0$ so that $P \longmapsto \mathcal{O}_X(D)$.*
- (5) *If $\deg(D) \geq 2g(X) + 1$, then $\mathcal{O}_X(D)$ is very ample. Thus $\mathcal{O}_X(D)$ is ample if and only if $\deg(D) > 0$.*

4 Ample Vector Bundles

4.1 Ample line bundles

Proposition 4.1. *Let A be a ring, X an A -scheme, $Y = \text{Proj}(A[x_0, \dots, x_d]) = \mathbb{P}_A^d$. Then*

- (1) *Let $f : X \longrightarrow Y$ be an A -morphism. Then $f^*\mathcal{O}_Y(1)$ is an invertible sheaf on X , generated by $d + 1$ global sections.*
- (2) *Conversely, if $\mathcal{L}$ is an invertible sheaf on X , generated by global sections $s_0, \dots, s_d$, then there exists unique A -morphism $f : X \longrightarrow Y$ such that $\mathcal{L} \cong f^*\mathcal{O}_Y(1)$ and f^*x_i is identified with s_i .*

Proof. (1): $\mathcal{O}_Y(1)$ is generated by $d + 1$ global sections, inducing global sections $s_1, \dots, s_d$ of $f^*\mathcal{O}_Y(1)$. While for all $x \in X$ and $y = f(x)$, then $(f^*\mathcal{O}_Y(1))_x = \mathcal{O}_Y(1)_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x}$. Thus $s_{0x}, \dots, s_{dx}$ generate $(f^*\mathcal{O}_Y)_x$.

(2): As assumption, X is covered by $\{X_{s_i}\}$. For all i , want to define $f_i : X_{s_i} \rightarrow D_+(x_i)$. By Hartshorne Chapter II exercise 2.4, it is equivalent to give a ring homomorphism between global sections $A[\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_d}{x_i}] \rightarrow \mathcal{O}_{X_{s_i}}$, which can be defined by mapping $\frac{x_j}{x_i}$ to $\frac{s_j}{s_i}$, where $\frac{s_j}{s_i}$ is the unique element $a \in \mathcal{O}_X(X_{s_i})$ such that $s_j|_{X_{s_i}} = as_i|_{X_{s_i}}$. It is easy to check that f_i satisfy conditions for gluing lemma. Get $f : X \rightarrow Y$. $\square$

Remark 4.1. *Intuitively, f should be $x \mapsto (s_0(x) : \dots : s_d(x))$. In addition, f is a closed immersion if and only if for all i , X_{s_i} is affine and the A -algebra is generated by $\{s_j/s_i\}$.*

Definition 4.1 (Immersion). *Let $f : X \rightarrow Y$ be a morphism of schemes. We say that f is an immersion if f factors as $X \xrightarrow{\text{open immersion}} Z \xrightarrow{\text{closed immersion}} Y$.*

Proposition 4.2. *There are lots of properties of immersions.*

- (1) *Closed immersions are immersions.*
- (2) *Composition of immersions are immersions.*
- (3) *Immersion are stable under base change.*

Remark 4.2. *Since (1),(2),(3) are in fact the properties (a),(b),(c) in Hartshorne Chapter II exercise 4.8, by the exercise, we immediately get that (d),(e),(f) also holds for immersion.*

Definition 4.2 (Quasi-projective). *Let A be a ring, X an A -scheme. We say that X is quasi-projective over A if $X \rightarrow \text{Spec } A$ factors as*

$$\begin{array}{ccc} X & \xrightarrow{\text{immersion}} & \mathbb{P}_A^d \\ & \searrow & \downarrow \\ & & \text{Spec } A \end{array}$$

In Midterm24.pdf, there is also a definition of immersion which is a little different from here and we would rename it.

Definition 4.3 (Locally Closed Immersion). *Let $f : X \rightarrow Y$ be a morphism of schemes. We say that f is a locally closed immersion if f factors as $X \xrightarrow{\text{closed immersion}} Z \xrightarrow{\text{open immersion}} Y$.*

Proposition 4.3. *Let $f : X \rightarrow Y$ be a morphism of schemes. If f is a immersion, then f is a locally closed immersion. On the other hand, if f is a locally closed immersion and suppose that either X is noetherian or f is quasi-compact, then f is an immersion.*

Definition 4.4 (Very Ample Invertible Sheaves). *Let A be a ring, X an A -scheme, $\mathcal{L}$ an invertible sheaf on X . We say that $\mathcal{L}$ is very ample relative to A if there exists A -immersion $f : X \rightarrow \mathbb{P}_A^d$ such that $\mathcal{L} \cong \mathcal{O}_X(1) = f^*\mathcal{O}_{\mathbb{P}_A^d}(1)$.*

Remark 4.3. $\mathcal{L}$ is very ample if and only if $\mathcal{L}$ can be generated by finitely many global sections such that the associated A -morphism $f : X \rightarrow \mathbb{P}_A^d$ is an immersion. If moreover $X \rightarrow \text{Spec } A$ is proper, then f is a closed immersion. Thus X is projective over A if and only if X is proper over A and admits a very ample invertible sheaf.

Definition 4.5 (Ample Invertible Sheaves). *Let X be a noetherian scheme, $\mathcal{L}$ an invertible sheaf on X . We say that $\mathcal{L}$ is ample if for all coherent sheaf $\mathcal{F}$ on X , there exists $n_0 > 0$ such that for all $n \geq n_0$, $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n$ is generated by global sections.*

Remark 4.4. Serre's Theorem are saying that for $X \rightarrow \operatorname{Spec} A$ proper with A noetherian, a very ample invertible sheaf $\mathcal{L}$ on X is ample. In fact, can remove proper by reducing to proper case with the following nontrivial fact which is Hartshorne Chapter II exercise 5.15.

Fact: Let X be a noetherian scheme, $U \subseteq X$ open subset, $\mathcal{F}$ coherent sheaf on U . Then there exists coherent sheaf on X whose restriction to U is just $\mathcal{F}$.

Example 4.1. (1) Any invertible sheaf on an affine noetherian scheme is ample.

(2) If $\mathcal{L}$ is ample, then $\mathcal{L}^r$ is ample for all $r > 0$. Conversely, if $\mathcal{L}^r$ is ample for some $r > 0$, then $\mathcal{L}$ is ample.

(3) Let $X = \mathbb{P}_A^d$. Then $\mathcal{O}_X(n)$ is very ample/ample if and only if $n > 0$.

Lemma 4.1. Let X be a noetherian scheme, $\mathcal{L}$ ample invertible sheaf on X . For all $x \in X$, there exists $n > 0$ and $s \in \mathcal{L}^n(X)$ such that $x \in X_s \subseteq X$ and X_s is an affine open subset.

Proof. Let $U \subseteq X$ be an affine open neighbourhood of x such that $\mathcal{L}|_U$ is free. Our goal is to find an affine open subset inside U of the form X_s containing x . Consider $Z = X \setminus U$ with the reduced scheme structure and the ideal sheaf $\mathcal{I}$ on Z . For all $n > 0$, since $\otimes_{\mathcal{O}_X} \mathcal{L}^n$ is exact functor, $\mathcal{I}\mathcal{L}^n \subseteq \mathcal{L}^n$ is a subsheaf. As $\mathcal{L}$ is ample, for n large enough, $\mathcal{I}\mathcal{L}^n$ is generated by global sections. In particular, since $x \notin Z$, there exists global section $s \in \mathcal{I}\mathcal{L}^n(X) \subseteq \mathcal{L}^n(X)$ such that s_x generates $(\mathcal{I}\mathcal{L}^n)_x \cong \mathcal{O}_{X,x} \otimes_{\mathcal{O}_{X,x}} \mathcal{L}_x^n \cong \mathcal{L}_x^n$. Thus $x \in X_s$. On the other hand, $X_s \subseteq U$ since if $y \in X_s$, then $(\mathcal{I}\mathcal{L}^n)_y$ contains s_y which generates $\mathcal{L}_y^n$. Thus $X_s = X_s \cap U$ is affine. $\square$

Theorem 4.1. Let A be a noetherian ring and X be a scheme of finite type over A , $\mathcal{L}$ an invertible sheaf on X . If $\mathcal{L}$ is ample, then there exists $r > 0$ such that $\mathcal{L}^r$ is very ample. Conversely, if $\mathcal{L}^r$ very ample for some $r > 0$, then $\mathcal{L}$ is ample.

Proof. Since X is noetherian, we can cover X by finitely many $\{X_{s_i}\}$ as in Lemma 4.1. Can also assume $s_i \in \mathcal{L}^n(X)$ for same n and can even assume that $n = 1$ by replacing $\mathcal{L}$ by $\mathcal{L}^n$. Note that X is of finite type over A , $\mathcal{O}_X(X_{s_i})$ is generated as A -algebra by finitely many $\{g_{ij}\}$. By Lemma 1.2, there exists $r > 0$ such that $g_{ij} \otimes s_i|_{X_{s_i}}^r$ extends to a global section $s_{ij} \in (\mathcal{O}_X \otimes_{\mathcal{O}_X} \mathcal{L}^r)(X) = \mathcal{L}^r(X)$ for all i, j . Now $\{s_i^r\}$ generate $\mathcal{L}^r$ since $\{X_{s_i} = X_{s_i^r}\}$ cover X . Thus $\{s_i^r, s_{ij}\}$ also generate $\mathcal{L}^r$.

By Proposition 4.1, there is a morphism $f : X \rightarrow \operatorname{Proj}(A[x_i, x_{ij}]) = \mathbb{P}$ such that $f^*\mathcal{O}_{\mathbb{P}}(1) \cong \mathcal{L}^n$. Now for all i , $\mathcal{O}_{\mathbb{P}}(D_+(x_i)) \rightarrow \mathcal{O}_X(X_{s_i})$ $x_{ij}/x_i \mapsto g_{ij}$ is surjective, inducing by $g_{ij} \otimes s_i|_{X_{s_i}}^r$ extends to s_{ij} . Then f is a closed immersion from X to $\cup_i D_+(x_i) \subseteq \mathbb{P}$, since X_{s_i} and $D_+(x_i)$ are both affine. Get f is locally closed immersion. As X is Noetherian, by Proposition 4.3, get f is immersion. Thus $\mathcal{L}^r$ is very ample by remark 4.2.

Conversely, first to show very ampleness implies ampleness. Assume $\mathcal{L}^r$ corresponds to immersion $f : X \rightarrow \mathbb{P}$ and $\mathcal{L}^r = f^*\mathcal{O}_{\mathbb{P}}(1)$. Then for all $\mathcal{F} \in \mathbf{Coh}(X)$, $f_*\mathcal{F}$ is a coherent sheaf on $\mathbb{P}$. As $\mathcal{O}_{\mathbb{P}}(1)$ is ample, $f_*\mathcal{F}(n)$ is globally generated for n large enough.

By Projection Formula, $f_*(\mathcal{F} \otimes f^*\mathcal{O}_{\mathbb{P}}(n)) \cong f_*\mathcal{F}(n)$. And there is a commutative diagram

for each $x \in X$,

$$\begin{array}{ccc} H^0(\mathbb{P}, f_*(\mathcal{F} \otimes f^*\mathcal{O}_{\mathbb{P}}(n))) \otimes \mathcal{O}_{\mathbb{P}, f(x)} & \longrightarrow & f_*(\mathcal{F} \otimes f^*\mathcal{O}_{\mathbb{P}}(n))_{f(x)} \\ \downarrow & & \parallel \\ H^0(X, \mathcal{F} \otimes \mathcal{L}^{rn}) \otimes \mathcal{O}_{X,x} & \xrightarrow{\alpha} & (\mathcal{F} \otimes \mathcal{L}^{rn})_x \end{array} \quad (65)$$

where the right arrow is an equality because f is immersion. Then α is surjective and conclude that $\mathcal{L}^r$ is ample and so is $\mathcal{L}$. $\square$

Corollary 4.1. *Let A be a noetherian ring and X be a scheme of finite type over A , $\mathcal{L}$ an ample invertible sheaf on X . Then X is quasi-projective (resp. projective) over A if and only if X admits an ample invertible sheaf (resp. X admits an ample invertible sheaf and X is proper).*

Remark 4.5. *Let A be a noetherian ring and X be a scheme of finite type over A , $\mathcal{L}$ an ample invertible sheaf on X . If X admits an ample invertible sheaf, then X is separated.*

Corollary 4.2. *Let A be a noetherian ring, X proper over $\text{Spec } A$, $\mathcal{L}$ ample invertible sheaf on X . Then for all $\mathcal{F}$ coherent sheaf on X , there exists n_0 relative to $\mathcal{F}$ such that $H^i(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n) = 0$ for all $i > 0$ and $n \geq n_0$.*

Reason 4.1. *Note that there exists $m > 0$ such that $\mathcal{L}^m$ is very ample. Since X is proper, get closed immersion $i : X \hookrightarrow \mathbb{P}_A^r$ such that $\mathcal{L}^m = \mathcal{O}_X(1)$. Apply Theorem 2.7 to $\mathcal{F}, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}, \dots, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{m-1}$.*

Theorem 4.2 (Serre Cohomological Criteria for Ampleness). *Let A be a noetherian ring, X proper over $\text{Spec } A$, $\mathcal{L}$ invertible sheaf on X . Then the following conditions are equivalent*

- (1) $\mathcal{L}$ is ample
- (2) for all $\mathcal{F}$ coherent sheaf on X , there exists n_0 relative to $\mathcal{F}$ such that $H^i(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n) = 0$ for all $i > 0$ and $n \geq n_0$.
- (3) for all $\mathcal{F}$ coherent sheaf on X , there exists n_0 relative to $\mathcal{F}$ such that $H^1(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n) = 0$.

Proof. Step 1. Since by Corollary 4.2, we have (1) $\Rightarrow$ (2), and (2) $\Rightarrow$ (3) is obvious. Only need to prove (3) $\Rightarrow$ (1). Moreover, suffices to prove that for all $\mathcal{F}$ coherent sheaf on X and closed point $x \in X$, there exists open neighbourhood $U_x \ni x$ and $n_0(x) > 0$ such that for all $n \geq n_0(x)$ and $y \in U_x$, the stalk $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)_y$ is generated by global sections of $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n$, since X noetherian, it can be covered by only finitely many U_x so that we can take a uniform upper bound of $n_0(x)$.

Step 2. Claim that it suffices to prove that for all $\mathcal{F}$ coherent sheaf on X , closed point $x \in X$ and large enough n , there exists open neighbourhood $U_x \ni x$ depending on n such that for all $y \in U_x$, the stalk $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)_y$ is generated by global sections of $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n$. Assume we have proved this. Apply it to $\mathcal{F} = \mathcal{O}_X$, there exists large enough n_1 and $V \ni x$ such that for all $y \in V$, $(\mathcal{L}^{n_1})_y$ is generated by global sections. Then apply this to $\mathcal{F}$, there exists large enough n_0 and open neighbourhoods $U_0, U_1, \dots, U_{n_1-1}$ of x such that for all i and $y \in U_i$,

$(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{n+i})_y$ is generated by global sections. Take $U_x = V \cap U_0 \cap \cdots \cap U_{n_1-1}$. For all $n \geq n_0$ and $y \in U_x$, assume $n = n_0 + mn_1 + i$, then $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)_y = (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{n_0+i})_y \otimes_{\mathcal{O}_{X,y}} (\mathcal{L}^{n_1})_y^m$ is generated by global sections.

Step 3. Let $x \in X$ be a closed point, $\mathcal{I}_x$ the ideal sheaf of $\{x\} \xrightarrow{i} X$, where $\{x\}$ has the same scheme structure as $\text{Spec}(k(x))$ with structure sheaf denoted by $k(x)$ for convenience. We have short exact sequence

$$0 \longrightarrow \mathcal{I}_x \longrightarrow \mathcal{O}_x \longrightarrow i_*k(x) \longrightarrow 0 \quad (66)$$

Tensor by $\mathcal{F}$, get

$$\mathcal{I}_x \mathcal{F} \longrightarrow \mathcal{F} \longrightarrow i_*k(x) \otimes_{\mathcal{O}_X} \mathcal{F} \longrightarrow 0 \quad (67)$$

Tensor by $\mathcal{L}^r$, get

$$\mathcal{I}_x \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n \longrightarrow \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n \longrightarrow i_*k(x) \otimes_{\mathcal{O}_X} \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n \longrightarrow 0 \quad (68)$$

Note that, by hypothesis, $H^1(X, \mathcal{I}_x \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n) = 0$ for all $n \geq n_0$. So $\Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n) \longrightarrow \Gamma(X, i_*k(x) \otimes_{\mathcal{O}_X} \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)$ is surjective for all $n \geq n_0$. Note that $i_*k(x) \otimes_{\mathcal{O}_X} \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n$ is only supported at x , we have that $\Gamma(X, i_*k(x) \otimes_{\mathcal{O}_X} \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n) = (i_*k(x) \otimes_{\mathcal{O}_X} \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)_x = k(x) \otimes_{\mathcal{O}_{X,x}} (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)_x$. Denote $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)_x$ by M . As $M/\mathfrak{m}_x M$ is generated by global sections, by Nakayama's Lemma, M is generated by global sections.

As $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n$ is coherent, take an affine open neighbourhood $U = \text{Spec } A$ of x so that $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)|_U = \tilde{N}$ for some finitely generated A -module. Assume x corresponds to maximal ideal $\mathfrak{m}$ and N is generated by $m_1, \dots, m_l$. Since $N_{\mathfrak{m}}$ is generated by global sections, $\frac{m_k}{1} = \frac{\sum_j a_{kj} s_{j,x}}{b_k}$, where $a_{kj} \in A$, $b_k \notin \mathfrak{m}$ and s_j are global sections. Then there exists $c_k \notin \mathfrak{m}$ such that $(b_k m_k - \sum_j a_{kj} s_{j,x}) c_k = 0$. Consider $b = \prod_{k=1}^l b_k c_k$, then $\mathfrak{m} \in D(b)$. Obviously, for each $\mathfrak{p} \in D(b)$, $\frac{m_k}{1}$ can be generated by global sections in $N_{\mathfrak{p}}$. Thus for all $n \geq n_0$, there exists open neighbourhood $U_x \ni x$ depending on n such that for all $y \in U_x$, $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)_y$ is generated by global sections, done! $\square$

Remark 4.6. In fact, all finiteness results in Theorem 2.7 still hold for X proper over A . Proofs can be seen in EGA or Stacks Project.

Proposition 4.4. Let X be a variety, $\mathcal{L}$ and $\mathcal{M}$ invertible sheaves on X . Then

- (1) If $\mathcal{L}$ and $\mathcal{M}$ are both ample, then $\mathcal{L} \otimes \mathcal{M}$ is ample.
- (2) If $\mathcal{L}$ is ample, then $\mathcal{L}^m \otimes \mathcal{M}$ is ample for m large enough.

Proof. (1): For all coherent sheaf $\mathcal{F}$, as $\mathcal{L}$ is ample, there exists n_1 such that $\mathcal{F} \otimes \mathcal{L}^n$ is globally generated for all $n \geq n_1$. In addition, since $\mathcal{M}$ is ample, there exists n_2 such that $\mathcal{L} \otimes \mathcal{M}^n$ is globally generated for all $n \geq n_2$. Set $n_0 = \max\{n_1 + 1, n_2\}$. Then for all $n \geq n_0$, $\mathcal{F} \otimes (\mathcal{L} \otimes \mathcal{M})^n = (\mathcal{F} \otimes \mathcal{L}^{n-1}) \otimes (\mathcal{L} \otimes \mathcal{M}^n)$ is globally generated. Thus $\mathcal{L} \otimes \mathcal{M}$ is ample.

(2): Since $\mathcal{L}$ is ample, there exists m_0 such that $\mathcal{L}^m \otimes \mathcal{M}$ is globally generated for all $m \geq m_0$. We may take $m \geq m_0 + 1$. For all coherent sheaf $\mathcal{F}$, there exists n_0 such that $\mathcal{F} \otimes \mathcal{L}^n$ is globally generated for all $n \geq n_0$. Then $\mathcal{F} \otimes (\mathcal{L}^m \otimes \mathcal{M})^n = (\mathcal{F} \otimes \mathcal{L}^n) \otimes (\mathcal{L}^{m-1} \otimes \mathcal{M})^n$ is globally generated. Thus $\mathcal{L}^m \otimes \mathcal{M}$ is ample for m large enough. $\square$

Proposition 4.5. *Let $f : X \rightarrow Y$ be a proper and affine morphism between noetherian schemes, $\mathcal{L}$ ample invertible sheaf on Y . Then $f^*\mathcal{L}$ is ample invertible sheaf on X .*

Proof. Since f is affine, the natural map $f^*f_*\mathcal{F} \rightarrow \mathcal{F}$ is surjective for all coherent sheaf $\mathcal{F}$, which can be affine locally defined as $M \otimes_A B \rightarrow M$. In addition, as f is proper, $f_*\mathcal{F}$ is a coherent sheaf on Y .

Then by Projection Formula, we know $f_*(\mathcal{F} \otimes f^*\mathcal{L}^n) \cong f_*\mathcal{F} \otimes \mathcal{L}^n$ is globally generated for n large enough. Hence we have a surjection $\mathcal{O}_Y^{\oplus I} \rightarrow f_*\mathcal{F} \otimes \mathcal{L}^n$. Take pull back, we get $\mathcal{O}_X^{\oplus I} \rightarrow f^*f_*(\mathcal{F} \otimes f^*\mathcal{L}^n) \rightarrow \mathcal{F} \otimes f^*\mathcal{L}^n$ for n large enough. Conclude that $f^*\mathcal{L}$ is ample. $\square$

Remark 4.7. *When f is not affine, then we cannot take affine cover such that $f^*f_*\mathcal{F}$ is of the form $M \otimes_A B$ because $f_*\mathcal{F}$ locally is not given by M as an A -module. So the natural map $f^*f_*\mathcal{F} \rightarrow \mathcal{F}$ is not surjective.*

Corollary 4.3. *Let X be a noetherian scheme, $\mathcal{L}$ ample invertible sheaf on X . Assume Y is a locally closed subscheme of X . Then $\mathcal{L}|_Y$ is ample.*

Proof. Since a locally closed immersion is composition of open immersion and closed immersion. It suffices to prove these two cases. When $f : Y \hookrightarrow X$ is open, by Hartshorne Chapter II exercise 5.15, for all $\mathcal{F} \in \mathbf{Coh}(Y)$, we can extend $\mathcal{F}$ to a coherent sheaf $\mathcal{F}'$ on X . Hence $\mathcal{F}' \otimes \mathcal{L}^n$ is globally generated for n large enough. While $\mathcal{F}' \otimes \mathcal{L}^n$ and $\mathcal{F} \otimes \mathcal{L}^n|_Y$ have same stalks at $y \in Y$, it is clear that $\mathcal{F} \otimes \mathcal{L}^n|_Y$ is globally generated so that $\mathcal{L}|_Y$ is ample. On the other hand, when $f : Y \rightarrow X$ is closed, since f is proper and affine, by Proposition 4.6, $\mathcal{L}|_Y$ is ample. $\square$

Proposition 4.6. *Let X be complete scheme, $\mathcal{L}$ invertible sheaf on X . Then*

- (1) $\mathcal{L}$ is ample on X if and only if $\mathcal{L}|_{X^{\text{red}}}$ is ample on X^{red} .
- (2) $\mathcal{L}$ is ample on X if and only if $\mathcal{L}|_{X_i}$ is ample on X_i for all irreducible component X_i of X .

Proof. Since $f : X^{\text{red}} \rightarrow X$ and $f_i : X_i \hookrightarrow X$ are closed immersions hence finite, by Proposition 4.5, if $\mathcal{L}$ is ample on X , then $\mathcal{L}|_{X^{\text{red}}}$ is ample on X^{red} and $\mathcal{L}|_{X_i}$ is ample on X_i .

For the converse, if $\mathcal{L}|_{X^{\text{red}}}$ is ample on X^{red} , consider the following commutative diagram for all $\mathcal{F} \in \mathbf{Coh}(X)$ and $x \in X$

$$\begin{array}{ccc} H^0(X, \mathcal{F} \otimes \mathcal{L}^n) \otimes \mathcal{O}_{X,x} & \xrightarrow{\alpha} & (\mathcal{F} \otimes \mathcal{L}^n)_x \\ \downarrow & & \parallel \\ H^0(X^{\text{red}}, f^*\mathcal{F} \otimes f^*\mathcal{L}^n) \otimes \mathcal{O}_{X^{\text{red}},x} & \twoheadrightarrow & (f^*\mathcal{F} \otimes f^*\mathcal{L}^n)_x \end{array} \quad (69)$$

Then α is surjective so that $\mathcal{L}$ is ample on X . Similarly, if $\mathcal{L}|_{X_i}$ is ample on X_i for all irreducible component X_i of X , then for all $x \in X$, we can find irreducible component X_i containing x . Consider the following commutative diagram for all $\mathcal{F} \in \mathbf{Coh}(X)$

$$\begin{array}{ccc} H^0(X, \mathcal{F} \otimes \mathcal{L}^n) \otimes \mathcal{O}_{X,x} & \xrightarrow{\alpha} & (\mathcal{F} \otimes \mathcal{L}^n)_x \\ \downarrow & & \parallel \\ H^0(X_i, f_i^*\mathcal{F} \otimes f_i^*\mathcal{L}^n) \otimes \mathcal{O}_{X_i,x} & \twoheadrightarrow & (f_i^*\mathcal{F} \otimes f_i^*\mathcal{L}^n)_x \end{array} \quad (70)$$

Thus $\mathcal{L}$ is ample on X . $\square$

4.2 Ample vector bundles

Definition 4.6. Let X be a noetherian scheme. A vector bundle $\mathcal{E}$ on X is said to be ample if for all $\mathcal{F} \in \mathbf{Coh}(X)$, the sheaf $\mathcal{F} \otimes S^n \mathcal{E}$ is globally generated for n large enough.

Proposition 4.7 (Local Criteria). Let X be a noetherian scheme, $\mathcal{E}$ vector bundle on X . Then $\mathcal{E}$ is ample if and only if for all $\mathcal{F} \in \mathbf{Coh}(X)$ and closed point $x \in X$, $H^0(X, \mathcal{F} \otimes S^n \mathcal{E})$ generate $(\mathcal{F} \otimes S^n \mathcal{E})_x$ as $\mathcal{O}_{X,x}$ -module.

Proof. First take $n_{1,x}$ such that $H^0(X, S^n \mathcal{E})$ generate $(S^n \mathcal{E})_x$ for all $n \geq n_{1,x}$. Then there is an open neighbourhood $U_{1,x}$ of x such that $S^{n_{1,x}} \mathcal{E}|_{U_{1,x}}$ is globally generated. And we can take $n_{2,x}$ large enough such that $H^0(X, \mathcal{F} \otimes S^i \mathcal{E} \otimes S^n \mathcal{E})$ generate $(\mathcal{F} \otimes S^i \mathcal{E} \otimes S^n \mathcal{E})$ for all $n \geq n_{2,x}$ and for all $0 \leq i \leq n_{1,x} - 1$. Then there is an open neighbourhood $U_{2,x}$ of x such that $\mathcal{F} \otimes S^i \mathcal{E} \otimes S^{n_{2,x}} \mathcal{E}|_{U_{2,x}}$ are globally generated.

Set $U_x = U_{1,x} \cap U_{2,x}$ and $n_{0,x} = n_{2,x}$. Then for all $n \geq n_{0,x}$, we can write $n = n_{2,x} + r \cdot n_{1,x} + i$, where $r \geq 0$ and $0 \leq i \leq n_{1,x} - 1$. As $\mathcal{F} \otimes S^i \mathcal{E} \otimes S^{n_{2,x}} \mathcal{E} \otimes (S^{n_{1,x}} \mathcal{E})^r|_{U_{0,x}}$ is globally generated and there is a surjection

$$\mathcal{F} \otimes S^i \mathcal{E} \otimes S^{n_{2,x}} \mathcal{E} \otimes (S^{n_{1,x}} \mathcal{E})^r \twoheadrightarrow \mathcal{F} \otimes S^n \mathcal{E} \quad (71)$$

so that $\mathcal{F} \otimes S^n \mathcal{E}|_{U_x}$ is globally generated. Note that X is quasi-compact so that U_x cover X and there is a finite subcovering $U_{x_1}, \dots, U_{x_m}$. Take $n_0 = \max\{n_{0,x_1}, \dots, n_{0,x_m}\}$. It is clear that $\mathcal{F} \otimes S^n \mathcal{E}$ is globally generated for all $n \geq n_0$. Conclude that $\mathcal{E}$ is ample. $\square$

Remark 4.8. Here the statement that $S^n \mathcal{E}|_{U_{1,x}}$ is globally generated might be a little misunderstood. What we mean is that $H^0(X, S^n \mathcal{E})$ generate $(S^n \mathcal{E})_y$ for all $y \in U_{1,x}$. But sometimes, it can be understood as the inverse image of $S^n \mathcal{E}$ along $U_{1,x} \hookrightarrow X$ is globally generated on $U_{1,x}$. And we cannot conclude $S^n \mathcal{E}$ is globally generated with the latter one.

Proposition 4.8. Let X be a noetherian scheme, $\mathcal{E}$ ample vector bundle on X . If $\mathcal{E}'$ is a vector bundle which is a quotient of $\mathcal{E}$, then $\mathcal{E}'$ is ample.

Reason 4.2. Note that there is a surjection $S^n \mathcal{E} \twoheadrightarrow S^n \mathcal{E}'$.

Proposition 4.9. Let X be a noetherian scheme, $\mathcal{E}_1$ and $\mathcal{E}_2$ ample vector bundles on X . Then for all $\mathcal{F} \in \mathbf{Coh}(X)$, $\mathcal{F} \otimes S^p \mathcal{E}_1 \otimes S^q \mathcal{E}_2$ is globally generated for $p + q$ large enough.

Proof. Firstly, we can take n_1 such that $S^n \mathcal{E}_1$ is globally generated for all $n \geq n_1$ and n_2 such that $S^n \mathcal{E}_2$ is globally generated for all $n \geq n_2$. Then take n_3 such that $\mathcal{F} \otimes S^i \mathcal{E}_2 \otimes S^n \mathcal{E}_1$ is globally generated for all $n \geq n_3$ and $0 \leq i \leq n_2 - 1$, and n_4 such that $\mathcal{F} \otimes S^j \mathcal{E}_1 \otimes S^n \mathcal{E}_2$ is globally generated for all $n \geq n_4$ and $0 \leq j \leq n_1 - 1$.

Put $n_0 = n_3 + n_4$. Then for all $p + q \geq n_0$, we know $p \geq n_3$ or $q \geq n_4$. We may assume that $p \geq n_3$ and write $q = r \cdot n_2 + i$, where $0 \leq i \leq n_2 - 1$. Note that there is a surjection

$$\mathcal{F} \otimes S^p \mathcal{E}_1 \otimes S^{r \cdot n_2} \mathcal{E}_2 \otimes S^i \mathcal{E}_2 \twoheadrightarrow \mathcal{F} \otimes S^p \mathcal{E}_1 \otimes S^q \mathcal{E}_2 \quad (72)$$

And by our assumption, $\mathcal{F} \otimes S^i \mathcal{E}_2 \otimes S^p \mathcal{E}_1$ and $S^{r \cdot n_2} \mathcal{E}_2$ are both globally generated. Conclude that $\mathcal{F} \otimes S^p \mathcal{E}_1 \otimes S^q \mathcal{E}_2$ is globally generated for $p + q$ large enough. $\square$

Corollary 4.4. *Let X be a noetherian scheme, $\mathcal{E}_1$ and $\mathcal{E}_2$ vector bundles on X . Then $\mathcal{E}_1$ and $\mathcal{E}_2$ are ample if and only if $\mathcal{E}_1 \oplus \mathcal{E}_2$ is ample.*

Proof. Note that $S^n(\mathcal{E}_1 \oplus \mathcal{E}_2) = \bigoplus_{i=0}^n S^i \mathcal{E}_1 \otimes S^{n-i} \mathcal{E}_2$. If $\mathcal{E}_1$ and $\mathcal{E}_2$ are ample, then by Proposition 4.9, $\mathcal{E}_1 \oplus \mathcal{E}_2$ is ample. For the converse, since $\mathcal{E}_\infty$ and $\mathcal{E}_2$ are both quotient of $\mathcal{E}_1 \oplus \mathcal{E}_2$. By Proposition 4.8, done! $\square$

Corollary 4.5. *Let X be a noetherian scheme, $\mathcal{E}_1$ ample vector bundle on X . Then for all globally generated vector bundle $\mathcal{E}_2$ on X , $\mathcal{E}_1 \otimes \mathcal{E}_2$ is ample.*

Proof. Since $\mathcal{E}_2$ is globally generated, there is an surjection

$$\mathcal{O}_X^{\oplus r} \twoheadrightarrow \mathcal{E}_2 \rightarrow 0 \quad (73)$$

where r is the dimension of global sections of $\mathcal{E}_2$. Tensoring by $\mathcal{E}_1$, we get

$$\mathcal{E}_1^{\oplus r} \twoheadrightarrow \mathcal{E}_1 \otimes \mathcal{E}_2 \rightarrow 0 \quad (74)$$

Then by Corollary 4.4, we know $\mathcal{E}_1^{\oplus r}$ is ample. Hence by Proposition 4.8, $\mathcal{E}_1 \otimes \mathcal{E}_2$ is ample. $\square$

Proposition 4.10. *Let X be a noetherian scheme, $\mathcal{E}$ vector bundle on X . If $\mathcal{E}$ is ample, then $S^n \mathcal{E}$ is ample for n large enough. Conversely, if $S^n \mathcal{E}$ is ample for some $n > 0$, then $\mathcal{E}$ is ample.*

Proof. Firstly note that when $\mathcal{E}$ is ample, $S^n \mathcal{E}$ is globally generated for n large enough. By Corollary 4.5, $\mathcal{E} \otimes S^n \mathcal{E}$ is ample for n large enough. As $S^{n+1} \mathcal{E}$ is a quotient of $\mathcal{E} \otimes S^n \mathcal{E}$, by Proposition 4.8, $S^{n+1} \mathcal{E}$ is ample for n large enough.

Conversely, if $S^n \mathcal{E}$ is ample for some $n > 0$, then for all coherent $\mathcal{F}$, consider $\mathcal{F} \otimes S^i \mathcal{E}$ for $0 \leq i \leq n-1$. We can take m_0 such that $\mathcal{F} \otimes S^i \mathcal{E} \otimes S^m(S^n \mathcal{E})$ is globally generated for all $m \geq m_0$. Note that there is a surjection $S^i \mathcal{E} \otimes S^m(S^n \mathcal{E}) \twoheadrightarrow S^{i+mn} \mathcal{E}$, we know $\mathcal{F} \otimes S^{i+mn} \mathcal{E}$ is globally generated. Thus $\mathcal{F} \otimes S^m \mathcal{E}$ is globally generated for all $m \geq m_0 n$. Conclude that $\mathcal{E}$ is ample. $\square$

Corollary 4.6. *Let X be a noetherian scheme, $\mathcal{E}$ vector bundle on X , $\mathcal{L}$ ample line bundle on X . Then $\mathcal{E}$ is ample if and only if $\mathcal{L}^\vee \otimes S^n \mathcal{E}$ is globally generated for some $n > 0$.*

Reason 4.3. *Immediately comes from Proposition 4.10 and Corollary 4.4.*

4.3 Serre criteria of ampleness

Given a vector bundle $\mathcal{E}$ on X . We claim that $\mathcal{E}$ is ample if and only if $\mathcal{L} := \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ is ample line bundle on $\mathbb{P}(\mathcal{E})$, where $\mathbb{P}(\mathcal{E}) \xrightarrow{\pi} X$ is the projective bundle on X associated to $\mathcal{E}$.

Lemma 4.2. *Let X be a noetherian scheme. For any coherent sheaf $\mathcal{F}$ on X and $n \geq 0$, we have an isomorphism*

$$\mathcal{F} \otimes S^n \mathcal{E} \xrightarrow{\sim} \pi_*(\pi^* \mathcal{F} \otimes \mathcal{L}^n) \quad (75)$$

In addition, $R^i \pi_(\pi^* \mathcal{F} \otimes \mathcal{L}^n) = 0$ for all $i > 0$.*

Proof. Note that there is a natural map $\mathcal{F} \rightarrow \pi_* \pi^* \mathcal{F}$ associated to identity $\pi^* \mathcal{F} \rightarrow \pi^* \mathcal{F}$. As $\pi_* \mathcal{L}^n \cong S^n(\mathcal{E})$, we get natural map $\mathcal{F} \otimes^{\text{pre}} S^n(\mathcal{E}) \rightarrow \pi_* \pi^* \mathcal{F} \otimes^{\text{pre}} \pi_* \mathcal{L}^n$ between presheaves. By universal property of sheafification, there is a natural map $\varphi : \mathcal{F} \otimes S^n \mathcal{E} \rightarrow \pi_*(\pi^* \mathcal{F} \otimes \mathcal{L}^n)$ such that the following diagram commutes

$$\begin{array}{ccc} \mathcal{F} \otimes S^n \mathcal{E} & \xrightarrow{\varphi} & \pi_*(\pi^* \mathcal{F} \otimes \mathcal{L}^n) \\ \uparrow & & \uparrow \\ \mathcal{F} \otimes^{\text{pre}} S^n(\mathcal{E}) & \longrightarrow & \pi_* \pi^* \mathcal{F} \otimes^{\text{pre}} \pi_* \mathcal{L}^n \end{array} \quad (76)$$

Now we can reduce to affine case. Assume $X = \text{Spec } A$ with A noetherian and $\mathcal{E} = \mathcal{O}_X^{\oplus r}$. Then $\mathbb{P}(\mathcal{E}) \cong \mathbb{P}_A^{r-1}$.

Now first to prove the second statement by descending induction on i . Note that $R^i \pi_*(\pi^* \mathcal{F} \otimes \mathcal{L}^n) = H^i(\mathbb{P}(\mathcal{E}), \widetilde{\pi^* \mathcal{F} \otimes \mathcal{L}^n})$ and $\dim \mathbb{P}(\mathcal{E}) = r - 1$. By Grothendieck Vanishing, $R^i \pi_*(\pi^* \mathcal{F} \otimes \mathcal{L}^n) = 0$ for all $i > r - 1$.

Suppose we have shown $R^{i+1} \pi_*(\pi^* \mathcal{F} \otimes \mathcal{L}^n) = 0$ for all coherent $\mathcal{F}$ on X . For all short exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$, apply push forward functor and we get a long exact sequence

$$\cdots \longrightarrow R^i \pi_*(\pi^* \mathcal{F}' \otimes \mathcal{L}^n) \longrightarrow R^i \pi_*(\pi^* \mathcal{F} \otimes \mathcal{L}^n) \longrightarrow R^i \pi_*(\pi^* \mathcal{F}'' \otimes \mathcal{L}^n) \longrightarrow 0 \quad (77)$$

Hence $R^i \pi_*(\pi^* \cdot \otimes \mathcal{L}^n)$ is a right exact functor on $\mathbf{Coh}(X)$. Since X is noetherian and affine, we can find a short resolution of $\mathcal{F}$

$$\mathcal{O}_X^{\oplus n_1} \longrightarrow \mathcal{O}_X^{\oplus n_0} \longrightarrow \mathcal{F} \longrightarrow 0 \quad (78)$$

Thus it suffices to show $R^i \pi_*(\pi^* \mathcal{O}_X \otimes \mathcal{L}^n) = 0$. Note that $R^i \pi_*(\pi^* \mathcal{O}_X \otimes \mathcal{L}^n) = R^i \pi_* \mathcal{L}^n = H^i(\mathbb{P}(\mathcal{E}), \mathcal{L}^n)$, which is 0 when $0 < i < r - 1$ or $i = r - 1$ and $n > -r$ by Theorem 2.6.

For the first statement, when $\mathcal{F} = \mathcal{O}_X$, the natural map exactly is the isomorphism $S^n \mathcal{E} \xrightarrow{\sim} \pi_* \mathcal{L}^n$. For general $\mathcal{F} \in \mathbf{Coh}(X)$, similarly consider the resolution, there is a commutative diagram with exact rows

$$\begin{array}{ccccccc} (S^n \mathcal{E})^{\oplus n_1} & \longrightarrow & (S^n \mathcal{E})^{\oplus n_0} & \longrightarrow & \mathcal{F} \otimes S^n \mathcal{E} & \longrightarrow & 0 \\ \downarrow \sim & & \downarrow \sim & & \downarrow \varphi & & \\ (\pi_* \mathcal{L}^n)^{\oplus n_1} & \longrightarrow & (\pi_* \mathcal{L}^n)^{\oplus n_0} & \longrightarrow & \pi_*(\pi^* \mathcal{F} \otimes \mathcal{L}^n) & \longrightarrow & 0 \end{array} \quad (79)$$

Then by 5 Lemma, φ is isomorphic. $\square$

Definition 4.7. Let $f : X \rightarrow Y$ be a morphism between noetherian schemes, $\mathcal{L}$ invertible sheaf on X . We say $\mathcal{L}$ is ample relative to f (or say f -ample) if there exists affine open covering U_i of Y such that $\mathcal{L}|_{f^{-1}(U_i)}$ is ample on $f^{-1}(U_i)$.

Remark 4.9. In fact, when Y is affine, by EGA II Corollary 4.6.6, $\mathcal{L}$ is f -ample if and only if $\mathcal{L}$ is ample on X .

Lemma 4.3. Let $f : X \rightarrow Y$ be a morphism between noetherian schemes, $\mathcal{L}$ f -ample invertible sheaf on X . Then for all coherent $\mathcal{F}$ on Y , the natural map $f^* f_*(\mathcal{F} \otimes \mathcal{L}^n) \rightarrow \mathcal{F} \otimes \mathcal{L}^n$ is surjective for n large enough.

Proof. Since the question is local, we may assume that Y is affine. Then $\mathcal{L}$ is ample. Hence $\mathcal{F} \otimes \mathcal{L}^n$ is globally generated for n large enough. Consider the following commutative diagram

$$\begin{array}{ccc} H^0(X, f^* f_*(\mathcal{F} \otimes \mathcal{L}^n)) \otimes \mathcal{O}_X & \longrightarrow & H^0(X, \mathcal{F} \otimes \mathcal{L}^n) \mathcal{O}_X \\ \downarrow & & \downarrow \\ f^* f_*(\mathcal{F} \otimes \mathcal{L}^n) & \xrightarrow{\alpha} & \mathcal{F} \otimes \mathcal{L}^n \end{array} \quad (80)$$

Thus the natural map α is surjective. $\square$

Proposition 4.11. *Let X be a noetherian scheme, $\mathcal{E}$ vector bundle on X with tautological line bundle $\mathcal{L} := \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$. Then $\mathcal{E}$ is ample on X if and only if $\mathcal{L}$ is ample on $\mathbb{P}(\mathcal{E})$.*

Proof. “ $\Rightarrow$ ”: For coherent $\mathcal{F}$ on $\mathbb{P}(\mathcal{E})$, want to show that $\mathcal{F} \otimes \mathcal{L}^n$ is globally generated. As $\pi_* \mathcal{L} = \mathcal{E}$, by definition, we know $\mathcal{L}$ is π -ample. Applying Lemma 4.3, $\pi^* \pi_*(\mathcal{F} \otimes \mathcal{L}^n) \rightarrow \mathcal{F} \otimes \mathcal{L}^n$ is surjective for n large enough. Hence it suffices to show that $\pi^* \pi_*(\mathcal{F} \otimes \mathcal{L}^n)$ is globally generated.

Denote $\mathcal{G} := \pi_*(\mathcal{F} \otimes \mathcal{L}^{n_1})$, want to show $\pi^* \mathcal{G}$ is globally generated. Again by Lemma 4.3, $\pi^* \pi_*(\pi^* \mathcal{G} \otimes \mathcal{L}^n) \rightarrow \pi^* \mathcal{G} \otimes \mathcal{L}^n$ is surjective for n large enough. It suffices to show that $\pi^* \pi_*(\pi^* \mathcal{G} \otimes \mathcal{L}^n)$ is globally generated. Then by Lemma 4.2, $\pi^* \pi_*(\pi^* \mathcal{G} \otimes \mathcal{L}^n) \cong \pi^*(\mathcal{G} \otimes S^n \mathcal{E})$, which is globally generated for n large enough since $\mathcal{E}$ is ample.

“ $\Leftarrow$ ”: By local criteria, suffices to show $H^0(X, \mathcal{F} \otimes S^n \mathcal{E})$ generate $(\mathcal{F} \otimes S^n \mathcal{E})_x$ for all closed point x . By Nakayama’s Lemma, it is equivalent to show $H^0(X, \mathcal{F} \otimes S^n \mathcal{E})$ generate $(\mathcal{F} \otimes S^n \mathcal{E})_x \otimes k(x) = (\mathcal{F} \otimes S^n \mathcal{E} \otimes k(x))_x$. Note that $\mathcal{F} \otimes S^n \mathcal{E} \otimes k(x)$ is a skyscraper sheaf, its global sections is same to the stalk at x . Hence we only need to show $H^0(X, \mathcal{F} \otimes S^n \mathcal{E}) \rightarrow H^0(X, \mathcal{F} \otimes S^n \mathcal{E} \otimes k(x))$ is surjective for all closed point $x \in X$.

By Lemma 4.2, $H^0(X, \mathcal{F} \otimes S^n \mathcal{E}) = H^0(\mathbb{P}(\mathcal{E}), p^* \mathcal{F} \otimes \mathcal{L}^n)$ and $H^0(X, \mathcal{F} \otimes S^n \mathcal{E} \otimes k(x)) = H^0(\mathbb{P}(\mathcal{E}), p^* \mathcal{F} \otimes \mathcal{L}^n \otimes p^* k(x))$. Denote $Y := \pi^{-1}(x)$, which is a closed subscheme of $\mathbb{P}(\mathcal{E})$. Consider the following exact sequence

$$0 \longrightarrow \mathcal{I}_Y \pi^* \mathcal{F} \longrightarrow \pi^* \mathcal{F} \longrightarrow \mathcal{O}_Y \otimes \pi^* \mathcal{F} \longrightarrow 0 \quad (81)$$

where $\mathcal{I}_Y \pi^* \mathcal{F}$ denotes the image of $\mathcal{I}_Y \otimes \pi^* \mathcal{F} \rightarrow \pi^* \mathcal{F}$. Tensoring by $\mathcal{L}^n$, get exact sequence

$$0 \longrightarrow \mathcal{I}_Y \pi^* \mathcal{F} \otimes \mathcal{L}^n \longrightarrow \pi^* \mathcal{F} \otimes \mathcal{L}^n \longrightarrow \mathcal{O}_Y \otimes \pi^* \mathcal{F} \otimes \mathcal{L}^n \longrightarrow 0 \quad (82)$$

Note that $\text{Spec } k(x) \rightarrow X$ is affine, by cohomology and base change, $\pi^* k(x) \cong \mathcal{O}_Y$. Hence it is equivalent to show that $H^1(\mathbb{P}(\mathcal{E}), \mathcal{I}_Y p^* \mathcal{F} \otimes \mathcal{L}^n) \rightarrow H^1(\mathbb{P}(\mathcal{E}), \mathcal{O}_Y \otimes p^* \mathcal{F} \otimes \mathcal{L}^n)$ is injective.

Now since $\mathcal{L}$ is ample, we find r large enough such that $\mathcal{L}^r$ very ample with an immersion $\ell : \mathbb{P}(\mathcal{E}) \hookrightarrow \mathbb{P}$ into some projective space. Take schematic closure of $\mathbb{P}(\mathcal{E})$ in $\mathbb{P}$, denoted by $\overline{\mathbb{P}(\mathcal{E})}$. Denote $\mathcal{G}_i := \pi^* \mathcal{F} \otimes \mathcal{L}^i$ for all $0 \leq i \leq r-1$. By Hartshorne Chapter II exercise 5.15, we can extend $\mathcal{G}_i$ to a coherent sheaf $\mathcal{G}'_i$ on $\overline{\mathbb{P}(\mathcal{E})}$. Denote $\overline{Y}$ to be the closure of Y in $\overline{\mathbb{P}(\mathcal{E})}$. Then $\mathcal{I}_{\overline{Y}} \mathcal{G}'_i|_{\mathbb{P}(\mathcal{E})} = \mathcal{I}_Y \mathcal{G}_i = \mathcal{I}_Y \pi^* \mathcal{F} \otimes \mathcal{L}^i$.

Now for $Z := \overline{\mathbb{P}(\mathcal{E})} \setminus \mathbb{P}(\mathcal{E})$, by Hartshorne Chapter III exercise 2.3, there is a commutative

diagram with exact rows

$$\begin{array}{ccccccc}
H^1(\overline{\mathbb{P}(\mathcal{E})}, \mathcal{I}_{\overline{Y}} \mathcal{G}'_i(m)) & \rightarrow & H^1(\mathbb{P}(\mathcal{E}), \mathcal{I}_{\overline{Y}} \mathcal{G}'_i(m)|_{\mathbb{P}(\mathcal{E})}) & \rightarrow & H^2_Z(\overline{\mathbb{P}(\mathcal{E})}, \mathcal{I}_{\overline{Y}} \mathcal{G}'_i(m)) & \rightarrow & \cdots \\
\downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \\
H^1(\overline{\mathbb{P}(\mathcal{E})}, \mathcal{G}'_i(m)) & \longrightarrow & H^1(\mathbb{P}(\mathcal{E}), \mathcal{G}'_i(m)|_{\mathbb{P}(\mathcal{E})}) & \longrightarrow & H^2_Z(\overline{\mathbb{P}(\mathcal{E})}, \mathcal{H}'_i(m)) & \longrightarrow & \cdots
\end{array} \tag{83}$$

As $\overline{\mathbb{P}(\mathcal{E})}$ is proper, by Serre Theorem, $H^1(\overline{\mathbb{P}(\mathcal{E})}, \mathcal{I}_{\overline{Y}} \mathcal{G}'_i(m))$ and $H^1(\overline{\mathbb{P}(\mathcal{E})}, \mathcal{G}'_i(m))$ are both zero for m large enough. On the other side, since Y is the fiber at x , Y is proper so that $\overline{Y} = Y$ in $\mathbb{P}$. Thus $\overline{Y} \cap Z = \emptyset$ and $\mathcal{I}_{\overline{Y}} \cong \overline{\mathbb{P}(\mathcal{E})}$ in a neighbourhood of Z . Again by Hartshorne Chapter III exercise 2.3, we may replace $\overline{\mathbb{P}(\mathcal{E})}$ by this neighbourhood of Z , which tells us that γ is isomorphic. Conclude that β is injective for m large enough and we are done! $\square$

Proposition 4.12 (Hartshorne's Cohomological Criteria for Ampleness). *Let X be a noetherian scheme, $\mathcal{E}$ vector bundle on X . If for all coherent $\mathcal{F}$, we have $H^i(X, \mathcal{F} \otimes S^n \mathcal{E}) = 0$ for all $i > 0$ and n large enough. Then $\mathcal{E}$ is ample.*

Conversely, if $\mathcal{E}$ is ample and X is proper, then for all coherent $\mathcal{F}$, we have $H^i(X, \mathcal{F} \otimes S^n \mathcal{E}) = 0$ for all $i > 0$ and n large enough.

Proof. “ $\Rightarrow$ ”: Similar to above argument, it suffices to show that $H^0(X, \mathcal{F} \otimes S^n \mathcal{E}) \rightarrow H^0(X, \mathcal{F} \otimes S^n \mathcal{E} \otimes k(x))$ is surjective for all closed point $x \in X$ and large enough n . Consider exact sequence

$$0 \rightarrow \mathcal{I}_{\{x\}} \mathcal{F} \otimes S^n \mathcal{E} \rightarrow \mathcal{F} \otimes S^n \mathcal{E} \rightarrow \mathcal{F} \otimes S^n \mathcal{E} \otimes k(x) \rightarrow 0 \tag{84}$$

Note that $\mathcal{I}_{\{x\}} \mathcal{F} \otimes S^n \mathcal{E}$ is coherent, we have $H^1(X, \mathcal{I}_{\{x\}} \mathcal{F} \otimes S^n \mathcal{E}) = 0$ for n large enough, done!

“ $\Leftarrow$ ”: By Proposition 4.11, the tautological line bundle $\mathcal{L}$ is ample. As X is proper, by Serre Theorem, for all coherent $\mathcal{F}$ on X , $H^i(\mathbb{P}(\mathcal{E}), \pi^* \mathcal{F} \otimes \mathcal{L}^n) = 0$ for all $i > 0$ and n large enough. By Lemma 4.2, $R^i \pi_*(\pi^* \mathcal{F} \otimes \mathcal{L}^n) = 0$. Note that $\Gamma(\mathbb{P}(\mathcal{E}), \cdot) = \Gamma(X, \cdot) \circ \pi_*$. By Grothendieck Spectral Sequence Theorem, we get

$$E_2^{pq} = H^p(X, R^q \pi_*(\pi^* \mathcal{F} \otimes \mathcal{L}^n)) \Rightarrow H^{p+q}(\mathbb{P}(\mathcal{E}), \pi^* \mathcal{F} \otimes \mathcal{L}^n) \tag{85}$$

Hence we get

$$\begin{aligned}
H^i(X, \mathcal{F} \otimes S^n \mathcal{E}) &\cong H^i(X, \pi_*(\pi^* \mathcal{F} \otimes \mathcal{L}^n)) \\
&\cong H^i(\mathbb{P}(\mathcal{E}), \pi^* \mathcal{F} \otimes \mathcal{L}^n) \\
&= 0, \text{ for all } i > 0 \text{ and } n \text{ large enough}
\end{aligned} \tag{86}$$

$\square$

Corollary 4.7. *Let X be a noetherian proper scheme. For short exact sequence of vector bundles on X , $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$, if $\mathcal{E}'$ and $\mathcal{E}''$ are both ample, then $\mathcal{E}$ is ample.*

Proof. First by Corollary 4.5, we know $\mathcal{E}' \oplus \mathcal{E}''$ is ample. Hence by previous proposition, for all coherent $\mathcal{F}$, $H^{X, \mathcal{F} \otimes S^p \mathcal{E}' \otimes S^q \mathcal{E}''} = 0$ for all $i > 0$ and $p + q$ large enough. By Hartshorne Chapter II exercise 5.16, there is a locally free filtration

$$S^r \mathcal{E} = E^0 \supseteq E^1 \supseteq \cdots \supseteq E^r \supseteq E^{r+1} = 0 \tag{87}$$

such that $E^j/E^{j+1} \cong S^j \mathcal{E}' \otimes S^{r-j} \mathcal{E}''$. Note that a short exact sequence of locally free sheaves is still exact after tensored by any $\mathcal{O}_X$ -module. One may check it locally. Then we get long exact sequence of cohomology

$$\cdots \rightarrow H^0(X, \mathcal{F} \otimes E^j) \rightarrow H^0(X, \mathcal{F} \otimes S^j \mathcal{E}' \otimes S^{r-j} \mathcal{E}'') \rightarrow H^1(X, \mathcal{F} \otimes E^{j+1}) \rightarrow \cdots \quad (88)$$

Since $H^i(X, \mathcal{F} \otimes S^j \mathcal{E}' \otimes S^{r-j} \mathcal{E}'') = 0$ for all $i > 0$ and r large enough, we get that $H^i(X, \mathcal{F} \otimes E^j) = H^i(X, \mathcal{F} \otimes E^{j+1})$ for all $i > 1$. As $E^r = S^r \mathcal{E}'$, we get $H^i(X, \mathcal{F} \otimes S^r \mathcal{E}) = H^i(X, \mathcal{F} \otimes S^r \mathcal{E}') = 0$ for all $i > 2$ and r large enough. For $i = 1$ case, note that there is an exact sequence

$$0 = H^1(X, \mathcal{F} \otimes E^r) \rightarrow H^1(X, \mathcal{F} \otimes E^{r-1}) \rightarrow H^1(X, \mathcal{F} \otimes S^{r-1} \mathcal{E}' \otimes \mathcal{E}'') = 0 \quad (89)$$

Hence $H^1(X, \mathcal{F} \otimes E^{r-1}) = 0$. Preceding step by step, we get $H^1(X, \mathcal{F} \otimes S^r \mathcal{E}) = 0$. Conclude that $\mathcal{E}$ is ample. $\square$

Proposition 4.13. *Let $f : X \rightarrow Y$ be a proper and affine morphism between noetherian schemes, $\mathcal{E}$ ample vector bundle on Y . Then $\mathcal{E}|_X$ is ample on X .*

Proof. Since f is affine, the natural map $f_* f^* \mathcal{F} \rightarrow \mathcal{F}$ is surjective for all coherent sheaf $\mathcal{F}$, which can be affine locally defined as $M \otimes_A B \rightarrow M$. In addition, as f is proper, $f_* \mathcal{F}$ is a coherent sheaf on Y .

Then by Projection Formula, we know $f_*(\mathcal{F} \otimes S^n(f^* \mathcal{E})) \cong f_* \mathcal{F} \otimes S^n \mathcal{E}$ is globally generated for n large enough. Hence we have a surjection $\mathcal{O}_Y^{\oplus I} \rightarrow f_* \mathcal{F} \otimes S^n \mathcal{E}$. Take pull back, we get $\mathcal{O}_X^{\oplus I} \rightarrow f^* f_*(\mathcal{F} \otimes S^n(f^* \mathcal{E})) \rightarrow \mathcal{F} \otimes S^n(f^* \mathcal{E})$ for n large enough. Conclude that $f^* \mathcal{E}$ is ample. $\square$

Corollary 4.8. *Let X be a noetherian scheme, $\mathcal{E}$ ample vector bundle on X . Assume Y is a locally closed subscheme of X . Then $\mathcal{E}|_Y$ is ample on Y .*

Proof. Since a locally closed immersion is composition of open immersion and closed immersion. It suffices to prove these two cases. When $f : Y \hookrightarrow X$ is open, by Hartshorne Chapter II exercise 5.15, for all $\mathcal{F} \in \mathbf{Coh}(Y)$, we can extend $\mathcal{F}$ to a coherent sheaf $\mathcal{F}'$ on X . Hence $\mathcal{F}' \otimes S^n \mathcal{E}$ is globally generated for n large enough. While $\mathcal{F}' \otimes S^n \mathcal{E}$ and $\mathcal{F} \otimes S^n \mathcal{E}|_Y$ have same stalks at $y \in Y$, it is clear that $\mathcal{F} \otimes S^n \mathcal{E}|_Y$ is globally generated so that $\mathcal{E}|_Y$ is ample. On the other hand, when $f : Y \rightarrow X$ is closed, since f is proper and affine, by Proposition 4.13, $\mathcal{E}|_Y$ is ample. $\square$

Proposition 4.14. *Let X be a noetherian scheme, $\mathcal{E}$ vector bundle on X . Then $\mathcal{E}$ is ample on X if $\mathcal{E}|_{X^{\text{red}}}$ is ample on X^{red}*

Reason 4.4. *Note that $\mathbb{P}(\mathcal{E}|_{X^{\text{red}}}) \cong \mathbb{P}(\mathcal{E})^{\text{red}}$. Then this result immediately comes from Proposition 4.11.*

5 Vector Bundles on Curves

Let C be a smooth projective curve of genus g .

5.1 Ample vector bundles on curves

Lemma 5.1. *For all invertible sheaf $\mathcal{L}$ on C with $\deg \mathcal{L} \geq 2g$, $\mathcal{L}$ is globally generated.*

Proof. Consider the following short exact sequence for all closed point $x \in C$

$$0 \longrightarrow \mathcal{L} \otimes \mathcal{O}_C(-x) \longrightarrow \mathcal{L} \longrightarrow \mathcal{L} \otimes \mathcal{O}_x \longrightarrow 0 \quad (90)$$

Note that by Serre duality, $\dim H^1(C, \mathcal{L} \otimes \mathcal{O}_C(-x)) = \dim H^0(C, \mathcal{L}^\vee \otimes \mathcal{O}_C(x) \otimes \omega_C)$. While $\deg(\mathcal{L}^\vee \otimes \mathcal{O}_C(x) \otimes \omega_C) \leq 2g - 2 + 1 - 2g < 0$, by Corollary 3.3, it has no nonzero global section. Hence $H^1(C, \mathcal{L} \otimes \mathcal{O}_C(-x)) = 0$. And we get an exact sequence

$$0 \longrightarrow \Gamma(C, \mathcal{L} \otimes \mathcal{O}_C(-x)) \longrightarrow \Gamma(C, \mathcal{L}) \longrightarrow \Gamma(C, \mathcal{L} \otimes \mathcal{O}_x) \longrightarrow 0 \quad (91)$$

As $\Gamma(C, \mathcal{L} \otimes \mathcal{O}_x) = \mathcal{L}_x$. We get $\mathcal{L}$ is globally generated. $\square$

Proposition 5.1. *For all coherent sheaf $\mathcal{F}$ on X , there exists n_0 such that for all $\mathcal{L}$ with $\deg \mathcal{L} > n_0$, we have that $\mathcal{F} \otimes \mathcal{L}$ is globally generated and $H^1(X, \mathcal{F} \otimes \mathcal{L}) = 0$.*

Proof. We first prove for invertible sheaf $\mathcal{M}$. Then by Lemma 5.1, we can take $n_1 = 2g - \deg M$. For a general coherent sheaf $\mathcal{F}$, since C is projective, $\mathcal{F}$ is quotient of some direct sum of finitely many line bundles, say $\mathcal{G}$. Then there exists n_0 such that $\mathcal{G} \otimes \mathcal{L}$ is globally generated and $H^1(C, \mathcal{G} \otimes \mathcal{L}) = 0$ for all $\mathcal{L}$ with $\deg \mathcal{L} > n_0$. Hence $\mathcal{F} \otimes \mathcal{L}$ is globally generated. Consider the following exact sequence

$$0 \longrightarrow \ker \otimes \mathcal{L} \longrightarrow \mathcal{G} \otimes \mathcal{L} \longrightarrow \mathcal{F} \otimes \mathcal{L} \longrightarrow 0 \quad (92)$$

By Grothendieck Vanishing, we get exact sequence

$$\cdots \longrightarrow H^1(C, \ker \otimes \mathcal{L}) \longrightarrow H^1(C, \mathcal{G} \otimes \mathcal{L}) \longrightarrow H^1(C, \mathcal{F} \otimes \mathcal{L}) \longrightarrow 0 \quad (93)$$

Hence we also get $H^1(C, \mathcal{F} \otimes \mathcal{L}) = 0$, done! $\square$

Corollary 5.1. *Let $\mathcal{L}$ be a line bundle on C . Then $\mathcal{L}$ is ample if and only if $\deg \mathcal{L} > 0$.*

Proof. Suppose $\mathcal{L}$ is ample and $\deg \mathcal{L} \leq 0$. Since $\mathcal{L}$ is ample, for a line bundle $\mathcal{M}$ with $\deg \mathcal{M} < 0$, we get $\mathcal{M} \otimes \mathcal{L}^n$ is globally generated for n large enough. While $\deg(\mathcal{M} \otimes \mathcal{L}^n) < 0$, hence by Corollary 3.3, it has no nonzero global section, contradiction! Hence $\deg \mathcal{L} > 0$.

For the converse, note that since C is projective, $\mathcal{L}$ is ample if and only if for all $\mathcal{F} \in \mathbf{Coh}(C)$, $H^1(X, \mathcal{F} \otimes \mathcal{L}^n) = 0$ for n large enough. Then by Proposition 5.1, we are done. $\square$

5.2 Filtration of vector bundles

Recall that for all vector bundle $\mathcal{E}$ on C , there is a filtration of subbundles by line bundles

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E} \quad (94)$$

such that each quotient $\mathcal{L}_i = \mathcal{E}_i / \mathcal{E}_{i-1}$ is a line bundle for all $1 \leq i \leq r$. We may use $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ to denote this filtration.

Lemma 5.2. *Let $\mathcal{E}$ be a vector bundle on C . Then for any nonzero section $s \in H^0(C, \mathcal{E})$, it naturally defines a line subbundle of $\mathcal{E}$, denoted by $[s]$.*

Proof. As s induces a rational section $\tau : C \dashrightarrow \mathbb{P}(\mathcal{E})$ and C is regular curve, the section is in fact regular. Note that there is a canonical inclusion $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(-1) \hookrightarrow \pi^*\mathcal{E}$, where $\pi : \mathbb{P}(\mathcal{E}) \rightarrow C$. Then $[s] := \tau^*\mathcal{O}_{\mathbb{P}(\mathcal{E})}(-1)$ is a line subbundle of $\mathcal{E}$. $\square$

Corollary 5.2. *Let $\mathcal{E}$ be a vector bundle on C , s nonzero section in $H^0(C, \mathcal{E})$. Then $[s] = \mathcal{O}_C(\text{div}(s))$.*

Proof. Suffice to show s is also a section of $[s]$ i.e. $s \in \Gamma(C, [s])$. By gluing property, we can reduce to affine case that $C = \text{Spec } A$ and $\mathcal{E} = \mathcal{O}_C^{\oplus r}$. Assume $s = (s_1, \dots, s_r)$. Then the canonical inclusion is given by

$$1 \mapsto \left(\frac{x_1}{x_i}, \dots, \frac{x_r}{x_i} \right) \quad (95)$$

And $(s_i, D_+(x_i)) \otimes 1 = (s_i \cdot \frac{s_1}{s_i}, \dots, s_i \cdot \frac{s_r}{s_i}) = (\frac{s_1}{1}, \dots, \frac{s_r}{1})$ is a section in $\Gamma(D(s_i), [s])$. Obviously, we can glue them up to section s in $\Gamma(C, [s])$, done! $\square$

Remark 5.1. *Then consider $\mathcal{E}' = \mathcal{E}/[s]$, which is also locally free. We have a filtration of $\mathcal{E}'$. Hence by lifting this filtration, we get a filtration of $\mathcal{E}$ starting with $[s]$.*

Definition 5.1. *Let $\mathcal{E}$ be a vector bundle on C , s nonzero section in $H^0(C, \mathcal{E})$. Define $\deg s$ to be $\deg([s])$.*

Lemma 5.3. *Let $\mathcal{E}$ be a vector bundle on C . Then for all nonzero section $s \in H^0(C, \mathcal{E})$, we have $\deg s$ bounded above.*

Proof. First fix a filtration of $\mathcal{E}$ by line bundles $(\mathcal{L}_1, \dots, \mathcal{L}_r)$. Then for any nonzero section $s \in H^0(C, \mathcal{E})$, there exists $1 \leq i \leq r$ such that $[s] \subset \mathcal{E}_i$ but $[s] \not\subset \mathcal{E}_{i-1}$. Hence we get a nonzero morphism $[s] \rightarrow \mathcal{L}_i$, which is injective as $[s]$ is locally free of rank one. Hence $\deg s = \deg([s]) \leq \deg \mathcal{L}_i$. Conclude that $\deg s \leq \max_{1 \leq i \leq r} \{\deg \mathcal{L}_i\}$. $\square$

Remark 5.2. *For $\mathcal{E}$ line bundle, we have $\deg s = \deg \mathcal{E}$ for all nonzero section s by Lemma 3.5.*

Definition 5.2. *Let $\mathcal{E}$ be a vector bundle on C . Define $\mu_{\max}(\mathcal{E}) := \sup\{\deg s : s \in H^0(C, \mathcal{E}) \setminus \{0\}\}$. For nonzero section $s \in H^0(C, \mathcal{E})$, we say s is maximal if $\deg s = \mu_{\max}(\mathcal{E})$ and the line subbundle $[s]$ is called maximal line subbundle of $\mathcal{E}$.*

Definition 5.3. *Let $\mathcal{E}$ be a vector bundle on C , $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ a filtration of $\mathcal{E}$. We say this filtration is maximal if $\deg \mathcal{L}_i = \mu_{\max}(\mathcal{E}/\mathcal{E}_{i-1})$ for all $1 \leq i \leq r$.*

Lemma 5.4. *Let $\mathcal{E}$ be a vector bundle on C of rank 2, $(\mathcal{L}_1, \mathcal{L}_2)$ a maximal filtration of $\mathcal{E}$. Then $\deg \mathcal{L}_2 - \deg \mathcal{L}_1 \leq 2g$.*

Proof. Consider the following exact sequence

$$0 \longrightarrow \mathcal{L}_1 \longrightarrow \mathcal{E} \longrightarrow \mathcal{L}_2 \longrightarrow 0 \quad (96)$$

Tensoring by $\mathcal{L}_1^\vee$, we get

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \otimes \mathcal{L}_1^\vee \longrightarrow \mathcal{L}_2 \otimes \mathcal{L}_1^\vee \longrightarrow 0 \quad (97)$$

Suppose that $\deg \mathcal{L}_2 - \deg \mathcal{L}_1 \geq 2g+1$. Then $\deg(\mathcal{L}_2 \otimes \mathcal{L}_1^\vee) \geq 2g+1$. By definition of degree, we have $\deg(\mathcal{L}_2 \otimes \mathcal{L}_1^\vee) = \chi(\mathcal{L}_2 \otimes \mathcal{L}_1^\vee) - \chi(\mathcal{O}_C)$. Hence $\chi(\mathcal{L}_2 \otimes \mathcal{L}_1^\vee) \geq g+2$ and $h^0(\mathcal{L}_2 \otimes \mathcal{L}_1^\vee) \geq g+2$. Consider the following long exact sequence of cohomology

$$0 \rightarrow H^0(C, \mathcal{O}_C) \rightarrow H^0(C, \mathcal{E} \otimes \mathcal{L}_1^\vee) \rightarrow H^0(C, \mathcal{L}_2 \otimes \mathcal{L}_1^\vee) \rightarrow H^1(C, \mathcal{O}_C) \rightarrow \cdots \quad (98)$$

Note that $h^0(C, \mathcal{O}_C) + h^0(C, \mathcal{L}_2 \otimes \mathcal{L}_1^\vee) \leq h^0(C, \mathcal{E} \otimes \mathcal{L}_1^\vee) + h^1(C, \mathcal{O}_C)$. Hence $h^0(C, \mathcal{E} \otimes \mathcal{L}_1^\vee) \geq 3$. Consider $H^0(C, \mathcal{E} \otimes \mathcal{L}_1^\vee) \otimes \mathcal{O}_C \xrightarrow{w_x} (\mathcal{E} \otimes \mathcal{L}_1^\vee)_x$. Since $\mathcal{E} \otimes \mathcal{L}_1^\vee$ is locally free of rank 2, we can find a nonzero section $s \in \ker w_x$. Then s defines a line subbundle $[s]$ of $\mathcal{E} \otimes \mathcal{L}_1^\vee$. As $\text{coeff}_x \text{div}(s) \geq 1$, we have $\deg[s] \geq 1$. Hence $\deg([s] \otimes \mathcal{L}_1) \geq \deg \mathcal{L}_1 + 1 > \deg \mathcal{L}_1$, contradicting the maximality of $\mathcal{L}_1$. $\square$

Corollary 5.3. *Let $\mathcal{E}$ be a vector bundle on C of rank r , $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ a maximal filtration of $\mathcal{E}$. Then $\deg \mathcal{L}_{i+1} - \deg \mathcal{L}_i \leq 2g$ for all $1 \leq i \leq r-1$.*

Proof. By induction on r . When $r = 2$, this is just previous lemma. For $r > 2$, consider the following exact sequence

$$0 \longrightarrow \mathcal{L}_1 \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}/\mathcal{L}_1 \longrightarrow 0 \quad (99)$$

The bundle $\mathcal{E}/\mathcal{L}_1$ has rank $r-1$, and the filtration $(\mathcal{L}_2/\mathcal{L}_1, \dots, \mathcal{L}_r/\mathcal{L}_1)$ is maximal. By induction hypothesis, we have $\deg(\mathcal{L}_{i+1}/\mathcal{L}_1) - \deg(\mathcal{L}_i/\mathcal{L}_1) \leq 2g$ for all $2 \leq i \leq r-1$. This implies $\deg \mathcal{L}_{i+1} - \deg \mathcal{L}_i \leq 2g$ for all $1 \leq i \leq r-1$. $\square$

5.3 Indecomposable vector bundles

Definition 5.4. *Let $\mathcal{E}$ be a vector bundle on C . We say $\mathcal{E}$ is decomposable if there exists a nontrivial decomposition $\mathcal{E} \cong \mathcal{E}' \oplus \mathcal{E}''$ for some vector bundles $\mathcal{E}'$ and $\mathcal{E}''$ on C . Otherwise, we say $\mathcal{E}$ is indecomposable.*

Lemma 5.5. *Let $\mathcal{E}$ be an indecomposable vector bundle on C of rank r . Assume there is a short exact sequence*

$$0 \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}'' \longrightarrow 0 \quad (100)$$

Then $\text{Hom}(\mathcal{E}', \mathcal{E}'' \otimes \omega_C) \neq 0$.

Proof. Since $\mathcal{E}$ is indecomposable, the above short exact sequence does not split. Hence $\text{Ext}^1(\mathcal{E}', \mathcal{E}'') \neq 0$. While

$$\begin{aligned} \text{Ext}^1(\mathcal{E}'', \mathcal{E}') &= H^0(C, \text{Ext}^1(\mathcal{E}'', \mathcal{E}')) \\ &\cong H^1(C, \text{Hom}(\mathcal{E}'', \mathcal{E}')) \\ &\cong H^0(C, \mathcal{E}'' \otimes \mathcal{E}'^\vee \otimes \omega_C) \\ &\cong \text{Hom}(\mathcal{E}', \mathcal{E}'' \otimes \omega_C) \end{aligned} \quad (101)$$

Conclude that $\text{Hom}(\mathcal{E}', \mathcal{E}'' \otimes \omega_C) \neq 0$. $\square$

Lemma 5.6. *Let $\mathcal{E}$ be a globally generated indecomposable vector bundle on C of rank r . Then $\mathcal{E}$ has a maximal filtration $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ with $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1 - (i-1)(2g-2)$ for all $1 \leq i \leq r$.*

Proof. By definition, we can pick a maximal section $s \in H^0(C, \mathcal{E})$. Then $[s]$ is a maximal line subbundle of $\mathcal{E}$. Suppose now we have constructed a filtration of $\mathcal{E}$ of length i starting with $[s]$.

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \dots \subseteq \mathcal{E}_i \subseteq \mathcal{E} \quad (102)$$

such that $\mathcal{L}_j$ is a maximal line subbundle of $\mathcal{E}/\mathcal{E}_{j-1}$ for all $1 \leq j \leq i$.

Set $\mathcal{E}' = \mathcal{E}/\mathcal{E}_i$. Since $\mathcal{E}$ is indecomposable, by previous lemma, we have $\text{Hom}(\mathcal{E}_i, \mathcal{E}' \otimes \omega_C) \neq 0$. Hence there is a nonzero morphism $\mathcal{E}_i \rightarrow \mathcal{E}' \otimes \omega_C$. Pick j_0 such that $f|_{\mathcal{E}_{j_0}}$ is nonzero and $f|_{\mathcal{E}_j} = 0$ for all $j < j_0$. Thus $\mathcal{L}_{j_0} \rightarrow \mathcal{E}' \otimes \omega_C$ is a nonzero morphism.

Note that by definition of maximal line bundle, $\mathcal{L}_{j_0}$ has nonzero global section. Hence we can pick nonzero $s' \in H^0(C, \mathcal{L}_{j_0})$. Note that s' is nowhere vanishing on $U := C \setminus \text{div}(s')$, for all $x \in U$, s'_x spans $\mathcal{L}_{j_0, x}$. As f is nonzero, $f_x : \mathcal{L}_{j_0, x} \rightarrow \mathcal{E}'_x \otimes \omega_{C, x}$ is a nonzero except for finitely many closed points. Hence $f_x(s'_x) \neq 0$ except for finitely many closed points so that $f_* s' \neq 0$ and $[f_* s'] \geq [s']$. On the other hand, since $\mathcal{E}$ globally generated, $\mathcal{E}'$ is also globally generated so that there exists a nonzero section in $s'' \in H^0(C, \mathcal{E}')$ such that $[s'']$ is maximal. Thus

$$\begin{aligned} \deg[s''] &\geq \deg([f_* s'] \otimes \omega_C^\vee) \\ &= \deg[f_* s'] - (2g-2) \\ &\geq \deg[s'] - (2g-2) \\ &= \deg \mathcal{L}_{j_0} - (2g-2) \\ &\geq \deg \mathcal{L}_1 - j_0(2g-2) \\ &\geq \deg \mathcal{L}_1 - i(2g-2) \end{aligned} \quad (103)$$

Now take pull back $[s'']$ along the quotient map $\mathcal{E} \twoheadrightarrow \mathcal{E}'$, we get a vector subbundle $\mathcal{E}_{i+1}$ of $\mathcal{E}$ containing $\mathcal{E}_i$ and with $\deg(\mathcal{E}_{i+1}/\mathcal{E}_i) = \deg[s''] \geq \deg \mathcal{L}_1 - i(2g-2)$. By induction, we get a filtration of $\mathcal{E}$ of length r starting with $[s]$ such that $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1 - (i-1)(2g-2)$ for all $1 \leq i \leq r$. $\square$

Corollary 5.4. *Let $\mathcal{E}$ be an indecomposable vector bundle on C of rank r . Then $\mathcal{E}$ has a filtration $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ with $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1 - (i-1)(2g-2)$ for all $1 \leq i \leq r$.*

Proof. Since $\mathcal{O}_C(1)$ is ample, there exists n_0 such that $\mathcal{E}(n)$ is globally generated for all $n > n_0$. And it is clear that indecomposability is preserved by twisting by line bundles. Hence $\mathcal{E}(n)$ is a globally generated indecomposable vector bundle on C for all $n > n_0$. By previous lemma, $\mathcal{E}(n)$ has a maximal filtration $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ with $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1 - (i-1)(2g-2)$ for all $1 \leq i \leq r$. Hence $\mathcal{E}$ has a filtration $(\mathcal{L}_1(-n), \dots, \mathcal{L}_r(-n))$ with $\deg \mathcal{L}_i(-n) \geq \deg \mathcal{L}_1(-n) - (i-1)(2g-2)$ for all $1 \leq i \leq r$. $\square$

Remark 5.3. *This is not a maximal filtration since $\mathcal{L}_i$ after twisting might be not given by a nonzero section.*

Lemma 5.7. *Let $\mathcal{E}$ be an indecomposable vector bundle on C of rank r and degree d . Then $\mathcal{E}$ has a filtration $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ with $\deg \mathcal{L}_i \geq \frac{d}{r} - (r-1)(3g-2)$ for all $1 \leq i \leq r$.*

Proof. Similar to Corollary 5.4, we may assume $\mathcal{E}$ is globally generated. Then by Corollary 4.5, $\mathcal{E}(1)$ is ample. By Lemma 5.6, $\mathcal{E}(1)$ has a maximal filtration $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ with $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1 - (i-1)(2g-2)$ for all $1 \leq i \leq r$. And by Corollary 5.3, $\deg \mathcal{L}_{i+1} - \deg \mathcal{L}_i \leq 2g$. Hence

$$-2g(i-1) \leq \deg \mathcal{L}_1 - \deg \mathcal{L}_i \leq (i-1)(2g-2) \quad (104)$$

Summing up, we get

$$-r(r-1)g \leq r \deg \mathcal{L}_1 - \sum_{i=1}^r \deg \mathcal{L}_i \leq r(r-1)(g-1) \quad (105)$$

Note that $\sum_{i=1}^r \deg \mathcal{L}_i = \deg \mathcal{E} = d$. Hence

$$-r(r-1)g \leq r \deg \mathcal{L}_1 - d \leq r(r-1)(g-1) \quad (106)$$

Moving terms and dividing by r , we get

$$\frac{r}{d} - (r-1)g \leq \deg \mathcal{L}_1 \leq \frac{d}{r} + (r-1)(g-1) \quad (107)$$

Therefore,

$$\begin{aligned} \deg \mathcal{L}_i &\geq \deg \mathcal{L}_1 - (r-1)(2g-2) \\ &\geq \frac{d}{r} - (r-1)g - (r-1)(2g-2) \\ &\geq \frac{d}{r} - (r-1)(3g-2) \end{aligned} \quad (108)$$

□

Remark 5.4. *In particular, when $g = 0$, we get $0 \leq \deg \mathcal{L}_1 - \frac{d}{r} \leq 1 - r$. Hence r must equal to 1 and $\mathcal{E}$ is a line bundle. This is consistent with the Grothendieck's theorem that all vector bundles on a rational curve are direct sum of line bundles.*

Definition 5.5. *Define $\mathcal{E}(r, d)$ to be the set of isomorphism classes of indecomposable vector bundles on C of rank r and degree d .*

Theorem 5.1 (Atiyah). *There is an integer N with respect to g , r and d such that $\mathcal{E}(n)$ is ample for all $\mathcal{E} \in \mathcal{E}(r, d)$.*

Proof. By Lemma 5.7, for all $\mathcal{E} \in \mathcal{E}(r, d)$, there is a filtration $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ of $\mathcal{E}$ with $\deg \mathcal{L}_i \geq \frac{d}{r} - (r-1)(3g-2)$ for all $1 \leq i \leq r$. Hence $\deg \mathcal{L}_i(n) > 0$ for all $1 \leq i \leq r$ when $n > (r-1)(3g-2) - \frac{d}{r}$. By Corollary 5.1, $\mathcal{L}_i(n)$ is ample for all $1 \leq i \leq r$ when $n > (r-1)(3g-2) - \frac{d}{r}$. Hence $\mathcal{E}(n)$ is ample for all $\mathcal{E} \in \mathcal{E}(r, d)$ when $n > (r-1)(3g-2) - \frac{d}{r}$ by Corollary 4.6. □

5.4 Vector bundles on elliptic curves

This section is based on the famous paper, *Vector Bundles over An Elliptic Curve* by Atiyah.

Let X be an elliptic curve over $\mathbb{C}$, i.e. a smooth projective curve of genus 1 with a chosen rational point x_0 . And $\mathcal{E}$ be a vector bundle on X of rank r and degree d . Set $A = \mathcal{O}_X(x_0)$.

Lemma 5.8. $\deg A = 1$.

Proof. Consider the following exact sequence

$$0 \longrightarrow A^\vee \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{x_0} \longrightarrow 0 \quad (109)$$

Then $\deg A = -\deg A^\vee = 0 - \deg \mathcal{O}_{x_0}$. While $\mathcal{O}_{x_0}$ is a skyscraper sheaf supported at x_0 , its degree is 1. Hence $\deg A = 1$. $\square$

Remark 5.5. By Lemma 3.8, $\deg(\mathcal{E} \otimes A) = \deg \mathcal{E} + r$. Hence there is a canonical map

$$\begin{aligned} \mathcal{E}(r, d) &\longrightarrow \mathcal{E}(r, d + nr) \\ \mathcal{E} &\longmapsto \mathcal{E} \otimes A^{\otimes n} \end{aligned} \quad (110)$$

Clearly this map is bijective as A is invertible. Hence we can identify $\mathcal{E}(r, d)$ with $\mathcal{E}(r, d + nr)$ for all $n \in \mathbb{Z}$. So we can just consider $\mathcal{E}(r, d)$ for $0 \leq d < r$.

Proposition 5.2. $\omega_X \cong \mathcal{O}_X$.

Proof. Note that $h^0(X, \omega_X) = g = 1$ and $\deg \omega_X = g - 1 - (1 - g) = 0$. Hence there is a nonzero section $s \in H^0(X, \omega_X)$. Then similar to proof of Corollary 3.3, $\omega_X \cong \mathcal{O}_X(\operatorname{div}(s))$ of degree 0. Hence $\omega_X \cong \mathcal{O}_X$. $\square$

Lemma 5.9. Let $\mathcal{E}$ be an indecomposable vector bundle of rank r with $h^0(X, \mathcal{E}) > 0$. Then $\mathcal{E}$ admits a maximal filtration $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ such that $\mathcal{L}_i \geq \mathcal{L}_1$.

Proof. The proof is similar to Lemma 5.6. By definition, we can pick a maximal section $s \in H^0(X, \mathcal{E})$. Then $[s]$ is a maximal line subbundle of $\mathcal{E}$. Suppose now we have constructed a filtration of $\mathcal{E}$ of length i starting with $[s]$.

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \dots \subseteq \mathcal{E}_i \subseteq \mathcal{E} \quad (111)$$

such that $\mathcal{L}_j$ is a maximal line subbundle of $\mathcal{E}/\mathcal{E}_{j-1}$ for all $1 \leq j \leq i$.

Set $\mathcal{E}' = \mathcal{E}/\mathcal{E}_i$. Since $\mathcal{E}$ is indecomposable, by Lemma 5.5, we have $\operatorname{Hom}(\mathcal{E}_i, \mathcal{E}') \neq 0$. Hence there is a nonzero morphism $\mathcal{E}_i \rightarrow \mathcal{E}'$. Pick j_0 such that $f|_{\mathcal{E}_{j_0}}$ is nonzero and $f|_{\mathcal{E}_j} = 0$ for all $j < j_0$. Thus $\mathcal{L}_{j_0} \rightarrow \mathcal{E}'$ is a nonzero morphism.

Note that by definition of maximal line bundle, $\mathcal{L}_{j_0}$ has nonzero global section. Hence we can pick nonzero $s' \in H^0(X, \mathcal{L}_{j_0})$. Note that s' is nowhere vanishing on $U := X \setminus \operatorname{div}(s')$, for all $x \in U$, s'_x spans $\mathcal{L}_{j_0, x}$. As f is nonzero, $f_x : \mathcal{L}_{j_0, x} \rightarrow \mathcal{E}'_x$ is a nonzero except for finitely many closed points. Hence $f_x(s'_x) \neq 0$ except for finitely many closed points so that $f_* s' \neq 0$ and $[f_* s'] \geq [s']$.

On the other hand, as f_*s' is a nonzero section in $H^0(X, \mathcal{E})$, $H^0(X, \mathcal{E}) \neq 0$ and we can pick an s'' maximal. If $[f_x(s'_x)]$ itself is maximal, then we may take $s'' = f_x(s'_x)$ and hence $[s''] = [f_x(s'_x)]$. Otherwise, $\deg[s''] \geq \deg[f_x(s'_x)] + 1$ and $g(X) = 1$ so that $[s''] \otimes [f_x(s'_x)]^\vee$ has positive degree and hence nonzero global section. Thus

$$[s''] \geq [f_*s'] \geq [s'] = \mathcal{L}_{j_0} \geq \mathcal{L}_1 \quad (112)$$

Now take pull back $[s'']$ along the quotient map $\mathcal{E} \twoheadrightarrow \mathcal{E}'$, we get a vector subbundle $\mathcal{E}_{i+1}$ of $\mathcal{E}$ containing $\mathcal{E}_i$ and with $\deg(\mathcal{E}_{i+1}/\mathcal{E}_i) = \deg[s''] \geq \deg \mathcal{L}_1$. By induction, we get a filtration of $\mathcal{E}$ of length r starting with $[s]$ such that $\mathcal{L}_i \geq \mathcal{L}_1$ for all $1 \leq i \leq r$. $\square$

Lemma 5.10. *Let $\mathcal{E} \in \mathcal{E}(r, d)$, $0 \leq d < r$ and $h^0(X, \mathcal{E}) > 0$. Then $\mu_{\max}(\mathcal{E}) = 0$ and $\mathcal{E}$ has a trivial line subbundle of rank $h = h^0(X, \mathcal{E})$, denoted by $\mathcal{I}_{\mathcal{E}}$.*

Proof. By Lemma 5.9, $\mathcal{E}$ has a maximal filtration $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ with $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1$ for all $1 \leq i \leq r$. If $\deg \mathcal{L}_1 > 0$, then $d = \deg \mathcal{E} = \sum_{i=1}^r \deg \mathcal{L}_i \geq r \cdot \deg \mathcal{L}_1 \geq r$, which contradicts the assumption that $0 \leq d < r$. Hence $\deg \mathcal{L}_1 = 0$.

Then for any $s \in H^0(X, \mathcal{L}_1)$, we have $0 \leq \deg s \leq \deg \mathcal{L}_1 = 0$. Consider $w_x : \Gamma(X, \mathcal{E}) \otimes \mathcal{O}_{X,x} \rightarrow \mathcal{E}_x$ for any $x \in X$. Then w_x is injective otherwise there is a nonzero section $s \in H^0(X, \mathcal{E})$ such that $\deg s \geq \text{coeff}_x \text{div}(s) \geq 1$. Hence $\Gamma(X, \mathcal{E})$ generates a trivial line subbundle of rank h . $\square$

Lemma 5.11. *Let $\mathcal{E} \in \mathcal{E}(r, r)$. Then $\mathcal{E}$ admits a maximal filtration $(\mathcal{L}, \dots, \mathcal{L})$ with $\deg \mathcal{L} = 1$.*

Proof. As $r > 0$, we know $h^0(X, \mathcal{E}) \neq 0$. By Lemma 5.9, $\mathcal{E}$ has a maximal filtration $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ with $\mathcal{L}_i \geq \mathcal{L}_1$ for all $1 \leq i \leq r$. Suppose $\deg \mathcal{L}_1 = 0$. Similar to proof of previous lemma, we can show that $\Gamma(X, \mathcal{E})$ generates a trivial subbundle $\mathcal{I}_{\mathcal{E}}$ of rank $h = h^0(X, \mathcal{E})$. Hence $r \geq h^0(X, \mathcal{E}) \geq \deg \mathcal{E} = r$ so that $h^0(X, \mathcal{E}) = r$ and $\mathcal{E} = \mathcal{I}_{\mathcal{E}}$, which contradicts the indecomposability of $\mathcal{E}$.

Now $\deg \mathcal{L}_1 \geq 1$. Note that $r = \deg \mathcal{E} \geq r \cdot \deg \mathcal{L}_1 \geq r$, which implies $\deg \mathcal{L}_i = 1$ for all i . Pick some nonzero global section s of $\mathcal{L}_1$. Then $\deg(\text{div}(s)) = \deg \mathcal{L}_1 = 1$. While there is a nonzero map $f_i : \mathcal{L}_1 \rightarrow \mathcal{L}_i$ for all $i \geq 1$, consider $f_i \circ s$. By argument similar to proof of Lemma 5.6, we know $f_i \circ s$ is a nonzero global section of $\mathcal{L}_i$. Note that $\deg(\text{div}(f \circ s)) = \deg \mathcal{L}_i = 1 = \deg(\text{div}(s))$ so that f is nowhere vanishing and hence f is invertible. Conclude that $\mathcal{L}_i \cong \mathcal{L}_1$ for all i . $\square$

Remark 5.6. *In fact, for $\mathcal{E} \in \mathcal{E}(r, d)$ with $d \geq r$, by similar argument, we can show that $\mathcal{E}$ admits a maximal filtration $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ with $\deg \mathcal{L}_1 \geq 1$.*

Definition 5.6. *An extension of vector bundles $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$ is called $\mathcal{E}''$ -partial if there is a commutative diagram*

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{p} & \mathcal{E}'' \\ & \searrow j & \downarrow q \\ & & \mathcal{E}_1'' \end{array} \quad (113)$$

such that $q \circ p \circ j = \text{id}_{\mathcal{E}_1''}$. Otherwise, we say this extension is $\mathcal{E}''$ -complete.

Remark 5.7. Dually, An extension of vector bundles $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$ is called $\mathcal{E}'$ -partial if the dual extension $0 \rightarrow \mathcal{E}''^\vee \rightarrow \mathcal{E}^\vee \rightarrow \mathcal{E}'^\vee \rightarrow 0$ is $\mathcal{E}'^\vee$ -partial.

Lemma 5.12. Given an extension $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \xrightarrow{p} \mathcal{E}'' \rightarrow 0$ with $\mathcal{E}''$ trivial, it is $\mathcal{E}''$ -complete if and only if $p_* : H^0(X, \mathcal{E}) \rightarrow H^0(X, \mathcal{E}'')$ is a zero map.

Proof. If the extension is $\mathcal{E}''$ -partial, then there is a commutative diagram

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{p} & \mathcal{E}'' \\ & \searrow j & \downarrow q \\ & & \mathcal{E}_1'' \end{array} \quad (114)$$

such that $q \circ p \circ j = \text{id}_{\mathcal{E}_1''}$. Hence $p_* \neq 0$ as $(q \circ p \circ j)_*$ is isomorphic.

Conversely, if $p_* \neq 0$, then we can pick a nonzero section $s \in H^0(X, \mathcal{E}'')$ such that $p_*s \neq 0$. Then p_*s defines a line subbundle $[p_*s]$ of $\mathcal{E}''$. As $H^0(X, \mathcal{E}'') \cong \oplus H^0(X, \mathcal{O}_X)$, we have p_*s is injective and $[p_*s]$ is a trivial line subbundle of $\mathcal{E}''$. And we can take orthogonal complement of $H^0(X, [p_*s])$ in $H^0(X, \mathcal{E}'')$, which implies that $[p_*s]$ is a direct summand of $\mathcal{E}''$. Take the projection $q : \mathcal{E}'' \rightarrow \mathcal{O}_X \cong [p_*s]$. Consider the following diagram

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{p} & \mathcal{E}'' \\ & \searrow j & \downarrow q \\ & & [p_*s] \end{array} \quad (115)$$

where j is the composition of $[p_*s] \cong \mathcal{O}_X \xrightarrow{s} \mathcal{E}$. Clearly this diagram is commutative and $q \circ p \circ j = \text{id}_{[p_*s]}$. $\square$

Corollary 5.5. Given an extension $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \xrightarrow{p} \mathcal{E}'' \rightarrow 0$ with $\mathcal{E}''$ trivial, it is $\mathcal{E}''$ -complete if and only if the coboundary map $\delta : H^0(X, \mathcal{E}'') \rightarrow H^1(X, \mathcal{E}')$ is injective.

Proof. Consider the following long exact sequence of cohomology

$$0 \rightarrow H^0(X, \mathcal{E}') \rightarrow H^0(X, \mathcal{E}) \xrightarrow{p_*} H^0(X, \mathcal{E}'') \xrightarrow{\delta} H^1(X, \mathcal{E}') \rightarrow \dots \quad (116)$$

If the extension is $\mathcal{E}''$ -complete, then by previous lemma, p_* is a zero map. Hence δ is injective. Conversely, if δ is injective, then p_* is a zero map. Hence the extension is $\mathcal{E}''$ -complete by previous lemma. $\square$

Proposition 5.3. Given a vector bundle $\mathcal{E}'$ and trivial vector bundle $\mathcal{E}''$ of rank r on X . The equivalence class of extensions of $\mathcal{E}''$ by $\mathcal{E}'$ is one-to-one corresponding with the set of coboundary maps $\delta : H^0(X, \mathcal{E}'') \rightarrow H^1(X, \mathcal{E}')$. In particular, if $h^1(X, \mathcal{E}') = r$, then there is a unique $\mathcal{E}''$ -complete extension of $\mathcal{E}''$ by $\mathcal{E}'$ up to equivalence relation that two extension $\mathcal{E}_1$ and $\mathcal{E}_2$ are equivalent if there is a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{E}' & \longrightarrow & \mathcal{E}_2 & \longrightarrow & \mathcal{E}'' \longrightarrow 0 \\ & & \parallel & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & \mathcal{E}' & \longrightarrow & \mathcal{E}_1 & \longrightarrow & \mathcal{E}'' \longrightarrow 0 \end{array} \quad (117)$$

which is more general than the isomorphism class of extensions.

Proof. It is well-known that the equivalence class of extensions of $\mathcal{E}''$ by $\mathcal{E}'$ is one-to-one corresponding with the set of extension classes in $\text{Ext}^1(\mathcal{E}'', \mathcal{E}')$. Note that $\omega_X \cong \mathcal{O}_X$. As $\mathcal{E}''$ is trivial, by Serre duality, we have that

$$\begin{aligned}
\text{Ext}^1(\mathcal{E}'', \mathcal{E}') &= H^0(X, \mathcal{E}xt^1(\mathcal{E}'', \mathcal{E}')) \\
&\cong H^1(X, \mathcal{H}om(\mathcal{E}'', \mathcal{E}')) \\
&\cong H^0(X, \mathcal{E}'' \otimes \mathcal{E}'^\vee \otimes \omega_X)^\vee \\
&\cong H^0(X, \mathcal{E}'' \otimes \mathcal{E}'^\vee) \\
&= H^0(X, \mathcal{E}''^\vee \otimes \mathcal{E}'^\vee) \\
&\cong H^0(X, \mathcal{H}om(\mathcal{E}'', \mathcal{H}om(\mathcal{E}', \mathcal{O}_X))) \\
&\cong \text{Hom}(H^0(X, \mathcal{E}''), H^0(X, \mathcal{E}'^\vee)) \\
&\cong \text{Hom}(H^0(X, \mathcal{E}''), H^1(X, \mathcal{E}'))
\end{aligned} \tag{118}$$

For the second part, as $h^1(X, \mathcal{E}') = r = h^0(X, \mathcal{E}'')$, the identity map gives $\mathcal{E}''$ -complete extension. On the other hand, given any $\mathcal{E}''$ -complete extension, consider the following long exact sequence of cohomology

$$0 \rightarrow H^0(X, \mathcal{E}') \rightarrow H^0(X, \mathcal{E}) \xrightarrow{p_*} H^0(X, \mathcal{E}'') \xrightarrow{\delta} H^1(X, \mathcal{E}') \rightarrow \dots \tag{119}$$

By Corollary 5.5, δ is injective. Since $h^1(X, \mathcal{E}') = r = h^0(X, \mathcal{E}'')$, δ is an isomorphism. Assume there are two different δ_1 and δ_2 . Then $\varphi := \delta_1^{-1} \circ \delta_2$ gives an automorphism of $H^0(X, \mathcal{E}'')$. As $\mathcal{E}''$ is trivial, φ in fact an automorphism of $\mathcal{E}''$. Assume $\mathcal{E}_1$ is the extension corresponding to δ_1 . Take pull back

$$\begin{array}{ccc}
\mathcal{E}_2 & \xrightarrow{q} & \mathcal{E}'' \\
\downarrow \psi & & \downarrow \varphi \\
\mathcal{E}_1 & \xrightarrow{p} & \mathcal{E}''
\end{array} \tag{120}$$

It is clear to check that $\mathcal{E}_2$ is locally free and ψ is isomorphic. As there is a commutative diagram

$$\begin{array}{ccc}
\mathcal{E}' & \xrightarrow{0} & \mathcal{E}'' \\
\downarrow & & \downarrow \varphi \\
\mathcal{E}_1 & \xrightarrow{p} & \mathcal{E}''
\end{array} \tag{121}$$

By universal property, there is a unique map $\mathcal{E}' \rightarrow \mathcal{E}_2$. Easy to check that $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E}_2 \rightarrow \mathcal{E}'' \rightarrow 0$ is exact and hence an extension. Consider the commutative diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & H^0(X, \mathcal{E}') & \longrightarrow & H^0(X, \mathcal{E}_2) & \longrightarrow & H^0(X, \mathcal{E}'') \xrightarrow{\delta} H^1(X, \mathcal{E}') \longrightarrow \dots \\
& & \parallel & & \downarrow \psi & & \downarrow \varphi & & \parallel \\
0 & \longrightarrow & H^0(X, \mathcal{E}') & \longrightarrow & H^0(X, \mathcal{E}_1) & \longrightarrow & H^0(X, \mathcal{E}'') \xrightarrow{\delta_1} H^1(X, \mathcal{E}') \longrightarrow \dots
\end{array} \tag{122}$$

Hence $\delta = \delta_1 \circ \varphi = \delta_2$. This implies that the extension is unique up to isomorphism. $\square$

Lemma 5.13. *Given an $\mathcal{E}'$ -complete extension $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \xrightarrow{p} \mathcal{E}'' \rightarrow 0$ with $\mathcal{E}'$ trivial, then $h^0(X, \mathcal{E}) = h^0(X, \mathcal{E}'')$.*

Proof. Take the dual extension $0 \rightarrow \mathcal{E}''^\vee \rightarrow \mathcal{E}^\vee \rightarrow \mathcal{E}'^\vee \rightarrow 0$. By Lemma 5.12, we know $H^0(X, \mathcal{E}^\vee) \cong H^0(X, \mathcal{E}'^\vee)$. Hence by Serre duality, $H^1(X, \mathcal{E}^\vee) \cong H^1(X, \mathcal{E}'^\vee)$. Take dimension, we get $h^1(X, \mathcal{E}) = h^1(X, \mathcal{E}')$. Note that $\deg \mathcal{E}' = 0$, by additivity of degree, $\deg \mathcal{E} = \deg \mathcal{E}''$. Hence by $h^0(X, \mathcal{E}) = h^0(X, \mathcal{E}'')$. $\square$

In the following, we would give “uniqueness theorem” and “existence theorem” of $\mathcal{E}(r, d)$.

Proposition 5.4 (uniqueness). *Let $\mathcal{E} \in \mathcal{E}(r, d)$, $d \geq 0$. Then*

- (1) $h^0(X, \mathcal{E}) = \begin{cases} 0 \text{ or } 1 & \text{if } d = 0 \\ d & \text{else} \end{cases}$
 (2) If $d < r$, then $\mathcal{E}' = \mathcal{E}/\mathcal{I}_{\mathcal{E}}$ is indecomposable with $h^0(X, \mathcal{E}') = h^0(X, \mathcal{E})$.

Proof. For (1), if $d \geq r$, then by remark of Lemma 5.11, $\mathcal{E}$ admits a maximal filtration $(\mathcal{L}_1, \dots, \mathcal{L}_r)$ with $\deg \mathcal{L}_1 \geq 1$. Hence for all i , $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1 > 0$. And by Serre duality, $h^1(X, \mathcal{L}_i) = h^0(X, \mathcal{L}_i^\vee) = 0$ for all i . Note that

$$h^0(X, \mathcal{E}) \leq \sum_{i=1}^r h^0(X, \mathcal{L}_i) = \sum_{i=1}^r \deg \mathcal{L}_i = \deg \mathcal{E} = d \leq h^0(X, \mathcal{E}) \quad (123)$$

We get $h^0(X, \mathcal{E}) = d$. Suppose now that $d < r$. If $d = 0$ and $\Gamma(X, \mathcal{E}) = 0$ there is nothing to prove. Hence we may suppose that $\Gamma(X, \mathcal{E}) \neq 0$ even if $d = 0$. If $d > 0$, we always have $\Gamma(X, \mathcal{E}) \neq 0$. Consider the following exact sequence

$$(\mathcal{E}) : 0 \longrightarrow \mathcal{I}_{\mathcal{E}} \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}' \longrightarrow 0. \quad (124)$$

Since $\mathcal{E}$ is indecomposable, $(\mathcal{E})$ is certainly $\mathcal{I}_{\mathcal{E}}$ -complete. Hence by Lemma 5.13, $h^0(X, \mathcal{E}') = h^0(X, \mathcal{E}) =: h$, and so by duality $h^1(X, \mathcal{E}'^\vee) = h$. By Proposition 5.3 the extension $(\mathcal{E})$ corresponds to a monomorphism

$$\delta : \Gamma(\mathcal{E}'^\vee) \longrightarrow H^1(X, \mathcal{E}^\vee), \quad (125)$$

since both terms here have same dimension, δ must be an isomorphism. Suppose that $\mathcal{E}'$ is decomposable and $\mathcal{E}' = \mathcal{F} \oplus \mathcal{G}$ with $\mathcal{F} \neq 0$ and $\mathcal{G} \neq 0$. Then we may write $\delta = \delta_1 \oplus \delta_2$ where

$$\delta_1 : \Gamma(\mathcal{F}^\vee) \longrightarrow H^1(X, \mathcal{F}^\vee) \text{ and } \delta_2 : \Gamma(\mathcal{G}^\vee) \longrightarrow H^1(X, \mathcal{G}^\vee) \quad (126)$$

are isomorphisms, $f = \dim H^1(X, \mathcal{F}^\vee)$, $g = \dim H^1(X, \mathcal{G}^\vee)$. By Lemma 13, δ_1 and δ_2 correspond to extensions

$$\begin{aligned} (\mathcal{E}_1) : 0 &\longrightarrow \mathcal{I}_r \longrightarrow \mathcal{E}_1 \longrightarrow \mathcal{F} \longrightarrow 0 \\ (\mathcal{E}_2) : 0 &\longrightarrow \mathcal{I}_r \longrightarrow \mathcal{E}_2 \longrightarrow \mathcal{G} \longrightarrow 0 \end{aligned} \quad (127)$$

Thus $\delta_1 \oplus \delta_2$ corresponds to $(\mathcal{E}_1) \oplus (\mathcal{E}_2)$, and also to $(\mathcal{E})$. Hence, again by Proposition 5.3, $\mathcal{E} \cong \mathcal{E}_1 \oplus \mathcal{E}_2$, contradiction!

We have now proved (2) and (1) for $d \geq r$. It remains to prove (1) for $0 \leq d < r$. We proceed by induction on r . For $r = 1$, (1) is certainly true. Suppose it is true for all $r' < r$, in particular then for $r - h$. Since $\mathcal{E}' \in \mathcal{E}(r - h, d)$, $\dim \Gamma(\mathcal{E}') = d$ if $d > 0$, and $= 0$ or 1 if $d = 0$. As we have already shown that $\dim \Gamma(\mathcal{E}') = \dim \Gamma(\mathcal{E})$, done! $\square$

Proposition 5.5 (existence). *Let $\mathcal{E}' \in \mathcal{E}(r', d)$ with $d \geq 0$, and if $d = 0$ we assume $h^0(X, \mathcal{E}') \neq 0$. Then there exists $\mathcal{E} \in \mathcal{E}(r, d)$ such that $\mathcal{E}' \cong \mathcal{E}/\mathcal{I}_{\mathcal{E}}$ and $h^0(X, \mathcal{E}) = h^0(X, \mathcal{E}')$, which is unique up to isomorphism.*

Proof. Set $h = h^0(X, \mathcal{E}')$. Then for trivial vector bundle $\mathcal{I}_h$ of rank h , by Proposition 5.3, there is an $\mathcal{E}'$ -complete extension

$$0 \longrightarrow \mathcal{I}_h \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}' \longrightarrow 0 \quad (128)$$

unique up to equivalence relation. By Lemma 5.13, $h^0(X, \mathcal{E}) = h^0(X, \mathcal{E}')$. Remains to show that $\mathcal{E}$ is indecomposable.

Suppose $\mathcal{E}$ is decomposable and $\mathcal{E} = \mathcal{F} \oplus \mathcal{G}$ with $\mathcal{F} \neq 0$ and $\mathcal{G} \neq 0$. Note that $\Gamma(X, \mathcal{I}_h) = \Gamma(X, \mathcal{E}) = \Gamma(X, \mathcal{F}) \oplus \Gamma(X, \mathcal{G})$, we get $\mathcal{I}_h$ is in fact $\mathcal{I}_{\mathcal{E}}$. Then $\Gamma(X, \mathcal{F})$ and $\Gamma(X, \mathcal{G})$ span trivial line subbundle $\mathcal{I}_{\mathcal{F}}$ and $\mathcal{I}_{\mathcal{G}}$ respectively. And $\mathcal{I}_{\mathcal{F}} \oplus \mathcal{I}_{\mathcal{G}} = \mathcal{I}_h$. While $\mathcal{E}' \cong \mathcal{E}/\mathcal{I}_h = \mathcal{E}/(\mathcal{I}_{\mathcal{F}} \oplus \mathcal{I}_{\mathcal{G}}) \cong \mathcal{F}/\mathcal{I}_{\mathcal{F}} \oplus \mathcal{G}/\mathcal{I}_{\mathcal{G}}$, as $\mathcal{E}'$ indecomposable, $\mathcal{F} = \mathcal{I}_{\mathcal{F}}$ or $\mathcal{G} = \mathcal{I}_{\mathcal{G}}$.

We may assume $\mathcal{F} = \mathcal{I}_{\mathcal{F}}$. There is a commutative diagram

$$\begin{array}{ccc} \mathcal{F} & & \\ \downarrow i & \nwarrow \pi & \\ \mathcal{I}_h & \xrightarrow{j} & \mathcal{E} \end{array} \quad (129)$$

such that $\pi \circ j \circ i = \text{id}_{\mathcal{F}}$, contradicting that the extension is $\mathcal{I}_h$ -complete. Conclude that $\mathcal{E}$ is indecomposable. $\square$

Theorem 5.2. *There exists a vector bundle $\mathcal{F}_r \in \mathcal{E}(r, 0)$ with nonzero global sections, called the Atiyah bundle, which is unique up to isomorphism. Moreover, there is an exact sequence for all $r \geq 1$*

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{F}_r \longrightarrow \mathcal{F}_{r-1} \longrightarrow 0 \quad (130)$$

Let $\mathcal{E} \in \mathcal{E}(r, 0)$. Then $\mathcal{E} \cong \mathcal{L} \otimes \mathcal{F}_r$ for some line bundle $\mathcal{L}$ of degree 0, which is also unique up to isomorphism. More precisely, $\mathcal{L}^r \cong \det \mathcal{E}$.

Proof. We first prove the existence of $\mathcal{F}_r$ by induction on r . For $r = 1$, assume $\mathcal{E} \in \mathcal{E}(1, 0)$ with nonzero global section s . Then $\mathcal{E} \cong \text{div}(s)$ is an effective divisor of degree 0, which must be isomorphic to $\mathcal{O}_X$. Suppose now we have proven the result for all $r' < r$. If $\mathcal{E} \in \mathcal{E}(r, 0)$ with nonzero global section, then $h^0(X, \mathcal{E}) = 1$ and by Proposition 5.4 there is an exact sequence

$$0 \longrightarrow \mathcal{I}_{\mathcal{E}} \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}' \longrightarrow 0 \quad (131)$$

where $\mathcal{E}' := \mathcal{E}/\mathcal{I}_{\mathcal{E}} \in \mathcal{E}(r-1, 0)$. By inductive hypothesis we must have $\mathcal{E}' \cong \mathcal{F}_{r-1}$. And by Proposition 5.5, we know extension of $\mathcal{F}_{r-1}$ by $\mathcal{O}_X$ is unique up to isomorphism. Hence the result also holds for r .

Now for any $\mathcal{E} \in \mathcal{E}(r, 0)$, then $\deg(\mathcal{E} \otimes A) = \deg \mathcal{E} + r \cdot \deg A = r$. Then by Lemma 5.11, $\mathcal{E} \otimes A$ admits a maximal filtration $(\mathcal{L}, \dots, \mathcal{L})$ with $\deg \mathcal{L} = 1$. Note that $\mathcal{O}_X$ is a line subbundle of $\mathcal{E} \otimes A \otimes \mathcal{L}^\vee$, $\mathcal{E} \otimes A \otimes \mathcal{L}^\vee \in \mathcal{E}(r, 0)$ with nonzero global section so that isomorphic to $\mathcal{F}_r$. Take the line bundle to be $A^\vee \otimes \mathcal{L}$ is enough. In particular, we get $(\mathcal{O}_X, \dots, \mathcal{O}_X)$ is a filtration of $\mathcal{F}_r$ and $\det \mathcal{E} \cong (\mathcal{A}^\vee \otimes \mathcal{L})^r$.

To show the uniqueness, suffices to prove if $\mathcal{F}_r \cong \mathcal{F}_r \otimes \mathcal{L}$, then $\mathcal{L}$ is trivial line bundle. As $(\mathcal{O}_X, \dots, \mathcal{O}_X)$ is a filtration of $\mathcal{F}_r$, if $\mathcal{L}$ is not a trivial line bundle, then $h^0(X, \mathcal{L}) = 0$ and hence $h^0(X, \mathcal{F}_r \otimes \mathcal{L}) \leq r \cdot h^0(X, A^\vee \otimes \mathcal{L}) = 0$, contradicting to $\Gamma(X, \mathcal{F}_r) \neq 0$ $\square$

Corollary 5.6. $\mathcal{F}_r \cong \mathcal{F}_r^\vee$.

Proof. According to proof of Theorem 5.2, we know $\mathcal{F}_r$ has a filtration $(\mathcal{O}_X, \dots, \mathcal{O}_X)$. Hence $\mathcal{O}_X$ is a quotient of $\mathcal{F}_r$. Dually, we get $\mathcal{O}_X$ is a subbundle of $\mathcal{F}_r^\vee$. Thus $\mathcal{F}_r^\vee$ also has nonzero global sections. Conclude $\mathcal{F}_r \cong \mathcal{F}_r^\vee$. $\square$

Corollary 5.7. For any $s < r$, there is an exact sequence $0 \rightarrow \mathcal{F}_s \rightarrow \mathcal{F}_r \rightarrow \mathcal{F}_{r-s} \rightarrow 0$.

Proof. First to show $\mathcal{F}_s$ is the unique subbundle of $\mathcal{F}$ of rank s , which is a successive extension of trivial line bundles. By Corollary 5.6, we know $\mathcal{F}_s \cong \mathcal{F}_s^\vee$ is such a subbundle of $\mathcal{F}_r \cong \mathcal{F}_r^\vee$. Suppose $\mathcal{E}_s$ is also such a subbundle. For $s = 1$, clearly $\mathcal{E}_1 \cong \mathcal{F}_1$. Induct on s and r , suppose we have proven for all $s' < s$ and $r' > s'$. Note that $\mathcal{E}_s/\mathcal{F}_1$ is a subbundle of $\mathcal{F}_r/\mathcal{F}_1 \cong \mathcal{F}_{r-1}$. By inductive hypothesis, $\mathcal{E}_s/\mathcal{F}_1 \cong \mathcal{F}_{s-1}$. Now we get an extension

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{E}_s \longrightarrow \mathcal{F}_{s-1} \longrightarrow 0 \quad (132)$$

As $\mathcal{F}_1$ is trivial line bundle, the extension corresponds a map $H^0(X, \mathcal{F}_1^\vee) \rightarrow H^1(X, \mathcal{F}_{s-1}^\vee)$. Or equivalently by Serre duality, a map $H^0(X, \mathcal{F}_1) \rightarrow H^0(X, \mathcal{F}_{s-1})^\vee$. Note that $h^0(X, \mathcal{F}_i) = 0$ for all i . The map is either a zero map or an isomorphism. If it is a zero map, then $\mathcal{E}_s \cong \mathcal{F}_1 \oplus \mathcal{F}_{s-1}$. But then $\mathcal{O}_X^{\oplus 2} \subseteq \mathcal{F}_1 \oplus \mathcal{F}_{s-1}$ is a trivial subbundle of $\mathcal{F}_r$, contradicting to $h^0(X, \mathcal{F}_r) = 1$. Hence the map is an isomorphism so that $\mathcal{E}_s \cong \mathcal{F}_s$.

Finally, consider $\mathcal{E}_{r-s} := \mathcal{F}_r/\mathcal{F}_s$. Dually, we get $\mathcal{E}_{r-s}^\vee$ is a subbundle of $\mathcal{F}_r$. Remains to check $\mathcal{E}_{r-s}^\vee$ is a successive extension of trivial line bundles. As there is a filtration $\mathcal{E}_1^\vee \subset \dots \subset \mathcal{E}_{r-s}^\vee$ and $\mathcal{E}_i^\vee/\mathcal{E}_{i-1}^\vee \cong \mathcal{F}_{r-i+1}/\mathcal{F}_{r-i} \cong \mathcal{F}_1$ is trivial line bundle, done! $\square$

Theorem 5.3. The fixed line bundle A determines a one-to-one correspondence

$$\alpha_{r,d} : \mathcal{E}(h, 0) \rightarrow \mathcal{E}(r, d) \quad (133)$$

where h is the greatest common divisor of r and d . And for all $\mathcal{E} \in \mathcal{E}(h, 0)$, $\alpha_{r,d}$ is defined uniquely by the following properties

- $\alpha_{r,0} = \text{id}$
- $\alpha_{r,r+d}(\mathcal{E}) \cong \alpha_{r,d}(\mathcal{E}) \otimes A$
- if $0 < d < r$, we have an exact sequence $0 \rightarrow \mathcal{I}_d \rightarrow \alpha_{r,d}(\mathcal{E}) \rightarrow \alpha_{r-d,d}(\mathcal{E}) \rightarrow 0$, where $\mathcal{I}_d$ is the trivial bundle of rank d .

In particular, $\det \alpha_{r,d}(\mathcal{E}) \cong \det \mathcal{E} \otimes A^d$.

Proof. With correspondence given by $\otimes A$, we may assume $d \geq 0$ first. When $d = 0$, we have $h = r$ so that we can just take $\alpha_{r,0}$ to be the identity map. If $d > 0$, note that in fact

Proposition 5.4 and Proposition 5.5 gives a one-to-one correspondence between $\mathcal{E}(r, d)$ and $\mathcal{E}(r - d, d)$, we have the following correspondence process that

$$\beta_{r,d} : \mathcal{E}(r, d) \longleftrightarrow \begin{cases} \mathcal{E}(r - d, d) & \text{if } r > d \\ \mathcal{E}(r, d - r) & \text{else} \end{cases} \quad (134)$$

Starting with $\mathcal{E}(r, d)$ and $\beta_{r,d}$ and repeating the process, we get $\mathcal{E}(r, d) \xleftrightarrow{\beta_{r,d}} \mathcal{E}(r', d') \xleftrightarrow{\beta_{r',d'}} \dots$. For each step, it is easy to see that $r' + d' < r + d$, $\gcd(r', d') = \gcd(r, d)$ and $d' \geq 0$. Hence the process would stop at $\mathcal{E}(h, 0)$. Combining these correspondence together, we get one-to-one correspondence $\beta_{r,d} : \mathcal{E}(r, d) \longleftrightarrow \mathcal{E}(r_0, 0)$. Take $\alpha_{r,d} = \beta_{r,d}^{-1}$. Then by definition, we know this correspondence satisfies those properties.

For uniqueness, suppose now we have another correspondence $\gamma_{r,d}$ satisfying those properties. Consider the correspondence $\mathcal{E}(r, d) \xleftrightarrow{\gamma_{r,d}^{-1}} \mathcal{E}(h, 0) \xleftrightarrow{\gamma_{r,d+r}} \mathcal{E}(r, d+r)$. Then $\gamma_{r,d+r} \circ \gamma_{r,d}^{-1}(\mathcal{E}) \cong \mathcal{E} \otimes A = \alpha_{r,d+r} \circ \alpha_{r,d}^{-1}(\mathcal{E})$. And the last property also tells us that $\gamma_{r+d,d} \circ \gamma_{r,d}^{-1} = \alpha_{r+d,d} \circ \alpha_{r,d}^{-1}$. Conclude that $\gamma \cong \alpha$. $\square$

Theorem 5.4. *We may regard X as an abelian variety with identity the fixed point x_0 . Then each $\mathcal{E}(r, d)$ can be identified with X in such a way that*

$$\det : \mathcal{E}(r, d) \rightarrow \mathcal{E}(1, d) \text{ corresponds to } H : X \rightarrow X \quad (135)$$

where $H(x) = h \cdot x$ and $h = \gcd(r, d)$.

Reason 5.1. *By Theorem 5.3, suffices to consider $\mathcal{E}(h, 0)$. Then for all $\mathcal{E} \in \mathcal{E}(h, 0)$, $\mathcal{E} \cong \mathcal{F}_h \otimes \mathcal{L}$ for some $\mathcal{L} \in \text{Pic}^0(X)$. Hence we can identify $\mathcal{E}(h, 0)$ with $\text{Pic}^0(X)$. While there is isomorphism $X \rightarrow \text{Pic}^0(X)$ sending x to $\mathcal{O}_X(x) \otimes A^\vee$, we can identify $\mathcal{E}(h, 0)$ with X . In particular, $\det \mathcal{E} \cong \det \mathcal{F}_h \otimes \mathcal{L}^h$ so that $\det$ corresponds to H .*

Corollary 5.8. *Notation as above, then for all $\mathcal{E} \in \mathcal{E}(r, d)$, we have that*

- (1) $\mathcal{E} \mapsto \det \mathcal{E}$ gives a one-to-one correspondence $\mathcal{E}(r, d) \longleftrightarrow \mathcal{E}(1, d)$.
- (2) $\mathcal{E} \cong \mathcal{E}_A(r, d) \otimes \mathcal{L}$ for some $\mathcal{L} \in \text{Pic}^0(X)$, where $\mathcal{E}_A(r, d) = \alpha_{r,d}(\mathcal{F}_h)$.
- (3) $\mathcal{E}_A(r, d) \otimes \mathcal{L} \cong \mathcal{E}_A(r, d)$ if and only if $\mathcal{L}^r \cong \mathcal{O}_X$.
- (4) $\mathcal{E}_A(r, d)^* \cong \mathcal{E}_A(r, -d)$.

6 Stable Vector Bundles on Curves

6.1 Stability

Let X be a regular projective curve over $\mathbb{C}$ of genus g .

Definition 6.1. *Let $f : \mathcal{E}_1 \rightarrow \mathcal{E}_2$ be a map between vector bundles on X . We say f is of maximal rank if the induced map $\det f : \det \mathcal{E}_1 \rightarrow \det \mathcal{E}_2$ is a nonzero map.*

Remark 6.1. *As $\deg \mathcal{E}_i = \deg(\det \mathcal{E}_i)$, if $f : \mathcal{E}_1 \rightarrow \mathcal{E}_2$ is of maximal rank, then $\deg \mathcal{E}_1 \leq \deg \mathcal{E}_2$.*

Definition 6.2. Let $\mathcal{F}$ be a coherent sheaf on X , $\mathcal{G} \subseteq \mathcal{F}$ subsheaf. We say $\mathcal{G}$ is saturated in $\mathcal{F}$ if $\mathcal{F}/\mathcal{G}$ is torsion free. Define the saturation of $\mathcal{G}$ in $\mathcal{F}$ is the minimal subsheaf $\mathcal{G}^{\text{sat}}$ of $\mathcal{F}$ containing $\mathcal{G}$ such that $\mathcal{G}^{\text{sat}}$ is saturated in $\mathcal{F}$.

Remark 6.2. As X is integral, we can directly construct $\mathcal{G}^{\text{sat}}$ by

$$\mathcal{G}^{\text{sat}} = \{s \in \mathcal{F} \mid \text{the image of } s \text{ in } \mathcal{F}/\mathcal{G}(U) \text{ is torsion}\} \quad (136)$$

which is clearly the minimal saturated subsheaf containing $\mathcal{G}$.

Lemma 6.1. Let $\mathcal{F}$ be a coherent sheaf on X , $\mathcal{G} \subseteq \mathcal{F}$ subsheaf with saturation $\mathcal{G}^{\text{sat}}$. Then $\text{rank } \mathcal{G} = \text{rank } \mathcal{G}^{\text{sat}}$.

Proof. As X is integral, by Proposition 3.1, there is an open subset $U \subset X$ such that $(\mathcal{F}/\mathcal{G})|_U$ is locally free and hence torsion free. By our construction of saturation, it is clear that $\mathcal{G}^{\text{sat}}|_U = \mathcal{G}|_U^{\text{sat}} = \mathcal{G}|_U$. Thus $\text{rank } \mathcal{G} = \text{rank } \mathcal{G}^{\text{sat}}$. $\square$

Given a nonzero map $f : \mathcal{E} \rightarrow \mathcal{F}$ between vector bundles. Consider $\ker f$ and $\text{im } f$, since they are subsheaves of $\mathcal{E}$ and $\mathcal{F}$ respectively, they are both torsion free. Moreover, as X is a regular curve, we get $\ker f$ and $\text{im } f$ are locally free. Take $\mathcal{E}' = \ker f$, $\mathcal{E}'' = \text{im } f$, $\mathcal{F}'$ the saturation of $\mathcal{E}''$ in $\mathcal{F}$ and $\mathcal{F}'' = \mathcal{F}/\mathcal{F}'$. Then we get a commutative diagram consisting of vector bundles, with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{E}' & \hookrightarrow & \mathcal{E} & \twoheadrightarrow & \mathcal{E}'' \longrightarrow 0 \\ & & & & \downarrow f & & \downarrow g \\ 0 & \longleftarrow & \mathcal{F}'' & \twoheadleftarrow & \mathcal{F} & \longleftarrow & \mathcal{F}' \longleftarrow 0 \end{array} \quad (137)$$

By Lemma 6.1, we also have $\text{rank } \mathcal{F}' = \text{rank } \mathcal{E}''$.

Definition 6.3. Let $\mathcal{F}$ be a coherent sheaf on X . Define the slope of $\mathcal{F}$ to be $\mu(\mathcal{F}) := \frac{\deg \mathcal{F}}{\text{rank } \mathcal{F}}$.

Definition 6.4. We say a vector bundle $\mathcal{E}$ on X is stable (resp. semi-stable) if for all proper subbundle $\mathcal{E}' \subset \mathcal{E}$, we have $\mu(\mathcal{E}') < \mu(\mathcal{E})$ (resp. $\mu(\mathcal{E}') \leq \mu(\mathcal{E})$).

Lemma 6.2. Let $\mathcal{E}$ be a semi-stable vector bundle of $\deg d$ and rank r . If $\gcd(r, d) = 1$, then $\mathcal{E}$ is stable.

Proof. Suffice to show for all subbundle $\mathcal{E}' \subseteq \mathcal{E}$, $\mu(\mathcal{E}') = \mu(\mathcal{E})$ if and only if $\mathcal{E}' = \mathcal{E}$. As $\mu(\mathcal{E}') = \mu(\mathcal{E})$, we get $\deg \mathcal{E}' \cdot \text{rank } \mathcal{E} = \deg \mathcal{E} \cdot \text{rank } \mathcal{E}'$. Since $\gcd(r, d) = 1$, we must have $r \mid \text{rank } \mathcal{E}'$ so that $\mathcal{E}' = \mathcal{E}$ and $\mathcal{E}' = \mathcal{E}$. $\square$

Proposition 6.1. There is an equivalent definition of stability (resp. semi-stability) that $\mathcal{E}$ is stable (resp. semi-stable) if for any nontrivial surjection $\mathcal{E} \twoheadrightarrow \mathcal{F}$ to vector bundle $\mathcal{F}$, we have $\mu(\mathcal{E}) < \mu(\mathcal{F})$ (resp. $\mu(\mathcal{E}) \leq \mu(\mathcal{F})$).

Proof. Note that for any nontrivial surjection $\mathcal{E} \twoheadrightarrow \mathcal{F}$ to vector bundle $\mathcal{F}$, there is a short exact sequence

$$0 \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0 \quad (138)$$

where the kernel $\mathcal{E}'$ is automatically a proper subbundle of $\mathcal{E}$. This gives a one-to-one correspondence between proper subbundles and nontrivial surjections. In addition, as $\deg \mathcal{E} = \deg \mathcal{E}' + \deg \mathcal{F}$ and $\text{rank } \mathcal{E} = \text{rank } \mathcal{E}' + \text{rank } \mathcal{F}$, we have

$$\begin{aligned} \deg \mathcal{E} \cdot \text{rank } \mathcal{E}' > \deg \mathcal{E}' \cdot \text{rank } \mathcal{E} &\iff \deg \mathcal{F} \cdot \text{rank } \mathcal{E}' > \deg \mathcal{E}' \cdot \text{rank } \mathcal{F} \\ &\iff \deg \mathcal{F} \cdot \text{rank } \mathcal{E} > \deg \mathcal{E} \cdot \text{rank } \mathcal{F} \end{aligned} \quad (139)$$

And respectively,

$$\begin{aligned} \deg \mathcal{E} \cdot \text{rank } \mathcal{E}' \geq \deg \mathcal{E}' \cdot \text{rank } \mathcal{E} &\iff \deg \mathcal{F} \cdot \text{rank } \mathcal{E}' \geq \deg \mathcal{E}' \cdot \text{rank } \mathcal{F} \\ &\iff \deg \mathcal{F} \cdot \text{rank } \mathcal{E} \geq \deg \mathcal{E} \cdot \text{rank } \mathcal{F} \end{aligned} \quad (140)$$

Conclude that the two definitions are equivalent. $\square$

Proposition 6.2. $\mathcal{E}$ is stable (resp. semi-stable) if and only if for all proper subbundle $\mathcal{E}' \subset \mathcal{E}$, $\deg(\mathcal{E}^\vee \otimes \mathcal{E}') < 0$ (resp. $\mathcal{E}' \subset \mathcal{E}$, $\deg(\mathcal{E}^\vee \otimes \mathcal{E}') \leq 0$).

Reason 6.1. Note that $\deg(\mathcal{E}^\vee \otimes \mathcal{E}') = \deg \mathcal{E}^\vee \cdot \text{rank } \mathcal{E}' + \deg \mathcal{E}' \cdot \text{rank } \mathcal{E}^\vee$. As $\text{rank } \mathcal{E}^\vee = \text{rank } \mathcal{E}$ and by Corollary 3.7, we have $\deg \mathcal{E}^\vee = -\deg \mathcal{E}$.

Proposition 6.3. Let $\mathcal{L}$ be a line bundle on X . Then $\mathcal{E}$ is stable (resp. semi-stable) if and only if $\mathcal{E} \otimes \mathcal{L}$ is stable (resp. semi-stable).

Proof. Since $\mathcal{L}^\vee$ is also a line bundle, suffices to show if $\mathcal{E}$ stable (resp. semi-stable) then $\mathcal{E} \otimes \mathcal{L}$ stable (resp. semi-stable). In fact, for all proper subbundle $\mathcal{F} \subset \mathcal{E} \otimes \mathcal{L}$, we get $\mathcal{F} \otimes \mathcal{L}^\vee$ a proper subbundle of $\mathcal{E}$. Hence $\mu(\mathcal{F} \otimes \mathcal{L}^\vee) < \mu(\mathcal{E})$ (resp. $\mu(\mathcal{F} \otimes \mathcal{L}^\vee) \leq \mu(\mathcal{E})$). Note that

$$\begin{aligned} \mu(\mathcal{F} \otimes \mathcal{L}^\vee) &= \frac{\deg(\mathcal{F} \otimes \mathcal{L}^\vee)}{\text{rank}(\mathcal{F} \otimes \mathcal{L}^\vee)} \\ &= \frac{\deg \mathcal{F} + \text{rank } \mathcal{F} \cdot \deg \mathcal{L}^\vee}{\text{rank } \mathcal{F}} \\ &= \mu(\mathcal{F}) + \deg \mathcal{L}^\vee \end{aligned} \quad (141)$$

and

$$\begin{aligned} \mu(\mathcal{E} \otimes \mathcal{L}) &= \frac{\deg(\mathcal{E} \otimes \mathcal{L})}{\text{rank}(\mathcal{E} \otimes \mathcal{L})} \\ &= \frac{\deg \mathcal{E} + \text{rank } \mathcal{E} \cdot \deg \mathcal{L}}{\text{rank } \mathcal{E}} \\ &= \mu(\mathcal{E}) + \deg \mathcal{L} \end{aligned} \quad (142)$$

Thus $\mu(\mathcal{F}) < \mu(\mathcal{E} \otimes \mathcal{L})$ (resp. $\mu(\mathcal{F}) \leq \mu(\mathcal{E} \otimes \mathcal{L})$). $\square$

Proposition 6.4. Let $\mathcal{E}$ and $\mathcal{F}$ be two semi-stable vector bundles of same rank and degree. Assume at least one of them is stable. If $f : \mathcal{E} \rightarrow \mathcal{F}$ is a nonzero map, then f is an isomorphism.

Proof. Suppose f is not isomorphism. By construction of the diagram 137, we get

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{E}' & \hookrightarrow & \mathcal{E} & \twoheadrightarrow & \mathcal{E}'' & \longrightarrow & 0 \\ & & & & \downarrow f & & \downarrow g & & \\ 0 & \longleftarrow & \mathcal{F}'' & \twoheadleftarrow & \mathcal{F} & \hookleftarrow & \mathcal{F}' & \longleftarrow & 0 \end{array} \quad (143)$$

Since $\mathcal{E}$ is semi-stable, we have $\deg(\mathcal{E}^\vee \otimes \mathcal{E}') \leq 0$. As $0 = \deg(\mathcal{E}^\vee \otimes \mathcal{E}) = \deg(\mathcal{E}^\vee \otimes \mathcal{E}') + \deg(\mathcal{E}^\vee \otimes \mathcal{E}'')$, we get $\deg(\mathcal{E}^\vee \otimes \mathcal{E}'') \geq 0$. On the other hand, since $\mathcal{F}$ is semi-stable, we also have $\deg(\mathcal{F}^* \otimes \mathcal{F}') \leq 0$. As g is of maximal rank, $\deg \mathcal{E}'' \leq \deg \mathcal{F}'$. While $\text{rank } \mathcal{E}'' = \text{rank } \mathcal{F}'$, we get $\deg(\mathcal{E}^* \otimes \mathcal{F}') \geq \deg(\mathcal{E}^\vee \otimes \mathcal{E}'')$. Since $\deg \mathcal{E} = \deg \mathcal{F}$ and $\text{rank } \mathcal{E} = \text{rank } \mathcal{F}$, we obtain

$$\begin{aligned} 0 &\geq \deg(\mathcal{F}^\vee \otimes \mathcal{F}') \\ &= \deg(\mathcal{E}^\vee \otimes \mathcal{F}') \\ &\geq \deg(\mathcal{E}^\vee \otimes \mathcal{E}'') \\ &\geq 0 \end{aligned} \tag{144}$$

As at least one of $\mathcal{E}$ and $\mathcal{F}$ is stable, at least one of the first and the last inequalities is strict, contradiction! Conclude that f must be isomorphic. $\square$

Definition 6.5. We say a vector bundle $\mathcal{E}$ on X is simple if $h^0(X, \text{End}(\mathcal{E})) = 1$. Or equivalently, the only elements in $\text{End}(\mathcal{E})$ are multiplications by scalar.

Lemma 6.3. A simple vector bundle $\mathcal{E}$ is indecomposable.

Proof. Suppose $\mathcal{E} \cong \mathcal{E}_1 \oplus \mathcal{E}_2$ for vector bundles $\mathcal{E}_1$ and $\mathcal{E}_2$. Then $\text{End}(\mathcal{E}_1) \oplus \text{End}(\mathcal{E}_2) \subseteq \text{End}(\mathcal{E})$ so that $h^0(X, \text{End}(\mathcal{E})) \geq 2$, contradiction! $\square$

Corollary 6.1. A stable vector bundle $\mathcal{E}$ is simple and hence indecomposable.

Reason 6.2. By Proposition 6.4, the only elements in $\text{End}(\mathcal{E})$ are zero map and isomorphisms. Hence $\text{End}(\mathcal{E})$ is a division algebra finite over $\mathbb{C}$, which must be $\mathbb{C}$ as $\mathbb{C}$ is algebraically closed.

Proposition 6.5. Let $\mathcal{E}$ be a semi-stable vector bundle and $\mathcal{F}$ be a vector bundle with $\mu(\mathcal{E}) \geq \mu(\mathcal{F})$ (resp. $\mu(\mathcal{E}) > \mu(\mathcal{F})$). If $f : \mathcal{E} \rightarrow \mathcal{F}$ is a nonzero morphism and $\mathcal{F}' := (\text{im } f)^{\text{sat}}$, then $\mu(\mathcal{F}') \geq \mu(\mathcal{F})$ (resp. $\mu(\mathcal{F}') > \mu(\mathcal{F})$).

Proof. By construction of the diagram 137, we get

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{E}' & \hookrightarrow & \mathcal{E} & \twoheadrightarrow & \mathcal{E}'' \longrightarrow 0 \\ & & & & \downarrow f & & \downarrow g \\ 0 & \longleftarrow & \mathcal{F}'' & \longleftarrow & \mathcal{F} & \longleftarrow & \mathcal{F}' \longleftarrow 0 \end{array} \tag{145}$$

Hence $\mu(\mathcal{F}') \geq \mu(\mathcal{E}'')$. Since $\mathcal{E}$ is semi-stable, we get $\mu(\mathcal{E}'') \geq \mu(\mathcal{E})$. As $\mu(\mathcal{E}) \geq \mu(\mathcal{F})$ (resp. $\mu(\mathcal{E}) > \mu(\mathcal{F})$), we are done! $\square$

Proposition 6.6. If $\mathcal{E}$ is not stable, then there exists stable subbundle $\mathcal{E}' \subset \mathcal{E}$ with $\mu(\mathcal{E}') \geq \mu(\mathcal{E})$. Moreover, if $\mathcal{E}$ is not semi-stable, then there exists stable subbundle $\mathcal{E}' \subset \mathcal{E}$ with $\mu(\mathcal{E}') > \mu(\mathcal{E})$.

Proof. Take $\mathcal{E}'$ to be the subbundle of $\mathcal{E}$ satisfying that $\mu(\mathcal{E}') \geq \mu(\mathcal{E})$ (resp. $\mu(\mathcal{E}') > \mu(\mathcal{E})$) with minimal rank. It is clear that $\mathcal{E}'$ is stable. $\square$

Proposition 6.7. Let $\mathcal{F}$ be a vector bundle of rank r . Then $\mathcal{F}$ is not stable (resp. semi-stable) if and only if there exists stable vector bundle $\mathcal{E}$ with $\mu(\mathcal{E}) \geq \mu(\mathcal{F})$ (resp. $\mu(\mathcal{E}) > \mu(\mathcal{F})$) and a nonisomorphic nonzero map $f : \mathcal{E} \rightarrow \mathcal{F}$.

Proof. By Proposition 6.6, suffices to show if there exists stable vector bundle $\mathcal{E}$ with $\mu(\mathcal{E}) \geq \mu(\mathcal{F})$ (resp. $\mu(\mathcal{E}) > \mu(\mathcal{F})$) and a nonisomorphic nonzero map $f : \mathcal{E} \rightarrow \mathcal{F}$, then $\mathcal{F}$ is not stable (resp. semi-stable). Suppose $\mathcal{F}$ is stable (resp. semi-stable).

Denote $\mathcal{F}' = (\text{im } f)^{\text{sat}}$. By Proposition 6.5, we have $\mu(\mathcal{F}') \geq \mu(\mathcal{F})$ (resp. $\mu(\mathcal{F}') > \mu(\mathcal{F})$). For the second case, this already contradicts to semi-stability of $\mathcal{F}$. For the first case, since $\mathcal{F}$ is stable, we get $\mathcal{F}'$ must be $\mathcal{F}$ itself. Consider the diagram 137

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{E}' & \hookrightarrow & \mathcal{E} & \twoheadrightarrow & \mathcal{E}'' \longrightarrow 0 \\ & & & & \downarrow f & & \downarrow g \\ 0 & \longleftarrow & \mathcal{F}'' & \twoheadleftarrow & \mathcal{F} & \longleftarrow & \mathcal{F}' \longleftarrow 0 \end{array} \quad (146)$$

Then $\mu(\mathcal{F}) = \mu(\mathcal{F}') \geq \mu(\mathcal{F}'') \geq \mu(\mathcal{E}) \geq \mu(\mathcal{F})$. Hence all equalities hold. As $\mu(\mathcal{E}) = \mu(\mathcal{E}'')$ and $\mathcal{E}$ is stable, we get $\mathcal{E}' = \ker f = 0$. Now $\mathcal{E}$ and $\mathcal{F}$ have same rank and degree. By Proposition 6.4, f is an isomorphism, contradiction! $\square$

6.2 Relation to ampleness

For this section, you may refer to the paper *Ample Vectotr Bundles on Curves*, Hartshorne.

Proposition 6.8. *The following conditions are equivalent:*

- (1) Every stable vector bundle of positive degree on X is ample.
- (2) Every vector bundle, all of whose quotient bundles have positive degree, is ample.

Proof. “ $\Rightarrow$ ”: Let $\mathcal{E}$ be a vector bundle, all of whose quotient bundles have positive degree. Induction on $\text{rank } \mathcal{E}$. When $\text{rank } \mathcal{E} = 1$, since line bundle of positive degree on X is ample, there is nothing to prove. Suppose $\text{rank } \mathcal{E} > 1$. If $\mathcal{E}$ is stable, then by assumption, $\mathcal{E}$ is ample.

Now $\mathcal{E}$ is not stable. By Proposition 6.6, there is a stable subbundle $\mathcal{E}' \subset \mathcal{E}$ with $\mu(\mathcal{E}') \geq \mu(\mathcal{E})$. Since $\mu(\mathcal{E}) > 0$, we get $\mu(\mathcal{E}') > 0$ so that $\mathcal{E}'$ has positive degree. By assumption, $\mathcal{E}'$ is ample. Consider the following exact sequence

$$0 \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E} \longleftarrow \mathcal{E}'' := \mathcal{E}/\mathcal{E}' \longrightarrow 0 \quad (147)$$

Note that quotient bundles of $\mathcal{E}''$ are also quotient bundles of $\mathcal{E}$ and hence have positive degree. By inductive hypothesis, we know $\mathcal{E}''$ is ample. Then by Corollary 4.7, we get $\mathcal{E}$ is ample.

“ $\Leftarrow$ ”: Let $\mathcal{E}$ be a stable vector bundle with positive degree. For all nontrivial quotient bundle $\mathcal{E} \twoheadrightarrow \mathcal{E}''$, as $\mathcal{E}$ is stable, we know $\mu(\mathcal{E}'') > \mu(\mathcal{E}) > 0$ so that $\mathcal{E}''$ has positive degree. Then by assumption, $\mathcal{E}$ is ample. $\square$

To relate stability and ampleness, we need a numerative criterion for ampleness.

Theorem 6.1 (Seshadri). *Let X be a complete scheme, D Cartier divisor on X . Then D is ample on X if and only if there exists $\varepsilon > 0$ such that for all integral curve $C \subset X$, we have that $\frac{D \cdot C}{m(C)} > \varepsilon$, where $m(C) := \max_{p \in C} \{\text{multi}_p(C)\}$.*

Proof. “ $\Rightarrow$ ”: Take m large enough such that mD is very ample. Then mD associates with an immersion $X \hookrightarrow \mathbb{P}^N$. Since X is proper, the immersion is closed. Take hyperplane $H \subseteq \mathbb{P}^N$

such that $H \cap X = mD$. Then $(mD \cdot C)_X = (H \cdot C)_{\mathbb{P}^N} = \deg C \geq m(C)$. Hence $\frac{(D \cdot C)_X}{m(C)} \geq \frac{1}{m}$ and we may take $\varepsilon = \frac{1}{2m}$.

“ $\Leftarrow$ ”: We may assume X is integral. Induct on $\dim X = n$. For $n = 1$, then there is no integral closed subscheme of higher dimension. As $D \cdot C > \varepsilon \cdot m(C) \geq 0$, by Nakai-Moishezon criterion, D is ample. When $n > 1$, by induction hypothesis, we know $D^s \cdot Y > 0$ for all integral closed subscheme $Y \subset X$ of dimension $s < n$. Again by Nakai-Moishezon criterion, suffices to show $D^n > 0$.

Take a smooth closed point of X and consider the blow up $f : X' \rightarrow X$ at p . Assume E is the exceptional divisor and $D' = f^*D$ the total transform of D . Then $f^{-1}(p) \cong \mathbb{P}^{n-1}$ and $\mathcal{O}_X(E)|_E = \mathcal{O}_E(-1)$. Set $G := aD' - E$ where $a = \frac{1}{\varepsilon}$. Want to show G is nef.

For all integral curve $C' \subset X'$. If $C' \subseteq E$, then $G \cdot C' = aD' \cdot C' - E \cdot C' = -E \cdot C' = \deg C' > 0$. Or if $C' \cap E' \neq \emptyset$ but $C' \not\subseteq E$, then $G \cdot C' = aD' \cdot C' - E \cdot C' = aD \cdot C - \text{multi}_p(C) > 0$, where C is the image of C' under f . Otherwise, $C' \cap E = \emptyset$, then $G \cdot C' = aD' \cdot C' - E \cdot C' = aD \cdot C > 0$. Thus by numerical criterion for nefness, G is nef.

Now G is nef. Note that $(D')^n = D^n$, $E^n = (E^{n-1})_E = (-1)^{n-1}$ and $(D')^i \cdot E^{n-i} = 0$ for all $0 < i < n$.

$$\begin{aligned} 0 \leq G^n &= (aD' - E)^n \\ &= (aD')^n - E^n \\ &= a^n D^n - 1 \end{aligned} \tag{148}$$

which implies that $D^n > 0$. □

In Hartshorne's paper, he proved the following deep theorem with results of Narasimhan and Seshadri relating stable vector bundles to certain group representations. Here we quote it without proof.

Theorem 6.2. *Let X be a complete smooth complex curve of genus $g \geq 2$, $\mathcal{E}$ stable vector bundle on X . Then $S^n \mathcal{E}$ is semi-stable for all $n \geq 1$.*

Lemma 6.4. *Let $f : X \rightarrow Y$ be a finite surjection between integral noetherian schemes. sometimes called finite cover. Then we have*

- (1) $R^i f_* \mathcal{F} = 0$ for all coherent $\mathcal{F}$ and $i > 0$
- (2) $\text{rank}(f_* \mathcal{F}) = \deg f \cdot \text{rank } \mathcal{F}$ for all coherent $\mathcal{F}$, where $\deg f = [K(X) : K(Y)]$.
- (3) $\deg(f_* \mathcal{F}) = \deg \mathcal{F} + \text{rank } \mathcal{F} \cdot \deg(f_* \mathcal{F})$ for all coherent $\mathcal{F}$.

Proof. For (1), we may assume $Y = \text{Spec } A$ and $X = \text{Spec } B$ with A finite over B . Then $R^i f_* \mathcal{F} = H^i(X, \mathcal{F})$, which by Serre vanishing is zero.

For (2), we can also $Y = \text{Spec } A$ and $X = \text{Spec } B$ with A finite over B . Assume $\mathcal{F}$ is given by M finite over B . Then $\deg f = [\text{Frac}(B) : \text{Frac}(A)]$, $\text{rank}(f_* \mathcal{F}) = \dim_{\text{Frac}(A)}(M \otimes_A \text{Frac}(A))$ and $\text{rank } \mathcal{F} = \dim_{\text{Frac}(B)}(M \otimes_B \text{Frac}(B))$. As f is surjective, the corresponding $\varphi : A \rightarrow B$ is injective.

Note that there is a natural map $\psi : B \otimes_A \text{Frac}(A) \rightarrow \text{Frac}(B)$. It is clear that ψ is injective. And since φ is integral, for all $b \in B$, we have $b^n + a_1 b^{n-1} + \cdots + a_{n-1} b + a_n = 0$

with $a_i \in A$. Then $b(b^{n-1} + a_1b^{n-2} + \dots + a_{n-1}) = -a_n \in A$ so that $\psi((b^{n-1} + a_1b^{n-2} + \dots + a_{n-1}) \otimes \frac{1}{-a_n}) = \frac{1}{b}$. Thus ψ is isomorphic and we get

$$\begin{aligned}
\text{rank}(f_*\mathcal{F}) &= \dim_{\text{Frac}(A)}(M \otimes_A \text{Frac}(A)) \\
&= \dim_{\text{Frac}(A)}(M \otimes_B (B \otimes_A \text{Frac}(A))) \\
&= \dim_{\text{Frac}(A)}(M \otimes_B \text{Frac}(B)) \\
&= [\text{Frac}(B) : \text{Frac}(A)] \cdot \dim_{\text{Frac}(B)}(M \otimes_B \text{Frac}(B)) \\
&= \deg f \cdot \text{rank } \mathcal{F}
\end{aligned} \tag{149}$$

For (3), this immediately comes from (1) and (2) as following

$$\begin{aligned}
\deg(f_*\mathcal{F}) &= \chi(Y, f_*\mathcal{F}) - \text{rank}(f_*\mathcal{F}) \cdot \chi(Y, \mathcal{O}_Y) \\
&= \chi(X, \mathcal{F}) - \deg f \cdot \text{rank } \mathcal{F} \cdot \chi(Y, \mathcal{O}_Y) \\
&= \deg \mathcal{F} + \text{rank } \mathcal{F} \cdot \chi(X, \mathcal{O}_X) - \deg f \cdot \text{rank } \mathcal{F} \cdot \chi(Y, \mathcal{O}_Y) \\
&= \deg \mathcal{F} + \text{rank } \mathcal{F} \cdot \chi(Y, f_*\mathcal{O}_X) - \text{rank}(f_*\mathcal{O}_X) \cdot \text{rank } \mathcal{F} \cdot \chi(Y, \mathcal{O}_Y) \\
&= \deg \mathcal{F} + \text{rank } \mathcal{F} \cdot \deg(f_*\mathcal{F})
\end{aligned} \tag{150}$$

done! □

From now on, we again assume X to be a smooth projective curve over $\mathbb{C}$.

Proposition 6.9. *Let $\mathcal{E}$ be a vector bundle on X of positive degree satisfying that $S^n\mathcal{E}$ is semi-stable for all n . Then $\mathcal{E}$ is ample.*

Proof. Assume $\text{rank } \mathcal{E} = r$ and $\deg \mathcal{E} = d$. Consider the associated projective bundle $\pi : \mathbb{P}(\mathcal{E})$. Then by Proposition 4.11, suffices to show the tautological line bundle $\mathcal{L} := \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ is ample. And by Theorem 6.1, want to find $\varepsilon > 0$ such that $\mathcal{L} \cdot C > \varepsilon \cdot m(C)$ for all integral curve $C \subseteq \mathbb{P}(\mathcal{E})$.

If $\pi(C)$ is a single point, then C is contained in some fiber of π , which is isomorphic to $\mathbb{P}_{\mathbb{C}}^{r-1}$. Hence $\mathcal{L} \cdot C = \deg C \geq m(C)$. When $\pi(C)$ is not a single point, then $\dim(\pi(C)) = 1$ so that $\pi(C) = X$ and $\pi|_C : C \rightarrow X$ is a finite surjection. Then by Lemma 6.4, $\text{rank}((\pi|_C)_*(\mathcal{L}^m|_C)) = \deg(\pi|_C)$ and $\deg((\pi|_C)_*(\mathcal{L}^m|_C)) = \deg(\mathcal{L}^m|_C) + \deg((\pi|_C)_*\mathcal{O}_C)$. Consider the following surjection

$$S^m\mathcal{E} \cong (\pi|_C)_*(\mathcal{L}^m) \twoheadrightarrow (\pi|_C)_*(\mathcal{L}^m|_C) \tag{151}$$

Note that pushforward preserves torsion-freeness and X is a regular curve, $(\pi|_C)_*(\mathcal{L}^m|_C)$ is locally free. By assumption, $S^m\mathcal{E}$ is semi-ample so that $\mu(S^m\mathcal{E}) \leq \mu((\pi|_C)_*(\mathcal{L}^m|_C))$. Then we have

$$\begin{aligned}
\frac{dm}{r} &\leq \frac{\deg(\mathcal{L}^m|_C) + \deg((\pi|_C)_*\mathcal{O}_C)}{\deg(\pi|_C)} \\
\frac{dm \cdot \deg(\pi|_C)}{r} &\leq \deg(\mathcal{L}^m|_C) + \deg((\pi|_C)_*\mathcal{O}_C) \\
\frac{d \cdot \deg(\pi|_C)}{r} &\leq \frac{1}{m} \cdot (\deg(\mathcal{L}^m|_C) + \deg((\pi|_C)_*\mathcal{O}_C))
\end{aligned} \tag{152}$$

Note that $m \cdot \deg(\mathcal{L}|_C) = \deg(\mathcal{L}^m|_C) = \chi(\mathcal{L}^m|_C) - \chi(\mathcal{O}_C)$. By definition of intersection number, $\mathcal{L} \cdot C = \deg(\mathcal{L}|_C)$. Hence

$$\frac{d \cdot \deg(\pi|_C)}{r} \leq \mathcal{L} \cdot C + \frac{1}{m} \cdot \deg((\pi|_C)_* \mathcal{O}_C) \quad (153)$$

Taking m converging to ∞ , we get

$$\frac{d \cdot \deg(\pi|_C)}{r} \leq \mathcal{L} \cdot C \quad (154)$$

Note that $\deg(\pi|_C) \geq m(C)$, we are done! $\square$

Remark 6.3. *It is not clear for me to see why $\deg(\pi|_C) \geq m(C)$.*

Theorem 6.3. *Vector bundle $\mathcal{E}$ is ample if and only if every quotient bundle of $\mathcal{E}$ has positive degree.*

Proof. If $\mathcal{E}$ is ample, then clearly every quotient bundle of $\mathcal{E}$ is also ample. And ample vector bundle automatically has positive degree. Conversely, by Proposition 6.8, suffices to prove every stable vector bundle of positive degree is ample. Consider stable vector bundle $\mathcal{E}$ of positive degree. When genus $g \geq 2$, by Theorem 6.2, $S^n \mathcal{E}$ is semi-stable for all $n \geq 1$. Then by Proposition 6.9, $\mathcal{E}$ is ample. For $g = 1$, this is a result by Atiyah. And for $g = 0$, X is a rational curve. Note that stable vector bundle is indecomposable, while the only indecomposable vector bundle is line bundle and line bundle of positive degree is ample, done! $\square$

Lemma 6.5. *Let $\mathcal{E}$ be a vector bundle on X , $s \in \Gamma(X, \mathcal{E})$ not vanishing at any singular point of X . Then $\mathcal{E}$ has a line subbundle $\mathcal{L}$ such that $s \in \Gamma(X, \mathcal{L})$. Moreover, if s is not invertible, $\mathcal{L}$ is ample.*

Proof. Note that s would give a rational section $\tau : X \dashrightarrow \mathbb{P}(\mathcal{E})$ defined on X minus finitely many regular points. Hence τ would extend to a regular section. Then just as what we did in Lemma 5.2, we get a line subbundle $\mathcal{L}$ of $\mathcal{E}$. In addition, s is a global section of $\mathcal{L}$. If s is not invertible, then $\deg \mathcal{L} = \deg(\text{div}(s)) \geq 1 > 0$ so that $\mathcal{L}$ is ample. $\square$

Proposition 6.10. *Let Y be a proper variety over algebraically closed field k , $\mathcal{E}$ vector bundle over Y globally generated. Then $\mathcal{E}$ is ample if and only if every quotient line bundle of $\mathcal{E}|_C$ is ample for all curve $C \subset Y$.*

Proof. If $\mathcal{E}$ is ample, since $C \hookrightarrow Y$ is closed immersion, $\mathcal{E}|_C$ is still ample by Corollary 4.3. Hence every quotient line bundle of $\mathcal{E}|_C$ is also ample. Conversely, suppose that every quotient line bundle of $\mathcal{E}|_C$ is ample for all curve $C \subset Y$.

First consider a line bundle $\mathcal{E}$. As $\mathcal{E}$ is globally generated, it induces a morphism $\varphi : Y \rightarrow \mathbb{P}^N$ such that $\mathcal{E} \cong \varphi^*(\mathcal{O}(1))$. Suppose φ is not quasi-finite, then there exists some fiber Y' of dimension ≥ 1 which would contain some curve C . Hence $\mathcal{E}|_C \cong \varphi^*(\mathcal{O}(1)|_{\varphi(C)})$ is trivial and of degree 0. While by assumption $\mathcal{E}|_C$ is ample, contradiction. Thus φ is quasi-finite. As Y is proper, we know φ is in fact finite so that by Proposition 4.5, we know $\mathcal{E}$ is ample.

When Y is a curve, we prove the result by induction on $\text{rank } \mathcal{E}$. By inductive hypothesis, every nontrivial quotient bundle of $\mathcal{E}$ is ample. Hence we suffice to find an ample line subbundle of $\mathcal{E}$. By Lemma 6.5, only need to find a section which does not vanish at any singular point of Y but not invertible.

Set $n = h^0(Y, \mathcal{E})$. As $\mathcal{E}$ is globally generated, there is a surjection $\psi : \mathcal{O}_Y^{\oplus n} \twoheadrightarrow \mathcal{E}$. Consider the Grassmannian scheme $\text{Gr}(\text{rank } \mathcal{E}, n)$ over k with universal quotient bundle $\mathcal{U}$. Then ψ would corresponds to a morphism $f : Y \rightarrow \text{Gr}(\text{rank } \mathcal{E}, n)$ satisfying that $\mathcal{E} \cong f^*\mathcal{U}$.

If $f(Y)$ is a single point, then similarly we get $\mathcal{E}$ trivial so that trivial line bundle is quotient line bundle of $\mathcal{E}$, contradicting to ample assumption. Then $\dim f(Y) = 1$ so that we can find $y \in Y$ such that $f(y)$ is not image of any singular point of Y . Note that for any two points z and w in $\text{Gr}(\text{rank } \mathcal{E}, n)$, $z = w$ if and only if

$$\Gamma(\text{Gr}(\text{rank } \mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_z) = \Gamma(\text{Gr}(\text{rank } \mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_w) \quad (155)$$

Thus $\Gamma(\text{Gr}(\text{rank } \mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_{f(y)}) \cap \Gamma(\text{Gr}(\text{rank } \mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_{f(y')}) \subsetneq \Gamma(\text{Gr}(\text{rank } \mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_{f(y')})$ for singular $f(y')$. Since k is infinite, we get

$$\Gamma(\text{Gr}(\text{rank } \mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_{f(x)}) \subsetneq \cup_{y' \text{ singular}} \Gamma(\text{Gr}(\text{rank } \mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_{f(y')}) \quad (156)$$

So we may pick global section s of $\mathcal{U}$ not vanishing at any $f(y')$. Pull back to Y , we get global section f^*s of $\mathcal{E}$ not vanishing at any singular y' .

Now if Y is an arbitrary proper variety, consider the associated projective bundle $\pi : \mathbb{P}(\mathcal{E}) \rightarrow Y$. By Proposition 4.11, suffices to show the tautological line bundle $\mathcal{L} := \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ is ample. Let C' be any curve in $\mathbb{P}(\mathcal{E})$. If C' is contained in some fiber, then $\mathcal{L}|_{C'}$ is ample on C' . Otherwise, $\pi|_{C'} : C' \rightarrow \pi(C')$ is a finite surjection and $\pi(C')$ is a curve in Y . By argument above, $\mathcal{E}|_{\pi(C')}$ is ample and hence $(\pi|_{C'})^*\mathcal{E}$ is ample on C' . As there is a surjection $\pi^*\mathcal{E} \twoheadrightarrow \mathcal{L}$, we also get $\mathcal{L}|_{C'}$ ample. Since $\mathcal{L}$ is globally generated, by argument above, $\mathcal{L}$ is ample and we are done! $\square$

7 Moduli Space of Vector Bundles

7.1 Hilbert scheme

Recall two local criterion for flatness.

Proposition 7.1. *Let $(A, \mathfrak{m})$ be a local ring with residue field k , M finitely generated A -module. Then the following conditions are equivalent:*

- (1) M is free
- (2) M is projective
- (3) M is flat
- (4) $\text{Tor}_1^A(M, k) = 0$.

Proof. (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) is clear. Suffices to prove (4) $\Rightarrow$ (1). As M is finitely generated, by Nakayama's Lemma, we can pick representative elements of a basis of $M/\mathfrak{m}M$ over k , which

also generate M . And these elements induces a surjection $A^{\oplus n} \rightarrow M$ with kernel N . Since $\mathrm{Tor}_1^A(M, k) = 0$, the following sequence is exact

$$0 \longrightarrow N \otimes_A k \longrightarrow A^{\oplus n} \otimes_A k \longrightarrow M \otimes_A k \longrightarrow 0 \quad (157)$$

Note that by our choice, $A^{\oplus n} \otimes_A k \rightarrow M \otimes_A k$ is an isomorphism. Hence $N \otimes_A k = 0$ so that by Nakayama's Lemma again, $N = 0$. Conclude that $M \cong A^{\oplus n}$ is free. $\square$

Proposition 7.2. *Let $f : (A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$ be a local homomorphism between local rings, M finitely generated B -module. Then M is flat over A if and only if $\mathrm{Tor}_1^A(M, A/\mathfrak{m}) = 0$.*

Remark 7.1. *This result is much more tricky. Proof can be seen in stacks.math.columbia.edu/tag/00MK.*

Corollary 7.1. *Let $f : (A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$ be a local homomorphism between local rings, M finitely generated B -module. Assume B is flat over A . Then M is flat over A if and only if $\mathrm{Tor}_1^A(M, B \otimes_A A/\mathfrak{m}) = 0$.*

Proof. By Proposition 7.2, suffices to show that if $\mathrm{Tor}_1^A(M, B \otimes_A A/\mathfrak{m}) = 0$, then $\mathrm{Tor}_1^A(M, A/\mathfrak{m}) = 0$. Consider the following exact sequence

$$0 \longrightarrow \mathrm{Tor}_1^A(M, A/\mathfrak{m}) \longrightarrow \mathfrak{m} \otimes_A M \longrightarrow M \longrightarrow A/\mathfrak{m} \otimes_A M \longrightarrow 0 \quad (158)$$

Since B is flat over A , tensoring by B , we get exact sequence

$$0 \longrightarrow \mathrm{Tor}_1^A(M, A/\mathfrak{m}) \otimes_A B \longrightarrow \mathfrak{m} \otimes_A M \otimes_A B \longrightarrow M \otimes_A B \longrightarrow B \otimes_A A/\mathfrak{m} \otimes_A M \longrightarrow 0 \quad (159)$$

On the other hand, we also have the following exact sequence

$$0 \longrightarrow \mathrm{Tor}_1^A(M, B \otimes_A A/\mathfrak{m}) \otimes_A B \longrightarrow \mathfrak{m} \otimes_A M \otimes_A B \longrightarrow M \otimes_A B \longrightarrow B \otimes_A A/\mathfrak{m} \otimes_A M \longrightarrow 0 \quad (160)$$

Hence we get $\mathrm{Tor}_1^A(M, A/\mathfrak{m}) \otimes_A B = 0$. While flat local homomorphism is automatically faithfully flat, by faithfully flat descent, we get $\mathrm{Tor}_1^A(M, A/\mathfrak{m}) = 0$. $\square$

Definition 7.1. *Let $f : X \rightarrow Y$ be a morphism, $\mathcal{F}$ sheaf module on X . We say $\mathcal{F}$ is flat over Y if for all $x \in Y$, $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$. In particular, if $\mathcal{O}_X$ is flat over Y , we say X is flat over Y .*

Given X flat over Y , $\mathcal{F}$ sheaf module on X . If there is a locally free resolution $\mathcal{L}^\bullet$ of $\mathcal{F}$, then for all $y \in Y$, restricting to fibered product $X(y) := X \times_Y \mathrm{Spec} \mathcal{O}_{Y,y}$, we get locally free resolution $\mathcal{L}^\bullet|_{X(y)}$ of $\mathcal{F}|_{X(y)}$ by flatness.

Let us introduce some results to better describe derived pushforward.

Lemma 7.1. *The derived pushforward $R^i f_* \mathcal{F}$ can be identified with the sheafification of the presheaf*

$$U \longmapsto H^i(f^{-1}(U), \mathcal{F}) \quad (161)$$

Proof. Take injective resolution $0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}^\bullet$. By definition, $R^i f_* \mathcal{F}$ is the i th cohomology of the following complex

$$0 \longrightarrow f_* \mathcal{I}^0 \xrightarrow{\psi_0} f_* \mathcal{I}^1 \xrightarrow{\psi_1} f_* \mathcal{I}^2 \xrightarrow{\psi_2} \dots \quad (162)$$

Hence $R^i f_* \mathcal{F}$ is the sheafification of presheaf $U \mapsto (\ker \psi_i)(U)/(\operatorname{im} \psi_{i-1})(U)$. Applying $\Gamma(U, \cdot)$, then we get

$$0 \longrightarrow \Gamma(f^{-1}(U), \mathcal{I}^0) \xrightarrow{\psi_0(U)} \Gamma(f^{-1}(U), \mathcal{I}^1) \xrightarrow{\psi_1(U)} \Gamma(f^{-1}(U), \mathcal{I}^2) \xrightarrow{\psi_2(U)} \dots \quad (163)$$

Note that $\operatorname{im} \psi_i$ is the sheafification of the presheaf $U \mapsto \operatorname{im} \psi_i(U)$. Checking stalkwise, it is clear that $R^i f_* \mathcal{F}$ is the sheafification of presheaf $U \mapsto (\ker \psi_i)(U)/\operatorname{im} \psi_{i-1}(U) = H^i(\Gamma(f^{-1}(U), \mathcal{I}^\bullet)) = H^i(f^{-1}(U), \mathcal{F})$. $\square$

Proposition 7.3.